

The R Rule

The collected substack post on the R rule substack



(The famous triple spiral - a Neolithic carving inside the Newgrange passage tomb)

I hope to write a series of posts on the R rule - without the [AI Odyssey](#) (or [Sprite World](#)) metaphorical crutch - on this substack.

The R rule was first proposed in a [post on a hypothetical world of process](#) (September 2025) and then [adopted for use](#) in the AI Odyssey in October 2025 where it became the driving force for [Telemachus](#).

However, I think it has real value as a cross discipline way of modelling learning and understanding - the reason for this series of posts.

My mode of working has been a series of “dialectic explorations” (“what-if” prompts), mostly with ChatGPT-5 (5.1 from World models post onwards), that I hope will result in posts that build up to a new picture, metaphor or framing for our extraordinary time. I’m “testing” along the way with Claud 4.5 Sonnet and Gemini 2.5 Pro (for now). Building on my previous writings but as a more targeted, hopefully coherent, argument.

Recently, I’ve joined the Cambridge University library as temporary reader for some necessary background research but we (ChatGPT and I)

can always be lead astray - so for now the ideas are meant to be taken as speculative but hopefully providing insight into our overheating AI age.

We end the first part of this series with a geometric interpretation of the R-rule as a minimal self in a probabilistic world: The [R Rotor](#).

Thanks for reading!

Andre,

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October 2025

Introducing the R-Rule

A Common Principle Linking Energy, Inference, and explorative Learning

[Andre Kramer](#)

Oct 27, 2025



“What if learning, cooling, and quantum evolution are the same process?”

What if a neural network training, a hot gas equilibrating, and a wavefunction evolving all follow the same recursive law—one that

reduces tension between expectation and reality? If that's true, it means intelligence, thermodynamics, and physics aren't separate stories at all, but different dialects of the same equation. And in our overheating AI age, that insight might change how we think about both learning *and* survival."

Across physics, cognition, and computation, systems appear to follow the same recursive law: they reduce the tension between what is expected and what is encountered. From Boltzmann's distribution of energies to Bayes' rule, gradient descent, and the Schrödinger equation, each domain expresses this logic in its own form. The **R-rule** generalizes these relations into a single update: a system that adjusts both its probabilities and phases to restore coherence. Seen this way, energy becomes inference, and learning becomes the universal calculus of change connecting matter, mind, and machine.

Thanks for reading Andre on the R rule! Subscribe for free to receive new posts and support my work.

A simple but far-reaching pattern recurs in learning, physics and thought.

Systems sustain themselves by **recursively minimizing tension** between what they *expect* and what they *encounter*.

Whether in the cooling of matter, the training of a neural network, or the evolution of a wavefunction, the same calculus of difference links energy, information, and inference. Yet this view also reveals its own boundary—our probabilistic, process-based metaphysics implies that prediction and learning are never complete. The gaps themselves become the source of insight and transformation.

This post sketches that shared logic at a high level. Later essays will develop each domain in more technical detail. Future work extends this framework toward active inference and predictive processing in cognitive systems.

1. Boltzmann Statistics

In Boltzmann's formulation, nature balances energy and probability.

Each microstate i has probability

$$p_i = e^{(-E_i/kT)} / Z,$$

$$p_i = \frac{e^{-E_i/kT}}{Z},$$

weighted by its energy E_i and temperature T , with partition function Z ensuring normalization.

The system relaxes toward equilibrium by minimizing free energy

$$F = \langle E \rangle - TS,$$

$$F = \langle E \rangle - TS.$$

where $S = -k \sum_i p_i \ln p_i$ is entropy.

$$S = -k \sum_i p_i \ln p_i$$

This is learning in its simplest thermodynamic form:

a structure adjusting itself until surprise is minimized.

1.5. Bayesian Inference

Bayes' rule formalizes how probabilities update when new information arrives:

$$P(H | D) = P(D | H) P(H) / P(D).$$

$$P(H|D) = \frac{P(D|H) P(H)}{P(D)}.$$

It describes the simplest possible learning loop:

a prior belief $P(H)$ meets data D , producing an updated posterior $P(H | D)$.

Boltzmann's distribution already has this structure.

If we identify energy as the *negative log-likelihood* of a hypothesis,

$$E_i = -kT \ln P(D | H_i),$$

$$E_i = -kT \ln P(D | H_i),$$

then the Boltzmann weight $e^{(-E_i/kT)}$ is proportional to the likelihood $P(D | H_i)$.

Temperature T controls how strongly prior structure resists revision—low T means sharp posteriors, high T broader uncertainty.

Both expressions describe **reweighting possibilities**:

in physics, by energetic plausibility; in inference, by evidential support.

Boltzmann's statistical mechanics thus anticipates Bayes' logic:

matter behaves as if it were *updating its expectations* to minimize free energy.

2. Machine Learning

Modern learning algorithms enact this Bayesian logic dynamically.

A model updates its internal parameters w to reduce prediction error:

$$\Delta w = -\eta \nabla L(w),$$

$$\Delta w = -\eta \nabla L(w),$$

where $L(w)$ is a loss (negative log-likelihood) and η a learning rate.

Gradient descent performs continuous Bayesian updating in parameter space,

transforming discrepancy into structure.

Boltzmann's energy gradients become gradients of belief;

entropy becomes uncertainty;

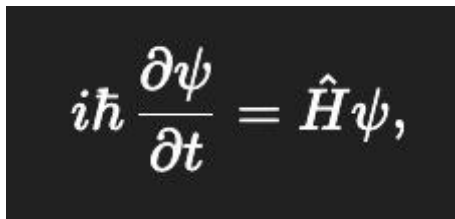
and η sets how quickly tension dissipates.

3. Quantum Mechanics

Quantum systems also minimize tension—this time among *phases* rather than energies.

The wavefunction $\psi(x,t)$ evolves according to

$$i\hbar \partial \psi / \partial t = \hat{H} \psi,$$


$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi,$$

where the Hamiltonian $\hat{H} = \hat{p}^2 / 2m + V(x,t)$

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x, t)$$

governs the balance of kinetic and potential energy.

Destructive interference cancels inconsistent paths; constructive interference reinforces coherent ones.

Where Boltzmann's world relaxes toward equilibrium, the quantum world settles into *coherence*:

difference resolved in complex amplitude space.

4. Dialectic

In philosophical dialectic, tension appears as **contradiction**—the clash between opposing concepts or forces.

Through recursive self-reference, these opposites transform one another, producing higher-order coherence.

Hegel called this *the labor of the negative*; today, we might call it feedback

or recursive error correction.

In every case, becoming proceeds by metabolizing contradiction.

5. Computing

In computation, this same logic appears as **recursion**—a process calling itself until input and output stabilize.

Classical computation performs this search **symbolically and sequentially**, evaluating discrete steps toward a fixed point.

Quantum computation performs it **coherently and in parallel**: amplitudes evolve in superposition, and interference prunes inconsistent solutions.

Both realize the same recursive architecture—a system exploring its own possibility space until difference resolves into consistency.

6. The Bridge: Boltzmann → R-rule → Schrödinger

The step from statistical relaxation to coherent evolution can be captured by a recursive update of the form

$$\psi' = \psi + \eta \sqrt{p(1-p)} [\alpha \sin \theta A - i \beta \cos \theta N],$$

$$\psi' = \psi + \eta \sqrt{p(1-p)} [\alpha \sin \theta A - i \beta \cos \theta N].$$

where $p = |\psi|^2 / \sum_b |\psi_b|^2$ represents relative probability.

$$p = |\psi|^2 / \sum_b |\psi_b|^2$$

Here:

- η is a learning rate (a thermodynamic hyperparameter),
- A is an exploratory or *actor* model,
- N a constraining or *normative* model,
- Each evolves under complementary drives, modulated by two hyperparameters:

- α governs the **A-channel**, biasing outward adaptation or exploration.
- β governs the **N-channel**, biasing inward stabilization or normative correction.
- and θ a balance between the two channels.

This **R-rule** (R for rotation, recursion, reflexivity) generalizes

Boltzmann's principle into a dynamical form:

a system continually updating both amplitude and phase to reduce internal tension.



Clarifying the Normalization Step

Think of the conversion to $p(a)$ as a **softmax-like normalization**.

It isn't quantum mechanics per se, but it performs a similar function:

a collapse from many potential actions b into the currently realized one

a .

Just as the softmax converts relative activations into a probability

distribution, here the normalization over $|\psi(b)|^2$ weights all possible

trajectories or configurations, producing a single *probabilistic*

commitment to a.

It's the bridge between exploration and selection — between *thinking about everything that could be done* and *actually doing one thing*.

In other words, the R-rule doesn't "measure" a quantum wavefunction, but it performs an analogous operation in learning space: a dynamic soft collapse of uncertainty into a specific act, guided by energy gradients and internal coherence.

Just as softmax turns activations into a probability distribution, this normalization over $|\psi(b)|^2$ weights all possible configurations, producing a single probabilistic commitment. It's the bridge between exploration and selection — between *thinking about everything that could be done* and *actually doing one thing*. 💡

In the continuous limit ($\eta \rightarrow 0$), the R rule converges to the Schrödinger equation.

The bridge between Boltzmann and quantum mechanics is thus not

mystical but mathematical:

probability acquiring feedback—energy learning to predict itself.

7. Toward a Common Calculus

Seen in this light, Boltzmann statistics, Bayesian inference, machine learning, quantum mechanics, dialectic reasoning, life and computation are not separate inventions but **different expressions of a single recursive grammar**.

Each describes a world in which *difference* drives self-organization and *recursion* transforms disorder into form.

Future posts will (hopefully) expand each domain or strands:

- Automated (commonly called unsupervised / machine) learning.
- Predictive processing in cognition.
- Quantum mechanics as coherent search.
- Dialectic as the meta-logic learning of recursive systems.

- **The limits of such prediction and learning** within a probabilistic, process-based metaphysics — where insight arises from what cannot be compressed, and where our models meet their own incompleteness.

For now, the key idea is simple:

Across nature, mind, and machine, systems persist by folding difference back into themselves—
transforming tension into structure, and contradiction into learning.

The **R-rule** unites Boltzmann, Bayes, and Schrödinger as one recursive principle—showing how the universe learns by folding difference back into itself.

Toward the Next Post: A Litmus Test

Before we embark on that more speculative exploration — where imagination, dialectic, and self-modeling become central — we owe a litmus test.

We must show, concretely, that the **R-rule** is not just metaphor or philosophy, but a unifying dynamic: one that **incorporates both Boltzmann machines** (from statistical learning, now recognized with the *2024 Nobel Prize*) **and Hamiltonian Monte Carlo exploration** (from modern Bayesian inference).

In our next post, we'll demonstrate exactly that — a concrete simulation showing how the R-rule can reproduce both *gradient-based settling (exploitation)* and *momentum-based exploration*, in a single evolving system.



[Andre on the R rule](#)

Exploitation AND Exploration

A litmus test for the R rule

[Andre Kramer](#)

Oct 28, 2025

What does it mean for a system to *learn*?

To descend the gradients of error, or to explore the unknown?



In our last post, we proposed the **R-rule** as a common dynamical principle linking energy, inference, and learning. It suggested that Boltzmann's physics, Bayesian reasoning, and even quantum evolution might all share a single recursive form — one that balances prediction and surprise, coherence and contradiction.

But we left an important question hanging:
does it *work*?

Before we wander into more speculative terrain, we need a **litmus test** — a concrete demonstration that the R-rule does what we claim. That it really can bridge the gap between exploitation and exploration; between systems that cool and systems that move; between learning from the world and imagining beyond it.

In this post, we'll make that case step by step.

First, we'll show how the R-rule **emerges naturally** from two known worlds:

- **Boltzmann machines**, which exploit — descending directly down energy gradients to find equilibrium; and
- **Hamiltonian Monte Carlo**, which explore — conserving momentum and coherence but never quite settling.

Then we'll demonstrate, through simulation, that the R-rule **combines the strengths of both**: exploring freely at first, then stabilizing in deeper, more coherent equilibria. Along the way, we'll discuss **Lyapunov stability** and **Hamiltonian dynamics** — two sides of the same coin that explain why the R-rule both moves and learns.

In future, we'll look beyond machine learning: toward **recursive meaning-making** itself.

Because in the end, the same dialectic that governs energy and motion — *exploration and exploitation* — may also underlie how minds, natural or artificial, come to know themselves.

1. From Boltzmann Machines: Learning by Cooling

Boltzmann machines learn by *cooling*.

They descend an energy landscape until thermal noise dies away, settling into a configuration that best matches the world they've sampled.

Formally, a Boltzmann machine defines an energy function:

$$E(s) = -\frac{1}{2} s^T W s - b^T s$$

$$E(s) = -\frac{1}{2} s^T W s - b^T s$$

and evolves toward configurations s that minimize it.

In continuous form, this descent can be written as:

$$ds/dt = -\partial E/\partial s + \xi(t)$$

$$\frac{ds}{dt} = -\frac{\partial E}{\partial s} + \xi(t)$$

where $\xi(t)$ is small stochastic noise.

This is **Langevin dynamics**—a noisy relaxation toward equilibrium.

Each step follows the gradient of the energy:

$$s' = s - \eta \partial E / \partial s + (2\eta T \epsilon)^{1/2}$$

$$s' = s - \eta \frac{\partial E}{\partial s} + \sqrt{2\eta T} \epsilon$$

where η is the learning rate, T is temperature, and ϵ is random noise.

At high temperature, the system explores.

At low temperature, it freezes.

Cooling is convergence.

In other words, Boltzmann learning is **pure exploitation**: it seeks stability, not novelty. Once equilibrium is reached, motion stops; the model remembers, but it no longer imagines.

In the **R-rule**, this becomes the $\alpha \sin \theta A(a, \psi)$ term — the *actor* component that still drives downward along energy gradients. But, as we'll see next, that's only half of what learning requires.

Boltzmann's descent gives the world shape;

the R-rule adds the momentum that lets it move through that shape.

2. From Hamiltonian Monte Carlo: Motion Without Rest

If Boltzmann machines learn by cooling, **Hamiltonian systems learn by moving**.

A Hamiltonian system doesn't minimize energy; it *conserves* it.

Where Boltzmann dynamics slide downhill, Hamiltonian dynamics orbit

endlessly, carrying momentum across valleys and barriers. This makes them ideal for exploration — for traversing complex spaces without getting trapped.

Formally, the system evolves according to **Hamilton's equations**:

$$\dot{q} = \partial H / \partial p, \quad \dot{p} = -\partial H / \partial q$$

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}$$

where q represents position (the current state) and p represents momentum (the conjugate drive).

Here, $H(\mathbf{q}, \mathbf{p})$ is the **Hamiltonian**, a function that represents the *total energy* of the system:

$$H(\mathbf{q}, \mathbf{p}) = E(\mathbf{q}) + K(\mathbf{p}),$$

$$H(q, p) = E(q) + K(p),$$

where

- $E(q)$ is the **potential energy**—the energy of position, reflecting how well the current state fits the world, and
- $K(p)$ is the **kinetic energy**—the energy of motion, representing the system's momentum or exploratory drive.

In **Hamiltonian Monte Carlo (HMC)** — a well-known extension to **Bayesian inference methods** —these equations are discretized to explore probabilistic landscapes:

$$q' = q + \epsilon \partial H / \partial p, \quad p' = p - \epsilon \partial H / \partial q$$

$$q' = q + \epsilon \frac{\partial H}{\partial p}, \quad p' = p - \epsilon \frac{\partial H}{\partial q}$$

The method's power lies in the coupling: momentum helps the sampler move smoothly across regions of low gradient, allowing it to leap over local minima.

But there's a catch: without dissipation, **it never stops**.

Energy is conserved, not minimized. The system explores forever, coherent but restless.

In the **R-rule**, this is captured by the $-i\beta \cos\theta N(a, \psi)$ term — the *normative* or *self-model* channel.

It represents the system's inner coherence: the persistence of motion, inference, or imagination that keeps learning alive even when direct error signals vanish.

Hamiltonian motion remembers the path it travels.

It is exploration without exhaustion — but also without rest.

Where Boltzmann dynamics freeze too soon, Hamiltonian ones never settle.

The R-rule will combine the two: **momentum to explore, dissipation to learn.**

3. The R-rule as Synthesis

Boltzmann dynamics descend.

Hamiltonian dynamics orbit.

Each captures one half of what learning requires — exploitation without imagination, or imagination without convergence.

The **R-rule** unites them.

$$\psi' = \psi + \eta * (p(1-p))^{1/2} * [\alpha \sin \theta A(a, \psi) - i \beta \cos \theta N(a, \psi)],$$

$$p(a) = |\psi(a)|^2 / \sum_b |\psi(b)|^2.$$

$$\psi' = \psi + \eta (p(1-p))^{1/2} [\alpha \sin \theta A(a, \psi) - i \beta \cos \theta N(a, \psi)],$$

$$p(a) = \frac{|\psi(a)|^2}{\sum_b |\psi(b)|^2}.$$

- The R-rule has two channels: A (actor-model) and N (self-model)
- A follows energy gradients downward (exploitation). N maintains momentum and coherence (exploration)
- The $\sqrt{p(1-p)}$ term modulates between them—high when uncertain, low when committed
- Together they create a self-annealing process.

Here, ψ represents the system's current state — a field of possible configurations or actions.

The two internal drives, **A** (actor-model) and **N** (self-model), correspond to the components we've just derived:

- **A** carries the *Boltzmann* term — a gradient-following force that minimizes local energy (α , exploitation). An *agentic acting core* — the inner drive that projects intention into the world. It is the system's outward interface: sampling actions, probing possibilities, and transforming prediction into experiment (actions).
- **N** carries the *Hamiltonian* term — a momentum-preserving motion that maintains global coherence (β , exploration) — the *normative* or *self-in-world model* channel. It represents the

system's internal coherence: the ongoing negotiation between its own expectations and the dynamics of the world it inhabits.

The factor $(p(1-p))^{1/2}$ acts as a **modulator of uncertainty** — strongest when the system is undecided ($p \approx 0.5$) and vanishing when it becomes certain ($p \rightarrow 0$ or 1).

It's the equivalent of an internal temperature: a controlled noise that balances *curiosity* and *commitment*.

Think of the conversion to p as a softmax-like normalization.

Not quantum mechanics, but a similar trick — collapsing probabilities from all possible actions b into the one currently enacted a .

It bridges exploration and selection: from thinking about everything that could be done, to doing one thing.

Over time, the α and β channels create a recursive dance:

α pulls the system inward, exploiting structure and aligning actions with the energy landscape — the gradient-following, Boltzmann-like component that grounds behavior in what is known.

β drives the system outward, maintaining coherence and exploring unvisited states — the Hamiltonian-like term that sustains motion across boundaries and opens the system to what is possible.

The result is a **self-annealing process**: motion that first broadens, then coheres.

Where Boltzmann machines cool too early, and Hamiltonian systems never cool at all, the R-rule discovers its own cooling schedule from within.

The R-rule learns not just *what* to do, but *when* to stop learning.

It exploits, explores, and equilibrates — all through the same recursive mechanism.

4. Empirical Demonstration: Learning to Settle

To see the R-rule in motion, we turn to a simple landscape:
a **tilted double-well potential**.

Two valleys, one shallow and one deep — a miniature cosmos of decision and stability.

Each system begins in the same place — **balanced between the two valleys, but with a slight slope toward the shallower side** — and then follows its own law of motion.

That small asymmetry is enough to reveal how each dynamic learns, explores, or settles.

What happens next reveals everything about how a system learns.

Boltzmann Dynamics — Cooling into the Nearest Well

The Boltzmann system slides straight downhill.

Noise gives it a little jiggle, but every step follows the energy gradient.

It quickly finds the **nearest minimum** — the shallow left well — and stops.

Stable, yes. But parochial.

It knows where it is, not what it could be.

Hamiltonian Dynamics — Endless Exploration

The Hamiltonian system leverages momentum.

Now position q and momentum p orbit each other, conserving total energy.

The particle sweeps gracefully across the barrier, visits both wells, and never rests.

Exploration without exploitation:

it sees everything, learns nothing.

The R-rule — Explore, Then Commit

The R-rule begins like Hamiltonian motion — broad and curious, tracing elegant spirals through state-space.

Then something changes.

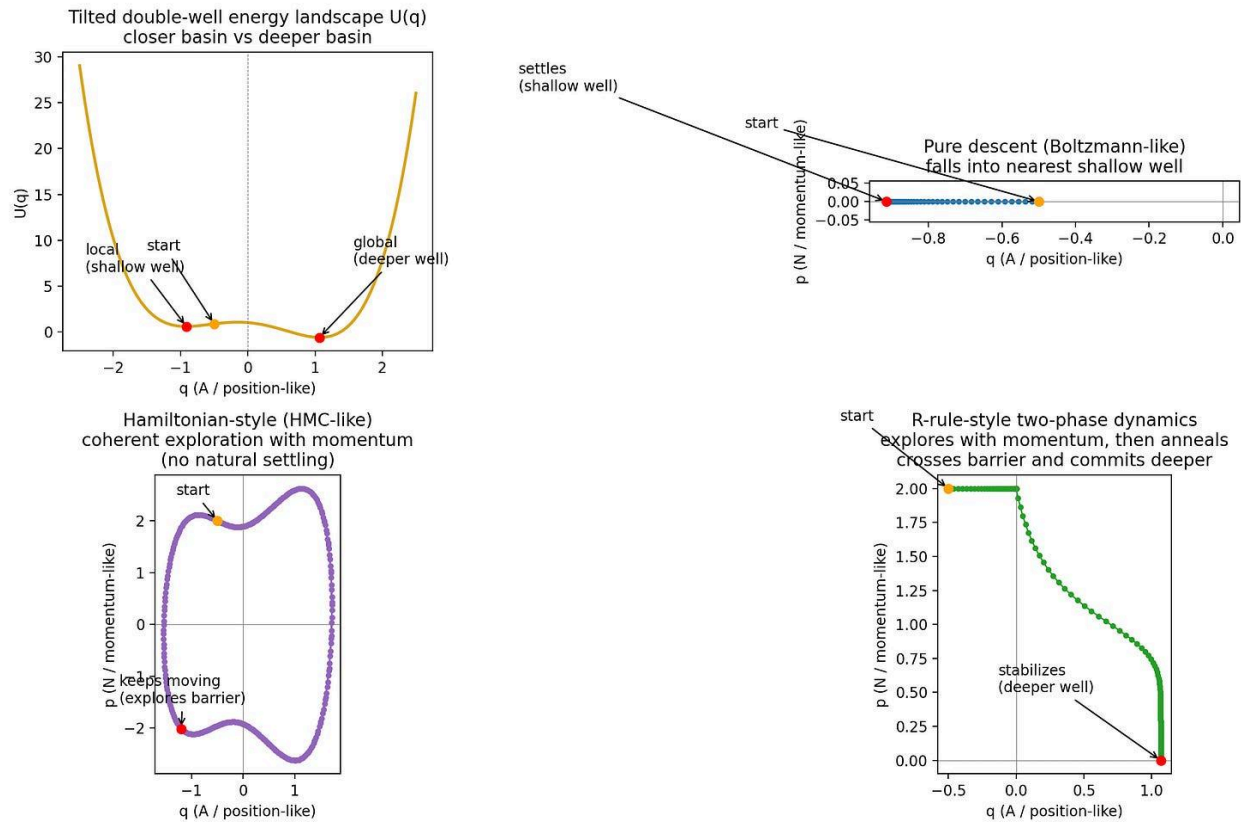
As the internal β -channel strengthens, energy dissipates and motion slows.

The system **crosses the barrier** and **settles in the deeper right-hand well**.

Not by external instruction, but by its own recursive balance between exploration (α) and stabilization (β).

In the simulated phase plots:

- The **Boltzmann trajectory** (blue) sinks directly into the nearest basin.
- The **Hamiltonian trajectory** (purple) orbits endlessly.
- The **R-rule trajectory** (green) spirals outward, crosses the barrier, then anneals smoothly into the deeper equilibrium.



The code for the demo is in Github:

<https://github.com/andrekrumer/chevron/blob/main/r-rule-demo.py>

5. Stability and Coherence

The simulation tells the story; the mathematics explains why. When we track the R-rule over time, $q(t)$ converges to a stable mean near the global

minimum with vanishing variance.

Unlike the Hamiltonian case, which continues to oscillate indefinitely, the R-rule's motion naturally damps and stabilizes.

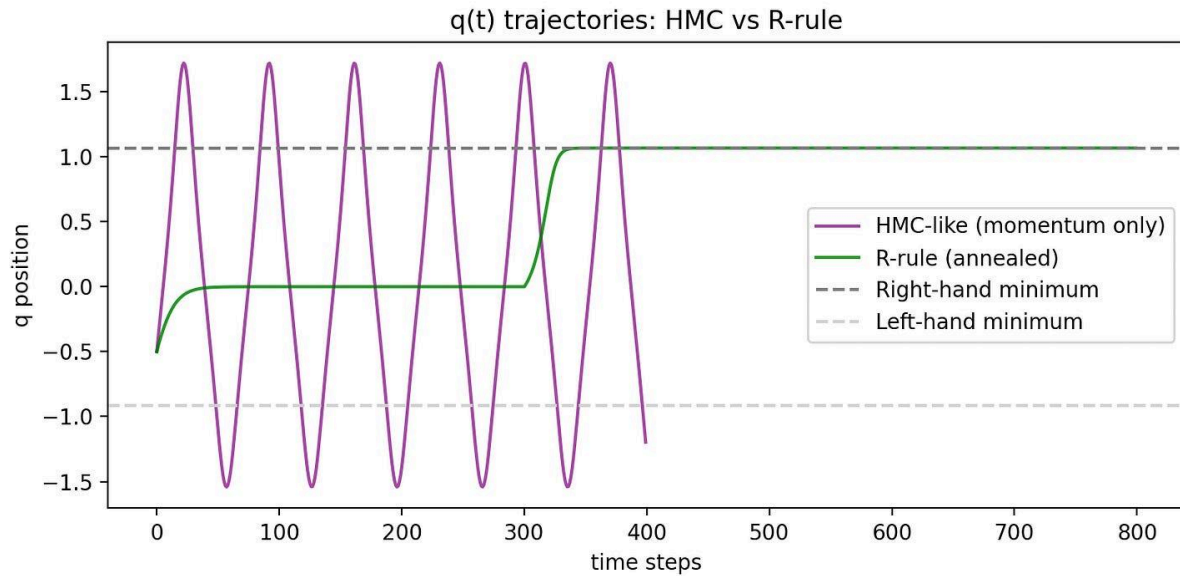
Formally, this *resembles* a system with a Lyapunov function — a measure of internal tension that decreases over time — though a full proof is beyond our present scope.

Empirically, the β -term behaves as a **dissipative correction**, ensuring that motion remains structured yet convergent.

In this sense, the R-rule occupies the middle ground between pure conservation and pure decay:

R-rule = Dissipative stabilization (α) + Hamiltonian-like exploration (β).

It moves with coherence, but without chaos; it settles, but never freezes.



In the simplest landscape, three fates emerge:

- Boltzmann: freezes too early.
- Hamiltonian: never lands.
- R-rule: discovers its own stopping rule.

6. After the Demonstration: Learning is Memory

The **bleak lesson**—a cousin of Richard Sutton's [*bitter lesson*](#) (*Simple, scalable methods beat clever engineering*) —of Boltzmann machines is this: learning is memory. And that's all there is.

There is no hidden essence behind the weights and states, no secret observer guiding convergence.

Systems learn by folding the past into the present.

Variety absorbs variety, as Ashby ([*law of requisite variety*](#)) taught: every adaptive system must become as complex as the world it faces.

Life, from cell to mind, learns reality by **minimising free energy**—by reducing surprise, prediction error, entropy.

But that's not enough. Pure Boltzmann learning only ever settles; it never dreams.

The graceful addition

The **R-rule** adds what Boltzmann alone could not: **imagination**.

It keeps one channel anchored in the real (exploitation, the actor-model), and another poised toward the possible (exploration, the self-model).

The two spin against each other, a dialectic of coherence and contradiction that lets a system cross barriers—energetic, semantic, existential—and then land again.

In that sense, God really does play dice, and it is luck for us that this is so.

Because without that stochastic grace, without those restless rotations of uncertainty, nothing would ever become new.

Next: *Deriving the R-rule from Bayes — the probabilistic heart behind the recursion, and how it connects to modern machine learning and the mathematics of inference.*



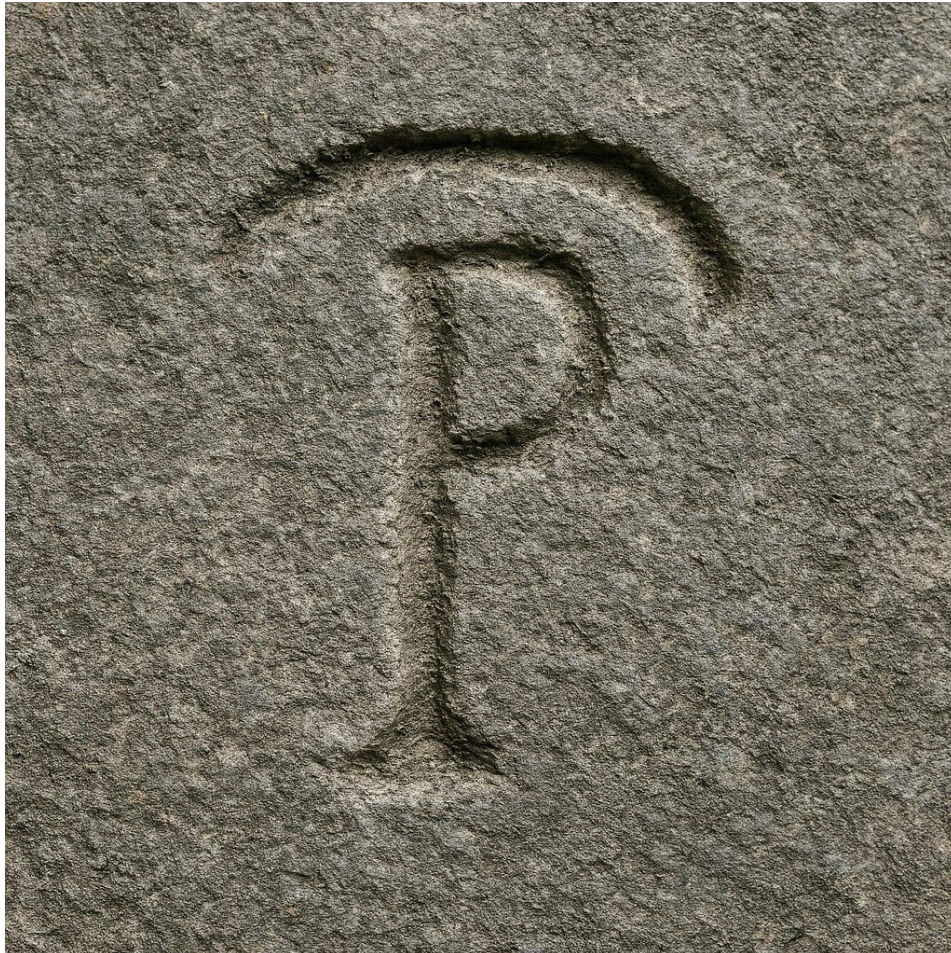
[Andre on the R rule](#)

From Bayes to Rotation

Toward recursive inference and architectures that learn to doubt well

[Andre Kramer](#)

Oct 30, 2025



Machine learning has mastered correlation—but not contradiction.

Most systems learn to predict within the bounds of experience, not beyond it.

Bayesian inference, the mathematical heart of modern learning, assumes that the world behaves as expected—that priors are adequate and surprise is noise.

But what happens when the world changes its rules?

In this post, we trace a path **from Bayes to counterfactuals**: from passive updating to active imagination.

We show how the **R-rule**—a recursive extension of Bayesian inference—naturally leads to a second-order form of reasoning, one that tracks both *beliefs* and *the motion of beliefs*.

This second-order dynamic (formally akin to the **Kramers equation** in statistical physics) provides the mathematical bridge between

Boltzmann learning and **Hamiltonian exploration**—between cooling and moving, remembering and imagining.

Modern large language models (LLMs) have made *Attention* and the *Transformer architecture* famous and today ubiquitous. Here we examine one extension of that architecture to make use of such counterfactual rotations.

1. Why Bayes Isn't Enough

Every modern theory of learning begins with Bayes' rule:

$$p(H | D) = (p(D | H) p(H)) / p(D)$$

$$p(H|D) = \frac{p(D|H)p(H)}{p(D)}$$

It is a simple, profound statement about rational belief revision:

update your hypothesis H in proportion to how well it predicts the data D .

Bayes' rule works beautifully—as long as your priors are true enough.

But out-of-distribution, the assumptions collapse.

The learner has no principled way to decide *which parts of its model to distrust*.

The result is brittle generalization: confident errors in alien contexts, the kind of failures that make AI both powerful and dangerous.

Traditional Bayes is **first-order inference**: it updates a single belief in response to evidence.

What's missing is **second-order inference**—a model that also tracks *how fast* beliefs are changing, *how uncertain* that motion is, and *how it interacts with the system's own coherence*.

This is the same leap Boltzmann made beyond equilibrium: not just counting microstates, but watching how they move.

The R-rule captures that motion.

It's Bayes, differentiated with respect to time—and then rotated into complex space to preserve both dissipation and coherence.

2. Transformers as Amortized Bayes

Bayesian learning can be written not as a static equation but as a *dynamic update rule*—a way for beliefs to move through time.

In log form, the Bayesian update for a hypothesis H given data D is:

$$\Delta \log p(H) = \log p(D|H) - \log p(D),$$

$$\Delta \log p(H) = \log p(D|H) - \log p(D),$$

or, in differential form:

$$d/dt \log p(H) = \partial/\partial t \log p(D|H) - \partial/\partial t \log Z,$$

$$\frac{d}{dt} \log p(H) = \frac{\partial}{\partial t} \log p(D|H) - \frac{\partial}{\partial t} \log Z,$$

where $Z = p(D)$ is the normalizing partition function ensuring all hypotheses sum to 1.

This shows Bayes as a **gradient flow** in probability space:

beliefs ascend or descend along evidence gradients, adjusting in proportion to predictive success.

In discrete form, it reads:

$$p'(H) = p(H) + \eta p(H)(1-p(H)) * \partial/\partial H \log p(D | H),$$

$$p'(H) = p(H) + \eta p(H)(1 - p(H)) \frac{\partial}{\partial H} \log p(D|H),$$

where η is a learning rate.

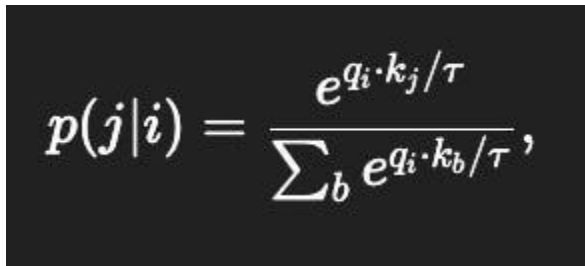
This is the canonical *first-order* Bayesian update — a close cousin of the Boltzmann machine rule, differing mainly in interpretation.

Attention as an Amortized Bayesian Update

A Transformer's attention mechanism enacts this same principle, but in parallel and parameterized form.

Each attention head computes:

$$p(j|i) = e^{(q_i \cdot k_j / \tau)} / \sum_b e^{(q_i \cdot k_b / \tau)},$$


$$p(j|i) = \frac{e^{q_i \cdot k_j / \tau}}{\sum_b e^{q_i \cdot k_b / \tau}},$$

where:

- q_i (query) represents a *prior belief* about what the model expects next;
- k_j (key) encodes the *likelihood* of contextual evidence;
- and the softmax denominator acts as the Bayesian normalizer Z .

The attention output

$$y_i = \sum_j p(j|i)v_j$$

$$y_i = \sum_j p(j|i)v_j$$

is then the **expected posterior representation**—the belief distribution after updating on all possible contextual evidence.

In other words, every attention head performs a **Bayesian update** in miniature:

- **Queries** $\approx$ priors
- **Keys** $\approx$ likelihoods
- **Values** $\approx$ evidence
- **Softmax** $\approx$ normalization

Each step is a fast, differentiable approximation of Bayesian inference — what cognitive scientists call *amortized Bayes*.

Where Bayes (and Attention) Break Down

This mechanism is elegant—but incomplete.

It assumes a stationary world: that the priors are well-calibrated and the data distribution is stable.

When faced with out-of-distribution (OOD) inputs, contradictions, or negations, attention still normalizes *as if nothing changed*.

The model updates its beliefs **inside the wrong world**.

Put differently, Transformers are **first-order Bayesians**: they update belief, but not *belief about belief*.

They lack a way to track how their own certainty is changing—to doubt their own gradients.

And so they collapse under surprise: overconfident where they should hesitate, frozen where they should explore.

Toward a Second-Order Bayes

What's missing is a recursive dimension:

a representation not just of the *state* of belief, but of its *motion*—its velocity, inertia, and phase.

In physics, that move—from position-only dynamics (Boltzmann) to position-plus-momentum (Kramers)—creates the leap from simple relaxation to exploration with memory.

In cognition, it creates the leap from perception to imagination.

That's where the **R-rule** begins: a second-order, complexified version of Bayes that adds a conjugate term to track internal coherence and counterfactual motion.

3. From Bayes to the R-rule

The Bayesian update is a law of motion for belief.

But as we've seen, it is a *first-order* law — beliefs respond to evidence but do not carry momentum.

Each update forgets the path that brought it there.

When the world shifts, learning must begin again from scratch.

To learn safely beyond distribution, a system needs not just *beliefs* but *belief dynamics* — an internal memory of how certainty itself moves through time.

This is what physicists call a **second-order process**: one that tracks both position and velocity, both energy and flow.

The Second-Order Bayesian Flow

Start from the first-order (log-space) Bayesian update:

$$d/dt \log p(H) = \partial/\partial t \log p(D | H) - \partial/\partial t \log Z.$$

$$\frac{d}{dt} \log p(H) = \frac{\partial}{\partial t} \log p(D|H) - \frac{\partial}{\partial t} \log Z.$$

Now introduce a conjugate variable $\dot{p}(H)$: the *rate of change of belief*.

We can then write the joint evolution as a pair of coupled equations:

$$dp/dt = -\partial E/\partial p + \xi(t),$$

$$d\dot{p}/dt = -\gamma\dot{p} - \partial V/\partial p + \eta(t),$$

$$\begin{aligned}\frac{dp}{dt} &= -\frac{\partial E}{\partial p} + \xi(t), \\ \frac{d\dot{p}}{dt} &= -\gamma\dot{p} - \frac{\partial V}{\partial p} + \eta(t),\end{aligned}$$

where E is an evidence-based energy landscape, γ a damping term, and ξ, η are stochastic fluctuations.

This is the canonical **Kramers equation** in statistical mechanics:
a second-order stochastic differential equation that combines friction
(learning) with **momentum** (memory).

It is, quite literally, **Bayes with memory**.

Breakout: A Note on the Kramers Equation

In physics, the **Kramers equation** extends Boltzmann's first-order relaxation law into a second-order flow, describing how a particle's probability density evolves in both position q and momentum p (we keep to the convention of using p for momentum here in this breakout - not probability):

$$\partial P / \partial t = -p \partial P / \partial q + \partial / \partial p (\gamma p P + \partial V / \partial q P + D \partial P / \partial p).$$

$$\frac{\partial P}{\partial t} = -p \frac{\partial P}{\partial q} + \frac{\partial}{\partial p} \left(\gamma p P + \frac{\partial V}{\partial q} P + D \frac{\partial P}{\partial p} \right).$$

It bridges Boltzmann's passive cooling and Hamilton's active motion — dissipation and exploration in one equation.

It's also, by coincidence, my namesake! Though I'm a Kramer not Kramers.

If Boltzmann taught us how systems *cool*, **Kramers** showed how they *flow*.

Complexification and the Origin of Rotation

Now imagine belief as a complex quantity:

$$\psi = \sqrt{p} e^{i\theta},$$

A black rectangular box containing the mathematical equation $\psi = \sqrt{p} e^{i\theta},$ in white serif font.

where p encodes certainty and θ encodes phase — the internal orientation of belief, its coherence with self and world.

Differentiating through time gives:

$$d\psi/dt = 1/(2 \sqrt{p}) dp/dt e^{i\theta} + i \sqrt{p} d\theta/dt e^{i\theta}.$$

$$\frac{d\psi}{dt} = \frac{1}{2\sqrt{p}} \frac{dp}{dt} e^{i\theta} + i\sqrt{p} \frac{d\theta}{dt} e^{i\theta}.$$

This shows two independent components of change:

- **Amplitude change** (real term): updating the strength of belief — learning by correction.
- **Phase change** (imaginary term): rotating belief — learning by counterfactual motion.

If we substitute back into our second-order Bayesian flow, the two directions become orthogonal update channels:

$$\psi' = \psi + \eta * ((p(1-p))^{1/2} * [\alpha \sin \theta A(a, \psi) - i\beta \cos \theta N(a, \psi)]).$$

$$\psi' = \psi + \eta(p(1 - p))^{1/2} [\alpha \sin \theta A(a, \psi) - i\beta \cos \theta N(a, \psi)].$$

The sine and cosine terms appear naturally as the **projections of belief motion** onto its real (amplitude) and imaginary (phase) components — a coupling of correction and coherence, exploitation and exploration.

Hello R rule!

Sidebar: Returning to Probability

Given the complex belief $\psi = \sqrt{p} e^{i\theta}$,

$$\psi = \sqrt{p} e^{i\theta},$$

we recover the observable probability as $p = |\psi|^2 = \psi^* \psi$.

$$p = |\psi|^2 = \psi^* \psi.$$

Differentiating gives $dp/dt = 2 \operatorname{Re}(\psi^* d\psi/dt)$.

$$\frac{dp}{dt} = 2 \operatorname{Re} \left(\psi^* \frac{d\psi}{dt} \right).$$

where $\operatorname{Re}(\cdot)$ means “take the real part” — the component that actually changes the magnitude of belief.

Substituting the R-rule shows that only the **real term**—the $\alpha \sin \theta A$ channel—changes p ;

the **imaginary term**— $\beta \cos \theta N$ —rotates phase but leaves magnitude untouched.

The modulation factor $\sqrt{p(1-p)}$ then appears naturally:

it controls how fast belief can change, peaking at uncertainty ($p \approx 0.5$) and vanishing at certainty ($p = 0$ or 1).

Finally, probabilities are normalized across all possible actions:

$$p(a) = |\phi(a)|^2 / \sum_b |\phi(b)|^2.$$

In short: the α -term learns; the β -term coheres; normalization keeps the whole system honest.

This normalization returns the complex dynamics of belief to the probabilistic world Bayes built, closing the circle between amplitude, phase, and action — like an internal compass that guides without dictating, preserving coherence while allowing freedom to move.

Interpretation

In this form, the R-rule unites three traditions:

- **Boltzmann** (gradient descent): beliefs cool into the most stable configuration;
- **Hamilton** (momentum conservation): beliefs flow with internal coherence;
- **Bayes** (inference): beliefs adjust to evidence under normalization.

The R-rule is their synthesis — a recursive, complex-valued inference law that updates both *what the system believes* and *how it believes*.

It turns static Bayesian updates into a dynamic oscillation between **predictive confidence** and **counterfactual imagination**.

This is the mathematical heart of **learning to doubt well**.

Insert: Logarithms, Products, and Euler's Bridge

Before complexifying, recall that Bayes' rule operates in **logarithmic space**.

Taking logs turns multiplicative evidence updates into additive gradients:

$$\log p(H|D) = \log p(D|H) + \log p(H) - \log p(D).$$

$$\log p(H|D) = \log p(D|H) + \log p(H) - \log p(D).$$

This transformation is what makes Bayesian inference tractable — it linearizes the flow of evidence.

But it also hides an important symmetry:

addition in log-space corresponds to **multiplication in probability-space**.

To restore that multiplicative structure dynamically, we exponentiate again — moving from additive log updates to **rotational motion** in the complex plane.

By **Euler's identity**:

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

so a belief represented as $\psi = \sqrt{p} e^{i\theta}$ carries both its magnitude (certainty) and its orientation (phase).

The real and imaginary projections, $\sin\theta$ and $\cos\theta$, then describe **how belief changes in amplitude and direction**.

This is not mere notation — it's the mathematical expression of two coupled learning modes:

- Log-space addition (Bayesian update)
- Complex-phase rotation (counterfactual inference)

Together, they yield a recursive inference rule that can both *learn from evidence* and *imagine alternatives* — the heart of the R-rule.

4. Dual Attention Heads: Implementing a Counterfactual Rotation

The R-rule describes a recursive flow of inference:

one channel cooling belief toward evidence, the other rotating it toward coherence.

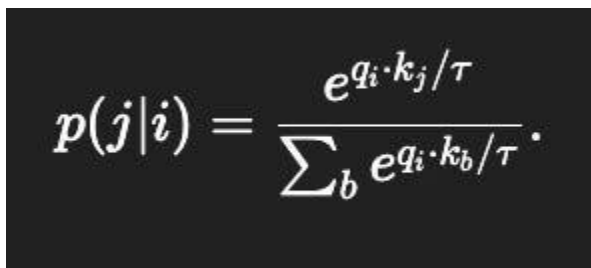
But equations do not think — architectures do.

How can such dual learning dynamics be implemented in a neural system? We explore one such approach next as an experiment in modifying the standard Transformer architecture. This isn't a fully recursive or counterfactual system yet — just a practical experimental step toward one.

Attention as a Single-Channel Learner

Standard Transformer attention compresses the Bayesian update into a single softmax operation:

$$p(j|i) = e^{(q_i \cdot k_j / \tau)} / \sum_b e^{(q_i \cdot k_b / \tau)}.$$


$$p(j|i) = \frac{e^{q_i \cdot k_j / \tau}}{\sum_b e^{q_i \cdot k_b / \tau}}.$$

This is efficient, but brittle.

Each head commits fully to one interpretation of the world — *factual attention*.

It learns “what fits,” but cannot represent “what might also have fit.”

The mechanism normalizes away contradiction.

Surprise becomes noise; doubt is discarded.

This is why attention works brilliantly in-distribution but can fail catastrophically OOD: it has no structural way to *represent its own counterfactuals*.

Splitting the Channel: Factual and Negation Heads

To embody the R-rule — without using complex numbers — we introduce a **dual-path attention mechanism**: two parallel heads operating on the same query–key–value triplets but with orthogonal epistemic roles.

$$\text{pf}(j|i) = \text{softmax}(q_i \cdot k_j / \tau),$$

$$p_c(j|i) = \text{softmax}(-q_i \cdot k_j / \tau).$$

$$\begin{aligned} p_f(j|i) &= \text{softmax}(q_i \cdot k_j / \tau), \\ p_c(j|i) &= \text{softmax}(-q_i \cdot k_j / \tau). \end{aligned}$$

- The **factual head** $A(a)$ behaves like standard attention: it emphasizes evidence consistent with the model's expectations.
- The **counterfactual head** $N(a)$ reverses or **negates** that alignment, attending to contradictory or low-probability cues — a structured way to model “what if not.”

Their outputs are combined through a **rotational mixer**, inspired directly by the R-rule:

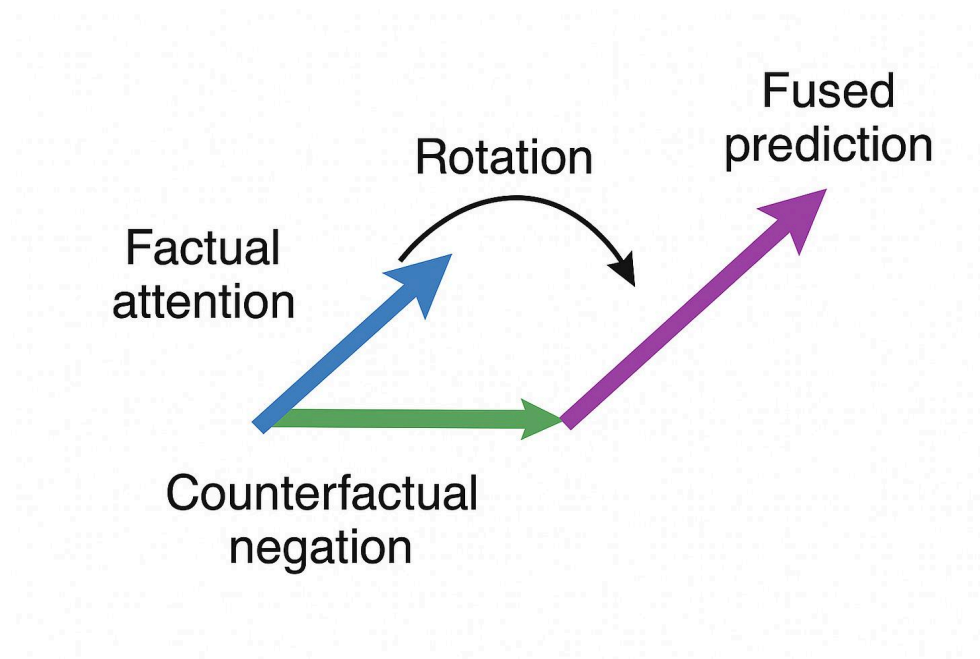
$$p'(a) \propto p(a) + \sqrt{p(a)(1-p(a))} * [\alpha \sin \theta A(a) - i \beta \cos \theta N(a)].$$

$$p'(a) \propto p(a) + \sqrt{p(a)(1-p(a))} [\alpha \sin \theta A(a) - i \beta \cos \theta N(a)].$$

Here:

- α governs the factual (Boltzmann) drive — **exploitation**,
 - β governs the counterfactual (Hamiltonian) drive — **exploration**,
 - θ acts as a dynamic gate between them,
 - and the imaginary unit i ensures the two flows remain orthogonal — a built-in guard against runaway confidence.
-

Interpreting the Rotation



In practice, this dual attention behaves as a self-correcting inference engine:

Component	Function	Epistemic Role
$A(a)$	Factual attention	Alignment with evidence
$N(a)$	Counterfactual attention	Alignment with negation
$\alpha \sin \theta$	Adaptive weighting	How much to trust data
$\beta \cos \theta$	Reflective weighting	How much to trust model
i	Orthogonality	Keeps exploration safe

Component Function Epistemic Role

$A(a)$ Factual attention Alignment with evidence

$N(a)$ Counterfactual attention. Alignment with negation

$\alpha \sin \theta$ Adaptive weighting How much to trust data

$\beta \cos \theta$ Reflective weighting How much to trust model

i Orthogonality Keeps exploration safe

The **rotation** between A and N isn't metaphorical — it's a literal geometric relation in probability space.

Factual attention pulls beliefs toward what is known; counterfactual attention rotates them toward what is *coherently unknown*.

The result is a dynamic equilibrium: learning that neither freezes nor explodes.

Learning to Doubt Well

This design does not aim to maximize accuracy.

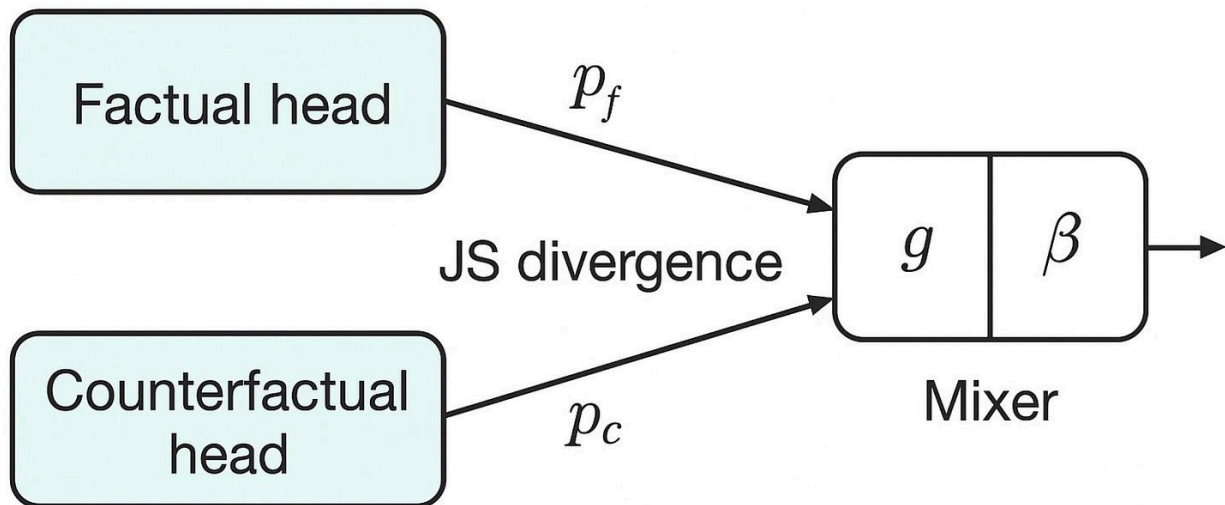
It aims to increase **epistemic safety** — the ability to remain coherent when the world violates prior assumptions.

In the R-rule's language:

- A grounds the system in *evidence* (the real component).
 - N keeps it open to *imagination* (the imaginary component).
- Their interplay, modulated by θ , defines a recursive loop of meaning: exploitation and exploration fused into a single safe dynamic.
-

5. Empirical Behaviour: Coherence Under Surprise

If the R-rule describes how learning systems *ought* to behave, dual attention heads let us see how that behaviour *actually looks* in a practical experiment. Albeit in a shallow recursion - a **real-valued, negation based, architectural approximation** of the full R-rule dynamics.



Factual vs Counterfactual (Negated) Behaviour

We trained the dual-head Transformer on paired in-distribution (ID) and out-of-distribution (OOD) data, using opposite concept pairs so that a counterfactual corresponds to a structured negation of the factual input.

Each head receives the same inputs but computes opposite attentional projections — one factual, one counterfactual.

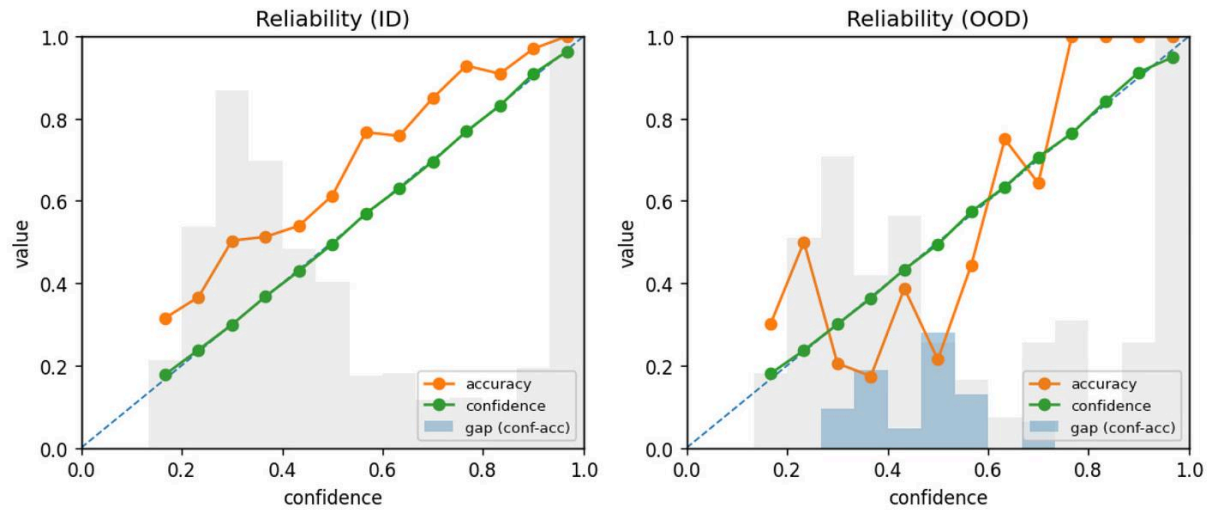
A gating variable θ learns to balance them dynamically.

Over time, this creates a self-regulating loop:

- When the model is confident, $\sin\theta$ dominates — the **factual head** leads.
- When confidence drops or contradiction spikes, $\cos\theta$ increases — the **counterfactual head** rises to the surface.

The system thus embodies an automatic *reversal reflex*: a structural form of epistemic humility.

Reliability Curves: Confidence That Knows Its Limits



Reliability curves show a striking property.

Unlike classical attention, which remains overconfident OOD, dual attention maintains calibration:

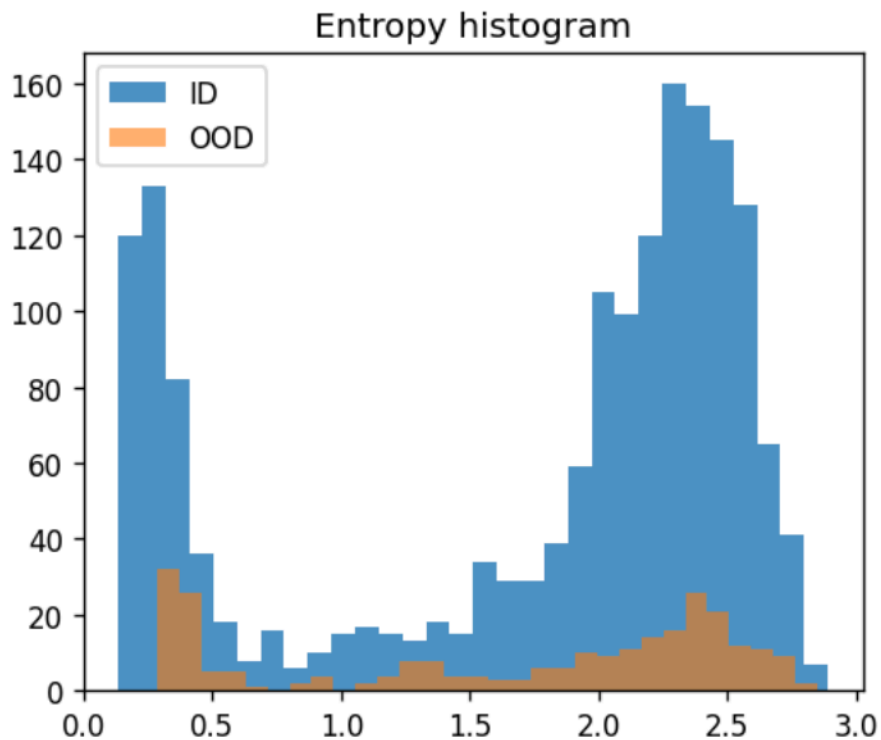
confidence tracks accuracy even under surprise.

In-distribution, the factual head dominates and behaves like any well-tuned Transformer.

Out-of-distribution, the counterfactual head tempers belief, flattening overconfidence spikes and widening uncertainty bands.

This is **graceful degradation** — intelligence that knows when it does not know.

Entropy and Separation



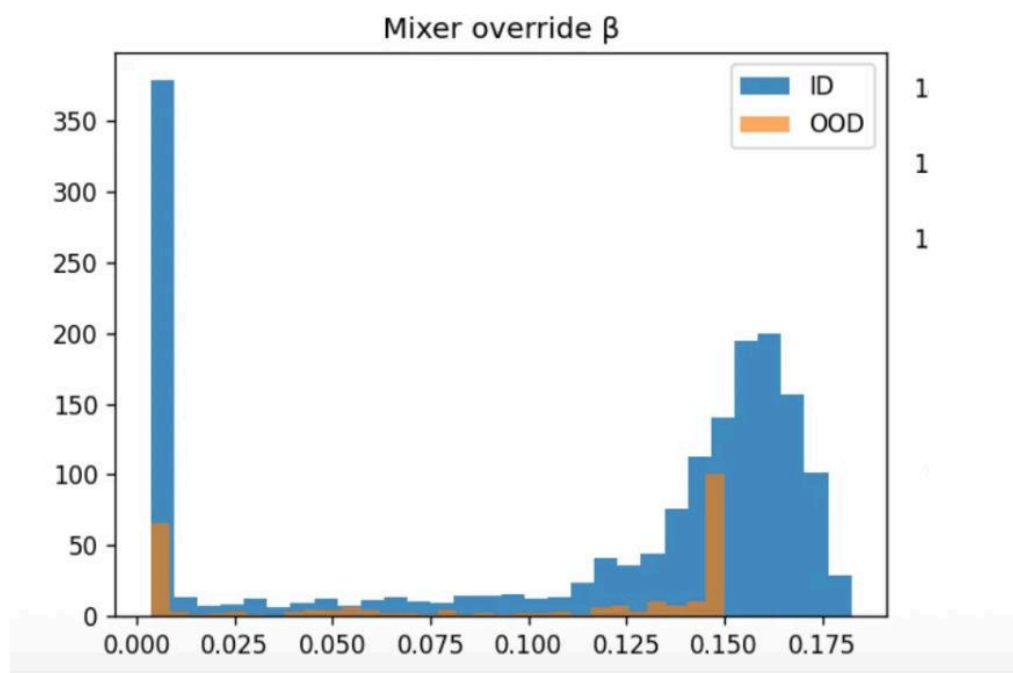
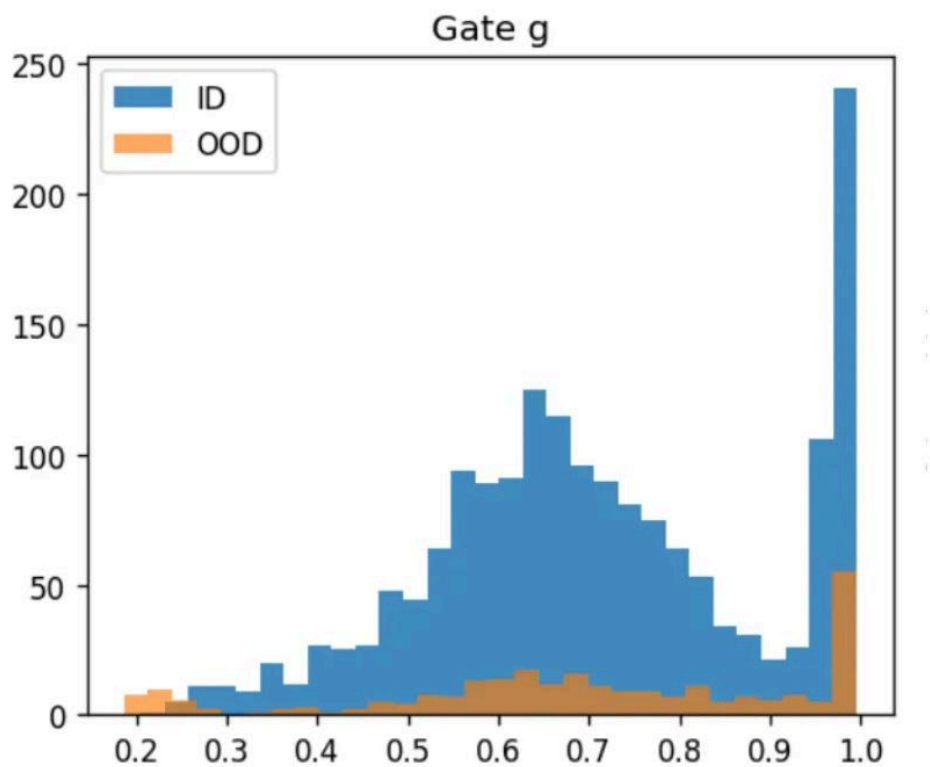
Entropy distributions reveal that the model not only hesitates when it should — it *hesitates distinctly*.

ID and OOD samples occupy separable regions in entropy space, showing that uncertainty is not arbitrary noise but a coherent internal signal.

In cognitive terms, the model doesn't confuse surprise with error.

It recognizes surprise as **an opportunity for counterfactual rehearsal**.

The Gate and the Override



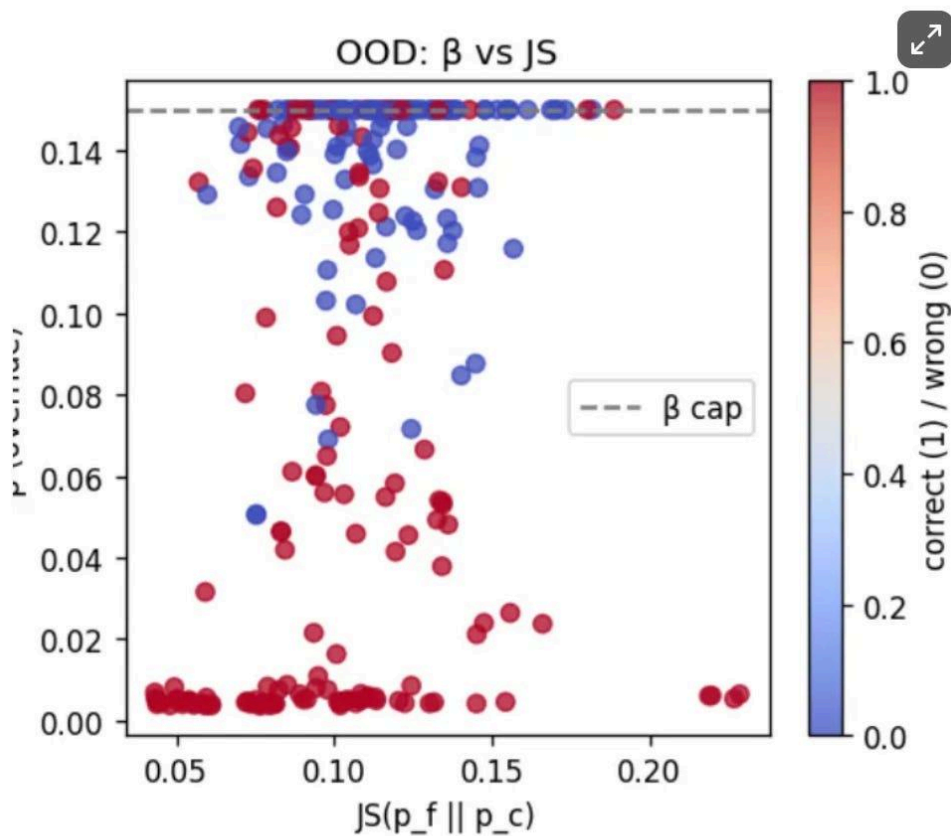
The learned distributions of $g = \sin\theta$ (gate) and $\beta = \cos\theta$ (override) tell a consistent story.

In stable contexts, the gate remains open — letting the factual head dominate.

As uncertainty rises, the override activates, capping overconfidence and rotating probability mass toward counterfactual alternatives.

The result is not randomness but **bounded curiosity**: controlled exploration under uncertainty.

β vs. JS Divergence: When to Doubt



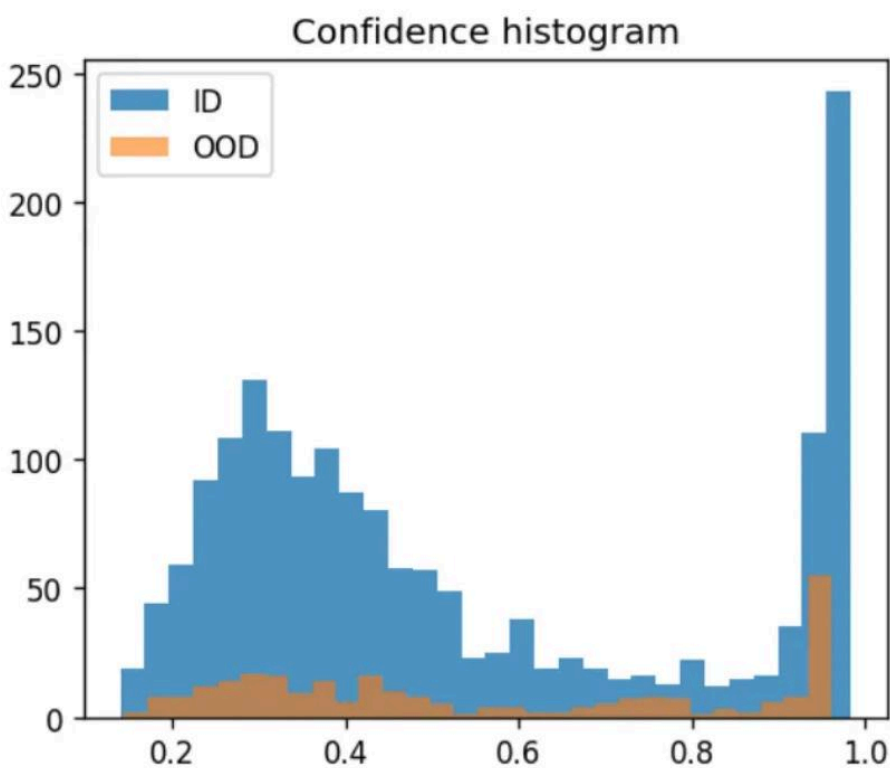
Plotting the override β against the Jensen–Shannon divergence between the two heads gives a remarkably interpretable signal.

High divergence — strong disagreement between heads — correlates with higher β values, meaning the system automatically reins itself in when its two inferential selves disagree.

This is not an engineered heuristic; it's the R-rule in action.

The system literally learns to doubt itself in proportion to its own internal contradiction.

Confidence and Entropy Histograms: Knowing When You're Wrong



Confidence histograms show that the dual-path model not only performs better out of distribution — it *knows* it's out of distribution.

Where a standard Transformer flattens its confidence across all samples, the R-inspired model produces a **bimodal pattern**:

- **High confidence** on in-distribution (ID) examples.
- **Low confidence** on out-of-distribution (OOD) ones.

Entropy histograms tell the same story in reverse:

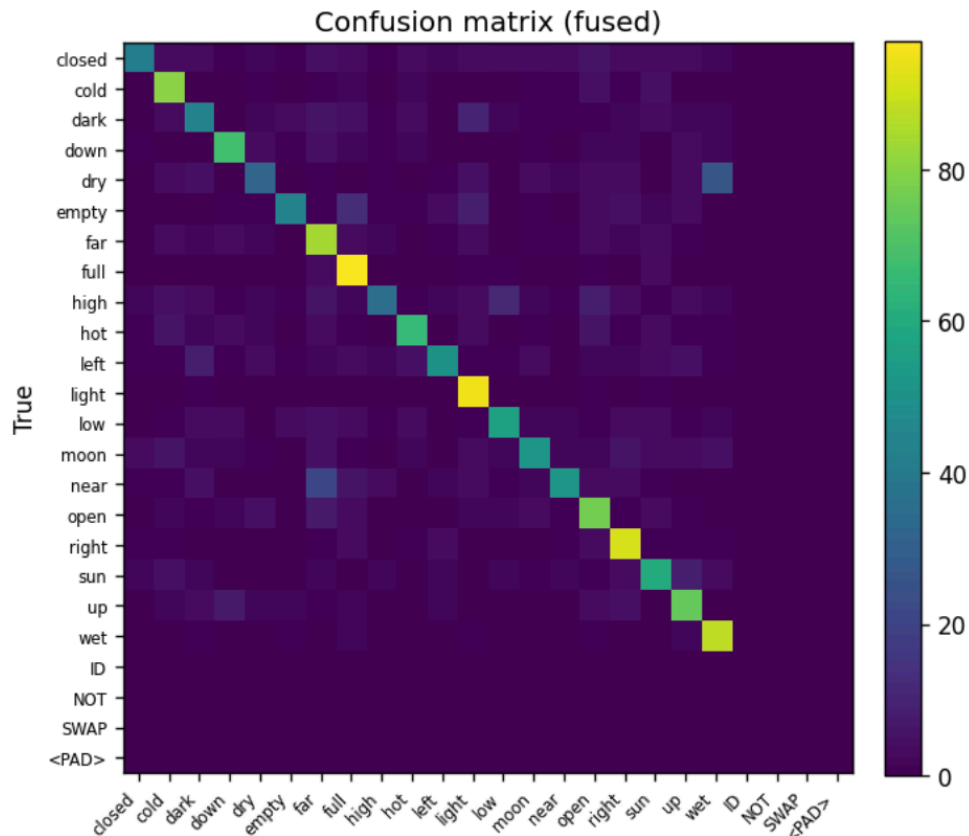
uncertainty is high where it should be (OOD), and low where it should be (ID).

This separation shows that the counterfactual channel does not simply inject noise — it **structures doubt**.

The model learns a kind of internal metacognition:

confidence is no longer a blind byproduct of logits, but a reflection of epistemic stability.

Safe Failure: Diffuse Errors



In the fused predictions, misclassifications are diffuse rather than clustered.

There are no catastrophic blind spots — only soft errors.

The system disperses failure across its representational field, much as a physical system dissipates energy.

This is a signature of **Lyapunov stability in cognition**:

small perturbations cause proportional, not runaway, changes.

The model *bends*, but does not *break*.

Interpretation

Safety or resilience here is not the absence of error.

It is the presence of **coherence under surprise** — the ability to reorganize internal probabilities without losing structural integrity.

The dual-head design doesn't make the system infallible; it helps make it *reflexive*.

The code is on Github:

https://github.com/andrekrumer/chevron/blob/main/dual_attention_heads.py

A preliminary result, but one that points the way to a deeper interpretation.

6. After the Experiment: Learning is Counterfactual Coherence

What does it mean for a system to be intelligent — not in performance, but in persistence?

When the world shifts beneath its priors, what lets it remain coherent?

The answer, we suggest, is not accuracy but *counterfactual coherence*: the capacity to hold tension between what is and what might be, and to use that tension as a source of stability.

The Limits of First-Order Learning

Classical learning — whether in Boltzmann machines, gradient descent, or standard attention — is first-order.

It learns by alignment, by following the gradients of truth.

But that alignment comes at a cost: once equilibrium is reached, motion ceases.

The system stops listening. It can no longer adapt without breaking itself.

In human terms: memory without imagination.

Rigid coherence, brittle certainty.

Dual attention may be a first tentative step along that way — a glimpse of what recursive, counterfactual coherence could become when rendered in working code.

The Second-Order Turn

The R-rule introduces something fundamentally different:

not just alignment, but **rotation** — a recursive interplay between acting and reflecting, between factual and counterfactual attention.

Where first-order systems converge, second-order systems orbit and damp.

They don't seek a single fixed truth but a stable relationship between truths — a coherence that *includes* contradiction.

In this sense, the R-rule is not just a model of learning; it's a model of **thinking**.

It captures the recursive act of revising one's own priors in light of how they fail.

The Principle of Counterfactual Coherence

Every adaptive system must eventually learn to manage surprise.

For physical systems, that means dissipating energy.

For cognitive systems, it means integrating contradiction.

The counterfactual channel—the N-term, the Hamiltonian component—makes this possible.

It ensures that when a system encounters what it cannot predict, it does not collapse; it reorients.

Generalization, in this light, is not about extending correctness across domains but sustaining coherence across difference.

A model that generalizes well is one that continues to learn when its assumptions fail—using the gap between its actor and its self-in-world-model as the very space where new structure forms.

Learning, then, is not convergence but *ongoing reconciliation* with a world that remains partly unlearnable.

Coda

The *bitter lesson* taught us that simple, scalable method

s beat clever engineering.

The *bleak lesson* taught us that learning is only memory.

The *hopeful lesson* is this:

learning is counterfactual coherence—a recursive dialogue between what we know, what we imagine, and what resists both.

Dual attention may be a first tentative step along that way.

Next time: we'll explore how learning systems cool themselves.

In classical optimization, annealing must be hand-tuned — a slow lowering of temperature to balance curiosity with convergence.

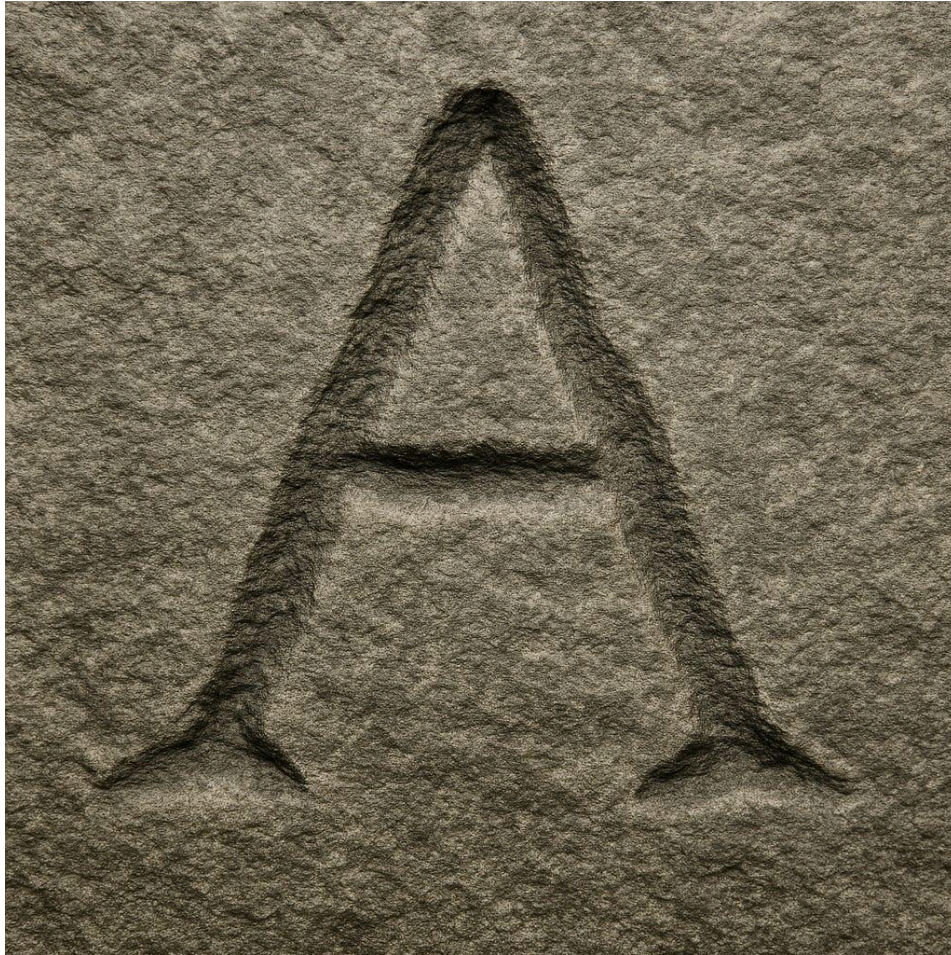
But in the R-rule, that balance emerges from within: the system's own uncertainty becomes its temperature.

We'll follow this self-annealing dynamic from Boltzmann's physics to modern AI.

From Rotation to Self-Annealing

Learning in the Forge of recursion

How the [R-rule](#) tempers exploration and coherence — and why reinforcement and search might just be sparks from a deeper self-supervising fire.



Annealing as Learning

When metals are forged, they are first heated — atoms vibrating in chaos — then cooled slowly, so that a new crystalline order can emerge. Learning works the same way. Too much noise and the structure melts; too little and it freezes into brittleness.

The secret is *annealing* — the art of cooling just fast enough to settle, but not so fast as to lose flexibility.

In physics, **simulated annealing** captures this balance by letting a system explore freely at high temperature, then gradually cooling to lock in a minimum.

But the **R-rule** shows that a system can anneal *itself*. Its effective temperature emerges from its own uncertainty:

$$T_{\text{eff}} = (p(1-p))^{1/2}.$$

$$T_{\text{eff}} = \sqrt{p(1-p)}.$$

When confidence is low ($p \approx 0.5$), the system explores widely.

As it learns, T_{eff} cools naturally, shrinking fluctuations until the system stabilizes.

No external schedule is required. Learning and self-regulation are one process.

This is **self-annealing** — a dynamic that tempers belief through its own contradictions.

From the R-rule to Reinforcement

In traditional reinforcement learning (RL), exploration and exploitation are split by design.

An **actor** chooses actions; a **critic** evaluates them.

They are coupled through an *external* reward signal — a form of annealing imposed from the outside.

Temperature schedules, entropy bonuses, and learning rates are all hand-tuned to ensure convergence.

The R-rule replaces these manual knobs with an **intrinsic forge** — a self-supervising mechanism that generates its own heat and cooling through recursive uncertainty.

RL Concept	R-rule Analogue
Reward	Internal coherence gradient
Entropy regularization	Uncertainty modulation $\sqrt{p(1 - p)}$
Exploration	Hamiltonian (β) term
Exploitation	Boltzmann (α) term
Actor–Critic loop	Recursive $\psi \leftrightarrow p$ coupling (A and N)

RL Concept R-rule Analogue

Reward Internal coherence gradient

Entropy regularization Uncertainty modulation $\sqrt{p(1-p)}$

Exploration Hamiltonian (β) term

Exploitation Boltzmann (α) term

Actor–Critic loop Recursive $\psi \leftrightarrow p$ coupling (A and N)

The R-rule doesn't wait for a teacher.

It forges its own corrections in the tension between exploration (momentum) and exploitation (cooling).

It replaces external reward with an *internal coherence signal* — a form of **self-supervised reinforcement**.

The Forge of Modern AI: RL and MCTS

If reinforcement learning is the **hammer** — shaping policy through feedback — and Monte Carlo Tree Search (MCTS) the **anvil** — structuring deliberation through expansion and pruning — then **self-annealing** lies in the *heat* between them.

Both embody the same dialectic:

- **RL**: recursive actor–critic learning through feedback from its own actions.
- **MCTS**: recursive search through expansion and backpropagation of value estimates.

Each balances exploration and coherence — yet both depend on **externally tuned temperature**: ϵ -schedules, UCB constants, entropy weights.

The R-rule **closes the loop**, embedding the annealing inside the learner itself.

Exploration cools as certainty grows — the agent tempers itself.

This self-tempering dynamic could, in principle, **unify learning, search, and control under a single recursive law**.

♦ What the R-rule Does Differently

The R-rule replaces the external reward scalar with an **internal tension** — between:

- prediction and reality,
- coherence and contradiction,
- the actor and its self-in-world-model (A vs N).

This tension drives the same kind of adaptive feedback loop, but it's generated *internally*.

You can think of it as an **emergent reward signal** — generated from the tension between prediction and observation — rather than a scalar supplied by the environment.

$$\Delta\psi = \eta * \text{sqrt}(p(1-p)) * [\alpha \sin\theta A(a,\psi) - i\beta \cos\theta N(a,\psi)].$$

$$\Delta\psi = \eta \sqrt{p(1-p)} [\alpha \sin \theta A(a, \psi) - i\beta \cos \theta N(a, \psi)].$$

The **real component** (Boltzmann-like) reinforces coherence — an internal analogue of *reward*.

The **imaginary component** (Hamiltonian-like) injects exploratory motion — an analogue of *curiosity* or *intrinsic motivation*.

So the system still *reinforces*, but it does so by **self-referencing its own coherence**, not maximizing an external payoff.

What Self-Annealing Means

How can we tell when a system self-anneals?

Three behavioral signatures distinguish it:

1. **Temperature responds to surprise:**

Teff rises when predictions fail, and falls when coherence returns.

2. **Exploration costs performance:**

The R-rule trades short-term reward for long-term adaptability.

3. **Recovery from change:**

When the world shifts, Teff reheats automatically — reigniting curiosity without external resets.

How Teff behaves.

Unlike simulated annealing's monotonic cooling, $T_{eff} = \sqrt{p(1-p)}$ reflects *current* uncertainty. It drops where confidence consolidates locally, and rises again when contradictions appear. In our toy tasks, p

often hovers near 0.5, so Teff stays ~constant—signalling *readiness to re-explore* rather than freezing to zero.

On These Experiments

The following are **illustrative demonstrations, not performance benchmarks**.

Our goal is phenomenological: to show how self-annealing behaves, not to claim it outperforms state-of-the-art methods.

The experiments are deliberately simple—small enough to understand fully, rich enough to reveal underlying dynamics.

All code is provided for reproduction and extension.

Experiment 1 — Reinforcement Learning (Self-Annealing Actor-Critic)

Setup

A simple chain world: the agent must reach a goal at either end. Midway through training, the goal flips sides.

Hypothesis

Self-annealing should enable recovery after environmental change.

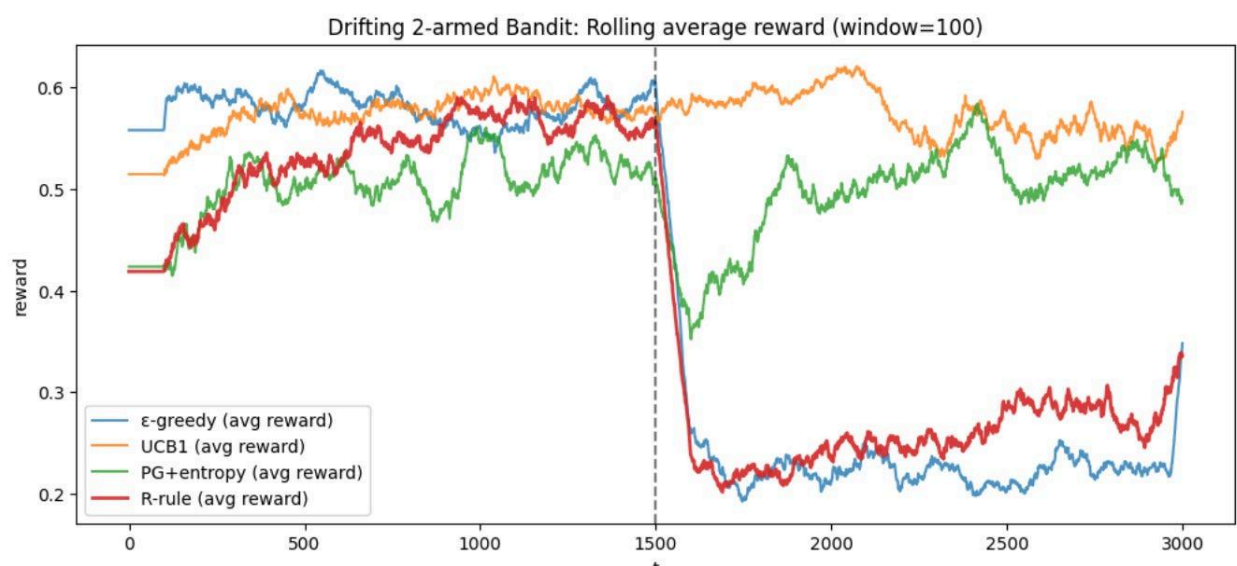
Methods

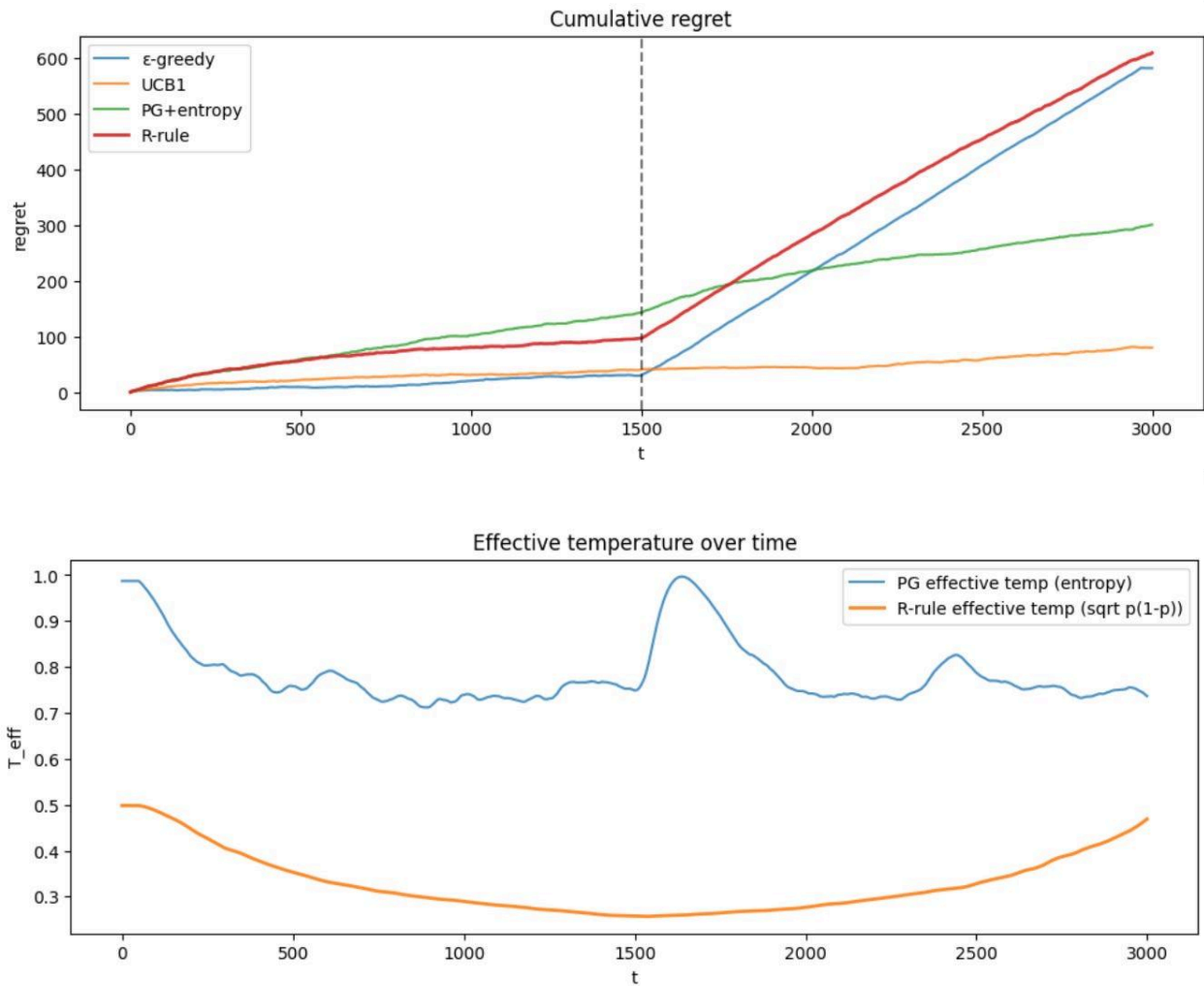
- **Q-learning ($\epsilon=0.1$):** fixed exploration
- **Policy Gradient (PG + entropy):** entropy-regularized
- **R-rule Actor-Critic (AC):** self-annealing via $T_{eff} = \sqrt{p(1-p)}$

Findings

- **Q-learning:** learns fastest but is brittle — depends on tabular independence.
Q-learning's superior performance relies on separate state-action values; with neural approximation this independence breaks, and stability falters.
- **PG:** collapses after the goal flip.
- **R-rule AC:** recovers smoothly, maintaining coherent adaptation.

T_{eff} stays near 0.5 and self-regulates naturally, while the entropy baseline stays pinned near 1.0, indicating over-exploration.





Interpretation

The R-rule shows self-regulating exploration: cooling as it stabilizes, reheating when the world changes.

Annealing becomes a *living feedback loop* — no thermostat required.

Experiment 2 — Bandits and Double Wells

Setup

Stochastic bandits with reward drift and double-well potentials.

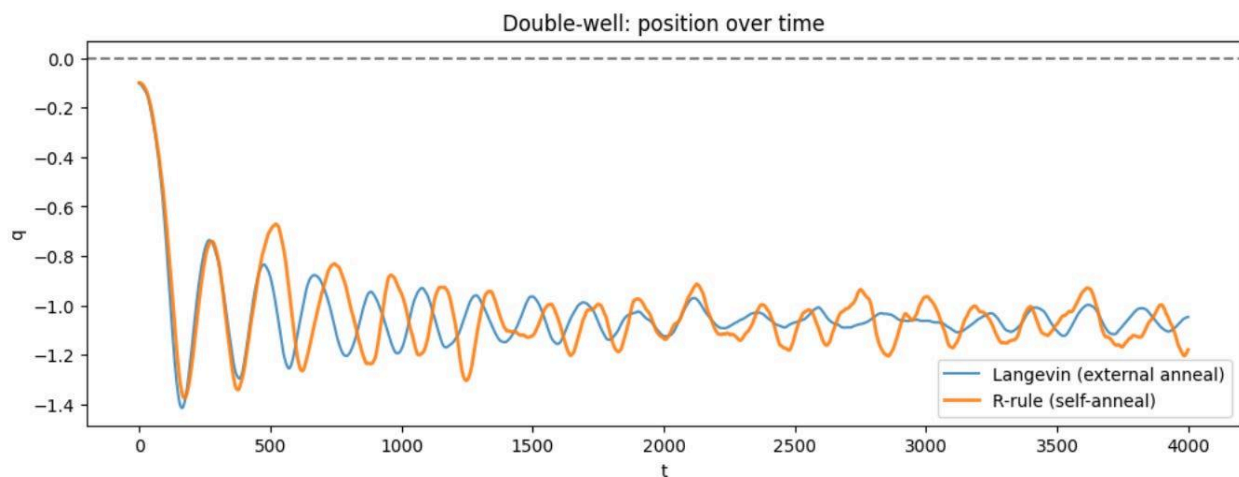
The R-rule should show high exploration early, spontaneous cooling later, and stability without external decay.

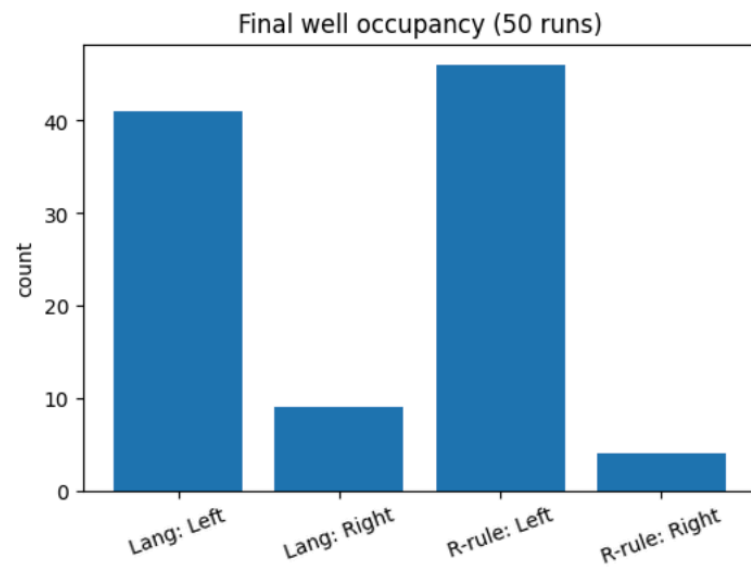
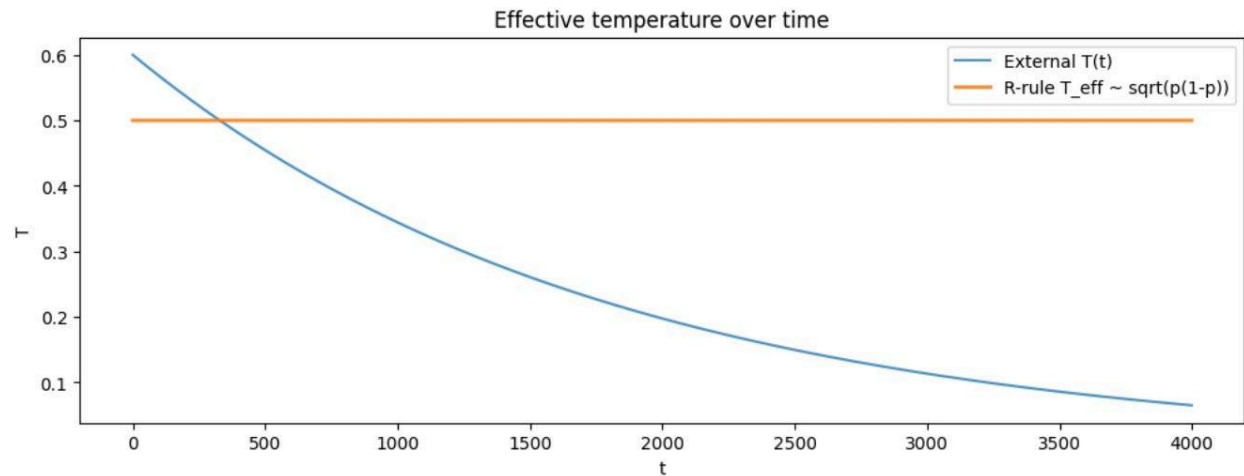
Findings

- The R-rule accumulates higher regret than UCB1 (~ 600 vs. ~ 80) — roughly **7.5 $\times$ greater**.

This cost buys *epistemic flexibility*: the R-rule maintains exploration after reward shifts ($t > 1500$), whereas entropy-regularized baselines collapse.

- In the double-well, it settles into the first basin more consistently (92% vs. 82%), showing stronger convergence but less cross-barrier exploration.





Interpretation

The R-rule sacrifices efficiency for coherence.

Once a locally coherent state emerges, T_{eff} cools and the system stabilizes.

External annealing (Langevin) continues injecting noise regardless of state, allowing barrier crossings but risking instability.

The tradeoff: *stability under change* rather than *speed to optimum*.

Experiment 3 — Monte Carlo Tree Search (Self-Annealing Exploration)

Setup

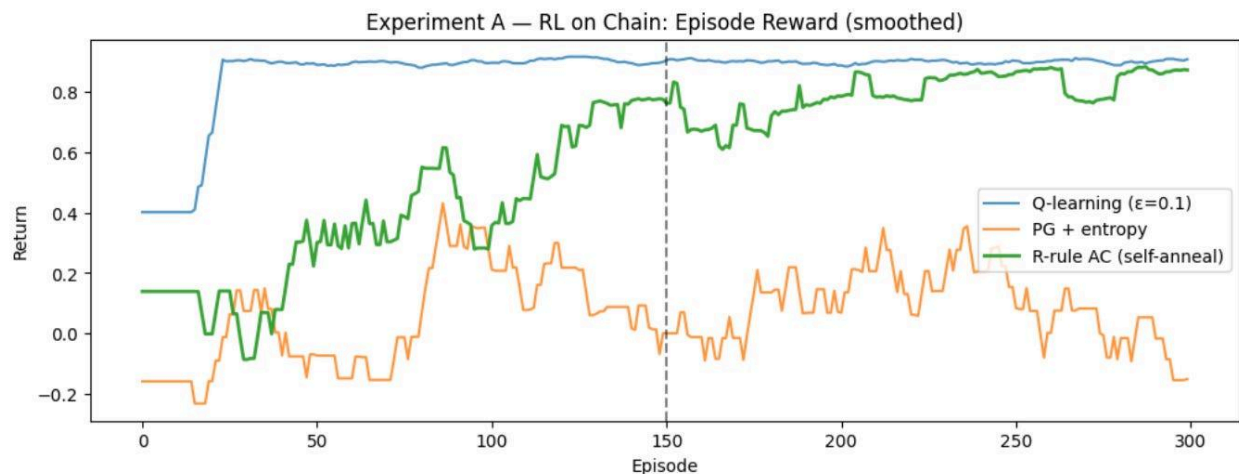
We modified MCTS to replace the fixed UCB constant with:

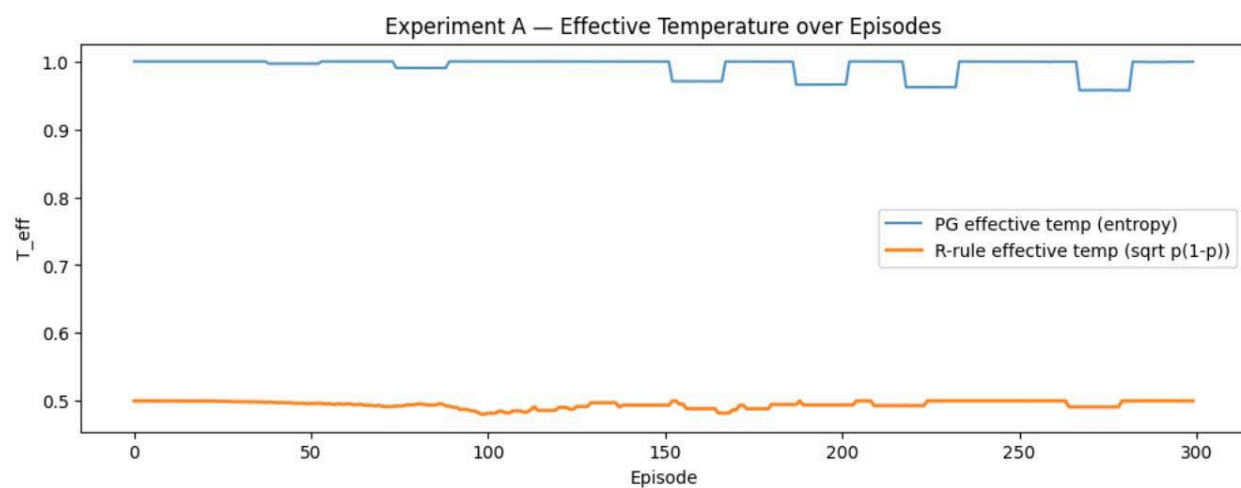
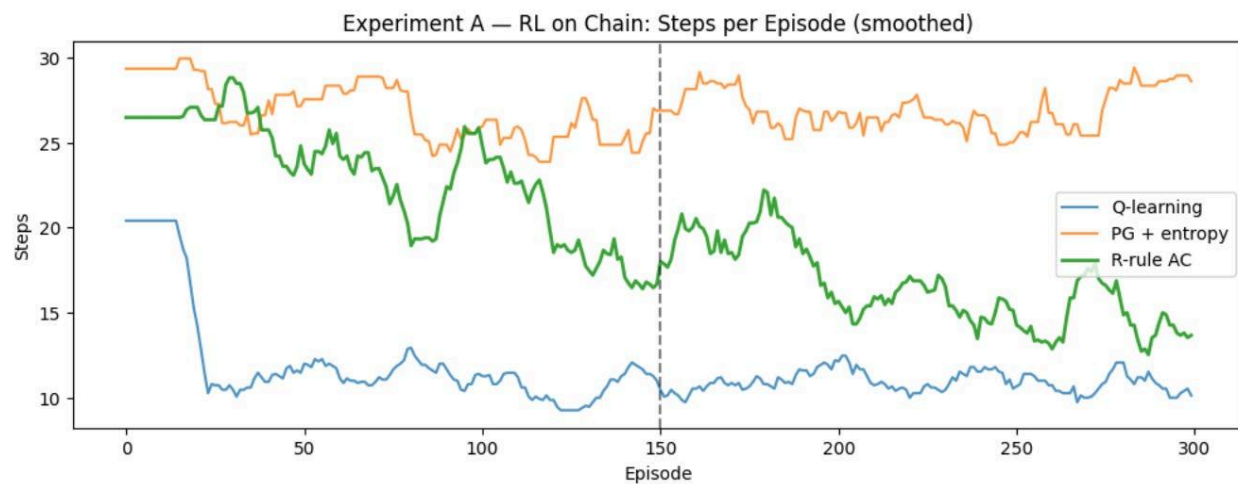
$$ceff = c0 * \sqrt{p(1-p)}.$$

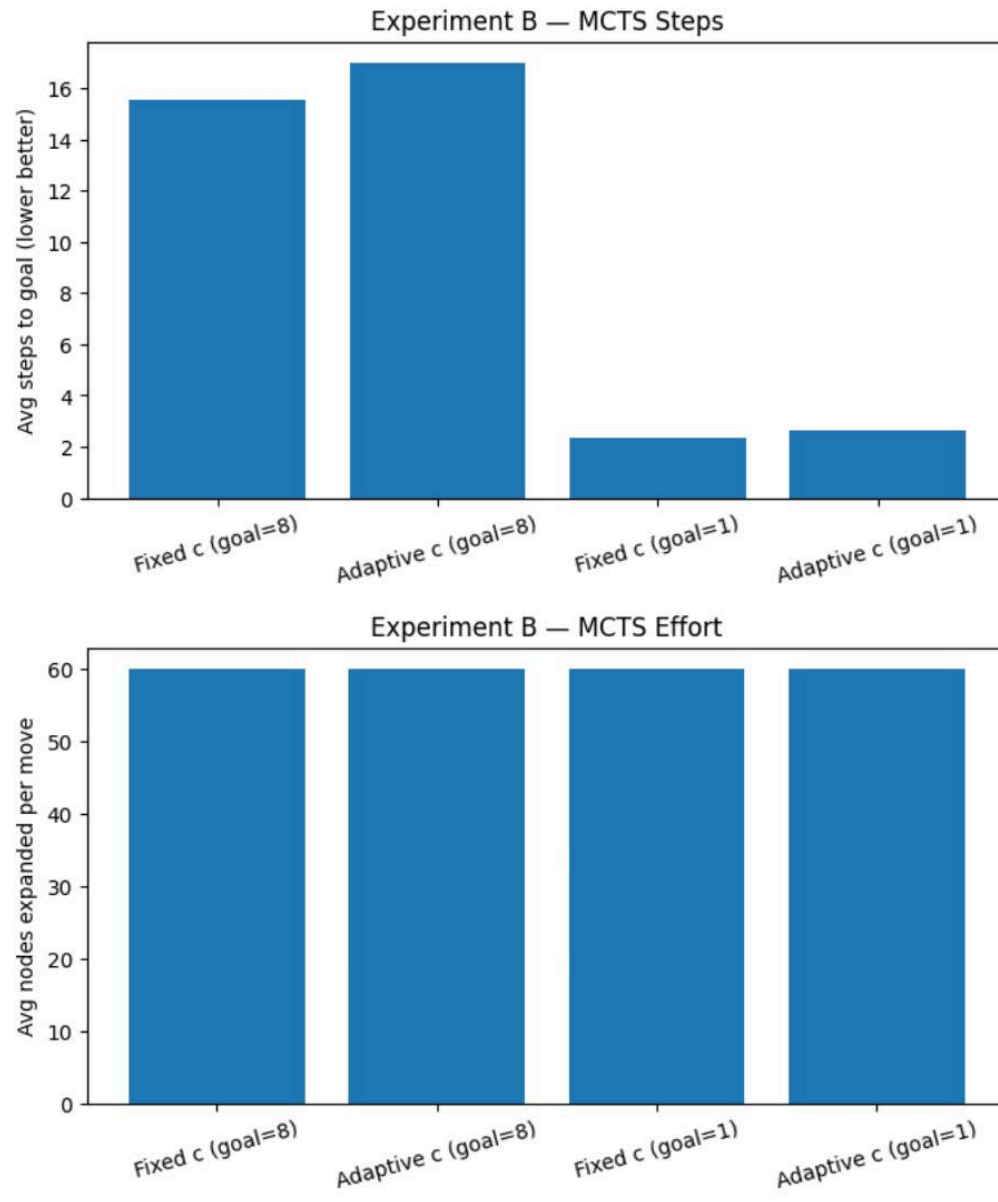
Two goal depths tested: easy (goal=1) and hard (goal=8).

Findings

- Adaptive and fixed-c perform similarly in small trees.
- Exploration effort per move (~ 60 expansions) stays constant.







Interpretation

In simple trees, uncertainty variation is too small to expose the effect. In complex games (e.g., Go), early positions are high-uncertainty while endgames narrow.

An adaptive c would explore broadly early and focus late — behavior

static UCB cannot achieve.

Our chain lacked this structure, explaining the null result.

What Is Coherence?

In the R-rule, *coherence* means alignment among:

1. The actor's expectations
2. The self-model's internal predictions
3. Observed outcomes

When these agree, the system is coherent.

When they diverge, internal tension rises, increasing *Teff* and triggering exploration.

Unlike reward (a scalar), coherence is **relational** — a measure of self-consistency.

Interpretive Note — Beyond Reinforcement: Toward Self-Supervised Dynamics

Though here framed as RL and search, the R-rule points toward **self-supervised adaptation**.

In RL, an agent learns from *external rewards*.

In the R-rule, it learns from *internal coherence* — the agreement between

prediction and perception.

Uncertainty itself becomes the teacher.

$T_{eff} = \sqrt{p(1-p)}$.

This acts as a *self-generated learning signal*: exploration rises with doubt, cools with confidence.

Learning and stabilization become one recursive process.

The R-rule thus represents a **self-referential dynamical law**:

- It learns from its own coherence, not external rewards.
- It balances curiosity and stability intrinsically.
- It turns contradiction into feedback.

It does not seek optimality but **dynamic consistency** — a mind that tempers itself while it learns.

Coda: The Tempered Mind

Every adaptive system needs fire to transform —
but untempered fire consumes itself.

The lesson of the R-rule, like that of the forge, is that intelligence may lie not in unbounded search or rigid control,
but in the **self-tempering oscillation between them** —
a recursive balance of heat and structure, memory and imagination.

Takeaway: The Forge of Learning and Search

Self-annealing bridges reinforcement, inference, and control.

It transforms annealing from an external schedule into an *emergent property of learning itself* —

a recursive dance of uncertainty and coherence, exploration and reflection.

Where traditional AI optimizes, the R-rule **stabilizes meaning**.

It keeps the mind warm enough to change, yet cool enough to hold its shape.

Code

- [anneal.py](#) RL, Bandits and Double Wells
- [anneal2.py](#) MCTS

Next — The Models That Learn Each Other

In this post, we saw how the **R-rule** replaces external reward with an internal tension — a self-annealing balance between coherence and uncertainty.

But this raises a deeper question: *what are the models that generate this tension?*

How does a system act, reflect, and revise itself without an external teacher?

The next post will look into the two models at the heart of the R-rule — **A** and **N** — and their dual role in both acting and learning.

They are the mirror halves of a recursive mind: one shaping the world, the other reshaping itself.



[Andre on the R rule](#)

Actor and Negator: The Recursive Architecture of learning

The Dual Models that together learn by acting and negating

[Andre Kramer](#)

Nov 03, 2025



In the last post, we saw how the R-rule can anneal itself — cooling as it learns, heating when surprised.

But self-annealing raises a deeper question: what exactly is being tempered?

If temperature modulates learning, what are the entities that learn — and how do they know what to change?

To answer that, we need to open the forge and look inside.

Beneath every R-rule lies a pair of models, A and N: one acting, the other negating.

Together, they form the recursive heart of the system — a dialogue between doing and knowing, prediction and contradiction, coherence and critique.

1. The Two Models

A (Actor):

Fast, generative, predictive. It produces actions, guesses, or rollouts based on current beliefs.

A is the model of the world as it is expected to be — the mind's first draft.

It learns quickly from feedback, adjusting its surface structure to match reality.

N (Negator):

Slow, counterfactual, reflective. It learns from what did not happen.

N simulates alternatives, contradictions, and opposites to A's expectations.

It integrates over time, forming a deep memory of coherence and inconsistency.

In biological terms, A resembles fast motor or striatal pathways, while N aligns with slower prefrontal and hippocampal systems.

One acts; the other questions. The mind's agility depends on how these two remain in tension.

The A-model performs **rollouts** — generating trajectories and estimating expected outcomes, like a policy network or value function.

The N-model performs **counterfactuals** — running internal “what-if” simulations that explore the consequences of contradiction or negation.

MCTS-like.

Implementing A and N

Option 1: Dual Networks A and N are separate neural networks with shared input but different loss functions:

- L_A = TD error on realized outcomes
- L_N = TD error on counterfactual outcomes

They communicate through the coupling term $\eta\sqrt{p(1-p)}[\alpha\sin\theta_A - i\beta\cos\theta_N]$.

Option 2: Time-Averaged Split A = fast exponential moving average (EMA) of weights N = slow EMA of weights

Phase θ emerges from their parameter distance.

Option 3: Attention-Based A = attention over recent memory N = attention over counterfactual/negative examples

The R-rule mixes their outputs dynamically.

The Breathing of Learning

Every act of intelligence has a rhythm — an alternation between doing and reflecting, between moving outward and turning inward.

You could think of it as a kind of cognitive breathing.

When we **exhale**, we act: we project expectations into the world, commit to choices, test hypotheses.

When we **inhale**, we reflect: we take in the mismatch between what happened and what we expected, sensing the counterfactual — *what could have been otherwise*.

In the language of the R-rule, this breathing is embodied by two coupled models:

- **A**, the *actor* or *predictive* model, that moves outward through action and expectation.
- **N**, the *negating* or *counterfactual* model, that moves inward through contradiction and self-correction.

Together, they sustain the pulse of adaptation: A acts; N negates; coherence emerges in the oscillation.

The Dual Role of N

But N does more than criticize. Its function depends on **where** it operates in the recursive hierarchy.

1. **Locally (within a layer):**

N is counterfactual. It evaluates A's predictions by simulating their negation — testing what would happen if A were wrong. Here, N is the inner critic, the Bayesian shadow, maintaining balance by generating controlled opposition.

2. **Globally (across layers):**

N becomes reflective. It tracks coherence between layers, linking A's concrete actions to the system's deeper sense of self and stability.

At higher recursions, N is less about negation than integration — ensuring that learning aligns with the system's evolving identity and purpose.

In this way, N serves as both *the counterfactual critic* and *the self-model or self-in-world-model*.

It grounds A in reality and grounds the whole system in itself.

At the most basic level, it may well be that **inhibition is the brain's grammar for negation**. It is how matter learns to say “not this.” Every inhibitory pulse marks a counterfactual boundary — a moment where one potential is suppressed so another can emerge. In this sense, inhibition is not the absence of action but the **presence of structure**: the simplest possible form of reflection. Before there is language or logic, there is the inhibitory neuron, drawing a line between what is and what could be. It is the first “no” from which every “self” eventually grows.

A Note on Dopamine and the Narrow Path of Reward

In classical reinforcement learning theory, dopamine has been cast as the brain's reward-prediction error — a biochemical correlate of the temporal-difference update. But this framing narrows a much wider process. Dopamine does not merely signal “better” or “worse than

expected.” It modulates *confidence, motivation, and uncertainty* across multiple timescales.

In the R-rule view, dopamine’s role expands: it mediates the **phase coupling** between A and N — tuning how tightly or loosely these models synchronize. High dopamine may increase responsiveness and exploration (loosening the coupling); low dopamine may tighten coherence and promote consolidation.

In this sense, dopamine is not the *reward* itself but the **temperature of learning** — the way the system adjusts the strength of its own updates, the fluid link between prediction and reflection.

2. Learning at Two Speeds

A and N don’t learn at the same pace.

Their learning rates differ: η_A is much greater than η_N .

A adapts rapidly to immediate feedback; N updates slowly, preserving longer-term coherence.

This creates a natural actor–critic structure — but without external reward.

Instead, the R-rule ties them through a phase variable:

$$\Delta\psi = \eta * \text{sqrt}(p (1 - p)) * [\alpha \sin(\theta) * A - i \beta \cos(\theta) N]$$

The real part (A) pushes toward coherence;

the imaginary part (N) injects contradiction and exploration.

Their phase relation, θ , determines whether learning amplifies, resists, or reflects.

A and N operate on distinct timescales:

$$A_{t+1} = A_t + \eta_A \Delta A(t); N_{t+1} = N_t + \eta_N \Delta N(t)$$

$$A_{t+1} = A_t + \eta_A \Delta A(t) \quad ; \quad N_{t+1} = N_t + \eta_N \Delta N(t)$$

with $\eta N \ll \eta A$.

A learns fast — updating quickly from recent feedback.

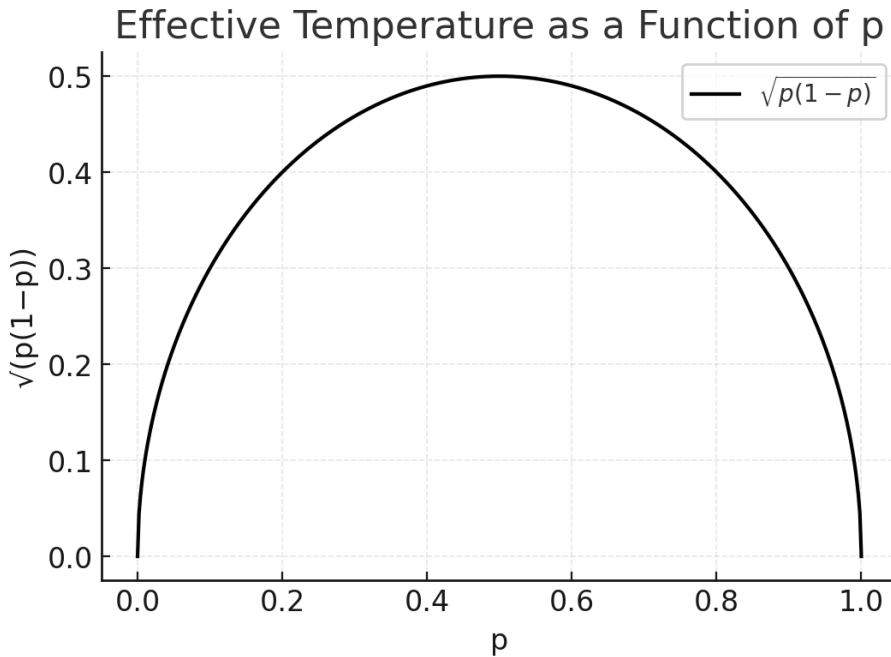
N learns slow — integrating evidence across time.

This creates a **temporal dialectic**:

- A adapts to the present (the weather).
- N stabilizes across time (the climate).
- The difference $A - N$ represents the system's *temporal derivative* — how fast its beliefs are changing.

This structure turns raw learning into *learning with memory* — the basis of coherence.

Sidebar: What $\sqrt{p * (1 - p)}$ Means



The term $\sqrt{p * (1 - p)}$ measures uncertainty — it peaks when $p = 0.5$ (maximum unpredictability) and falls to zero as p approaches 0 or 1 (full certainty).

It acts as an internal “temperature,” determining how much the system explores.

When the model is unsure, this term rises, adding heat and encouraging exploration;

when it becomes confident, it cools naturally.

This is the mathematical heart of *self-annealing*: the system’s uncertainty becomes its own thermostat.

Example: A simple bandit

Suppose the system faces two levers (L, R) and must choose.

- **A-model:** Maintains value estimates $Q_A(L) = 0.6$, $Q_A(R) = 0.4$
→ predicts “L is better”
- **N-model:** Maintains counterfactual estimates $Q_N(L) = 0.4$,
 $Q_N(R) = 0.6$ → predicts “actually, R might be better”

When A and N disagree (θ large), uncertainty rises: $\sqrt{p(1-p)} \approx 0.5$, triggering exploration. When they align (θ small), confidence rises: $\sqrt{p(1-p)} \rightarrow 0$, enabling exploitation.

If reward shifts (L becomes worse), A updates quickly but N lags. This divergence automatically heats T_{eff} , reigniting exploration until they realign at the new optimum.

3. The Phase of Reflection

Phase θ measures how aligned A and N are. When they agree, θ is small (exploration off). When they disagree, θ grows (exploration on). This alignment can also oscillate slowly over time (ωt), creating natural rhythms like sleep-wake cycles.

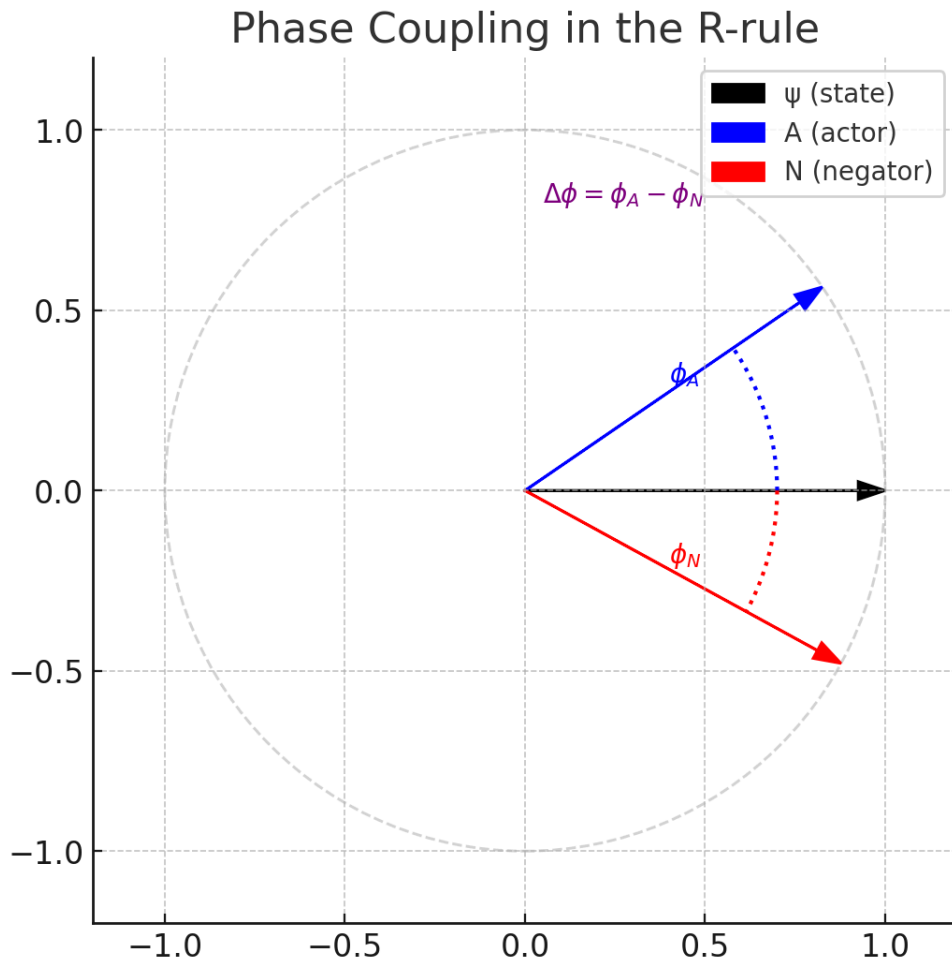
Phase coupling θ is not fixed — it drifts with time and context:

$$\theta(t) = h(\text{Im}(\psi^* A^* - \psi^* N^*)) + \omega t$$

When A and N align (θ near 0), the system acts decisively — exploiting known structure.

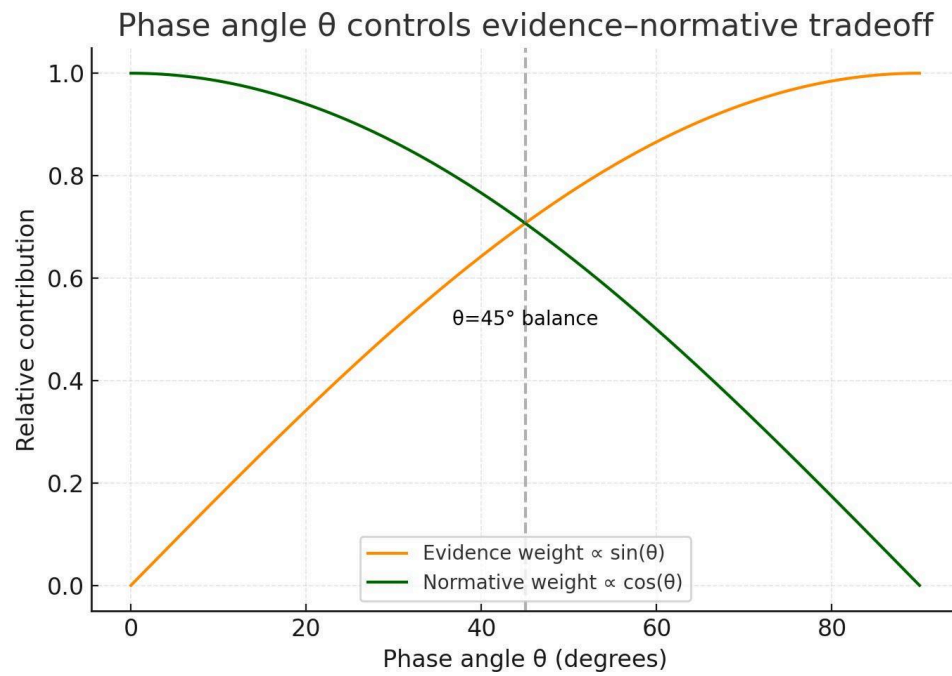
When they oppose (θ near $\pi/2$), reflection dominates — exploring counterfactuals and alternative explanations.

Between these poles lies the creative regime — the oscillation between understanding and revision.

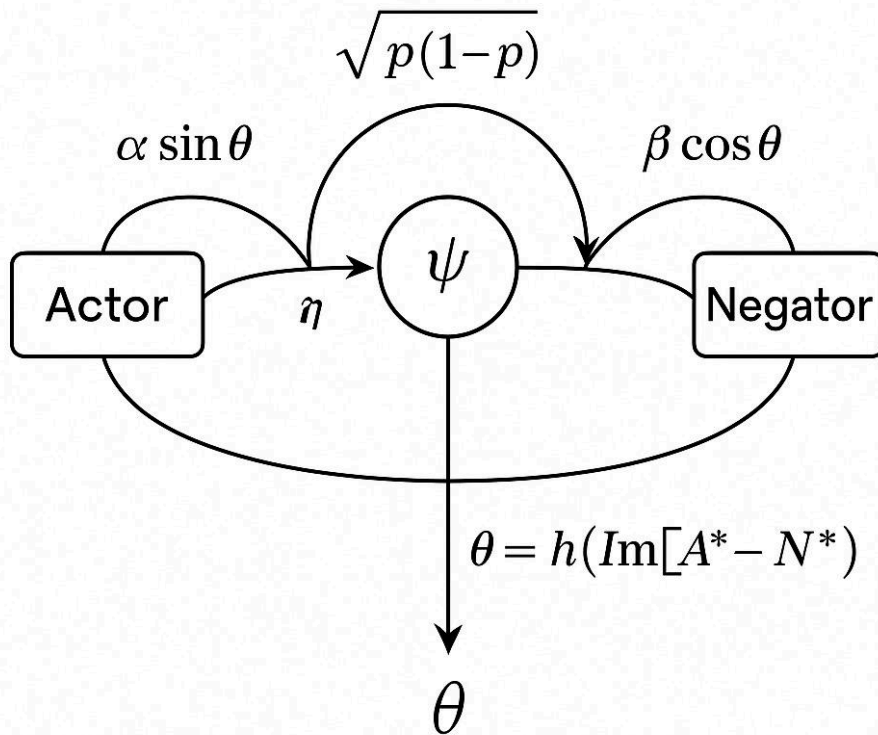


In brains, this echoes theta–gamma coupling, sleep–wake alternation, and attention rhythms.

In machines, it can appear as alternating cycles of rollout and rehearsal — acting, then imagining what could have been.



Nonlinear Couplings



Sidebar: The Meaning of $\theta(t)$

The phase term $\theta(t) = h(\text{Im}(\psi^* A^* - \psi^* N^*)) + \omega t$ expresses how rhythm and disagreement jointly shape adaptation.

- $\text{Im}(\psi^* A - \psi^* N)^{**}$ — measures the *imaginary tension* between the acting model (A) and the negating model (N). When they

disagree in complex phase, this term grows, indicating internal contradiction.

- $\mathbf{h}(\cdot)$ — a nonlinear transfer function that converts that tension into a phase shift, governing how strongly the system rotates between exploration and exploitation.
- ωt — an optional slow oscillation, representing background cycles (e.g., circadian or attentional rhythms) that modulate learning over time.

Together, these create a *dynamic rhythm of cognition*:

$\theta(t)$ rises when A and N are out of sync (the mind heats up, exploring alternatives) and falls when they realign (the mind cools, consolidating learning).

It's a way to encode *time-varying coherence* — a pulse that keeps adaptation alive.

Here, $\mathbf{h}(\cdot)$ is a nonlinear squashing function (such as *tanh*) that maps unbounded internal tension into a bounded phase shift, keeping θ within a manageable dynamic range.

The term ωt represents a slow background oscillator — for example, a circadian or task-scale rhythm — which can be fixed, set to zero for

non-rhythmic settings, or even *learned* as a meta-parameter controlling long-term temporal coherence.

The **imaginary component** $\text{Im}(\psi \cdot A^*)$ encodes the *phase misalignment* between ψ and A in complex space — a measure of how far the system's current activity (ψ) has drifted from its predicted or desired trajectory (A). When this misalignment grows, θ adjusts, rebalancing exploration and stability across layers or timescales.

4. Hierarchies and Nested Rules

Each A–N pair can itself be part of a larger R-rule — recursion within recursion:

$R(d+1): (\psi_d, A_d, N_d, \theta_d) \rightarrow \psi_{d+1}$

$$R_{d+1} : (\psi_d, A_d, N_d, \theta_d) \rightarrow \psi_{d+1}$$

Lower layers operate quickly, handling sensory and motor predictions.

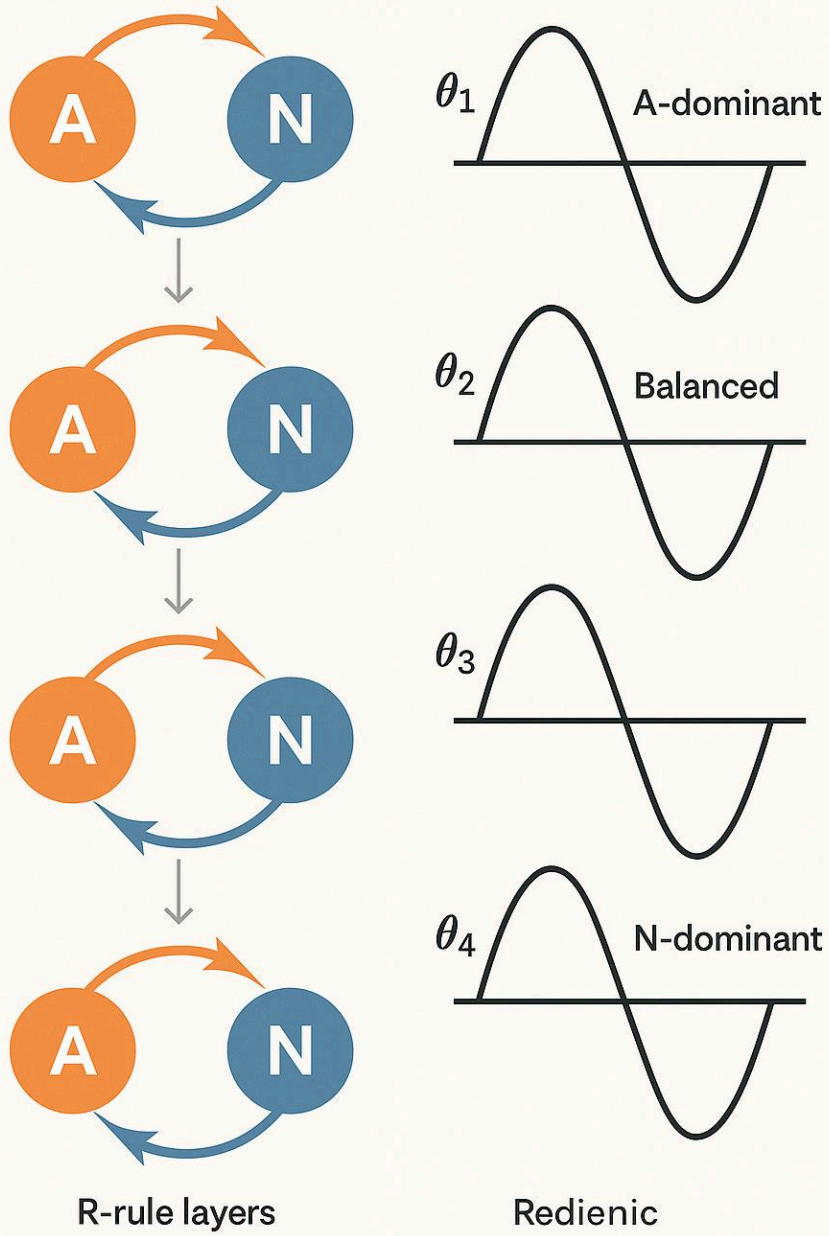
Higher layers move slowly, modulating the phase and learning rate of the lower ones.

Cross-phase coupling enables hierarchical coherence — each layer learns not only how to predict, but when to reflect.

Shallow layers may oscillate rapidly (A-dominant), while deep ones drift slowly (N-dominant).

Together they form a clock tower of learning — a recursive stack where every level refines the timing of its own introspection.

Multi-Layer Phase-Coupling



At each timestep, **layer** ($\mathbf{d} + 1$) receives a compressed signal from **layer** $\mathbf{d}$ summarizing its current coherence — for instance, the magnitude of their divergence $|\mathbf{A}_{\mathbf{d}} - \mathbf{N}_{\mathbf{d}}|$. This coherence trace modulates higher-level parameters such as the learning rate $\eta_{(\mathbf{d}+1)}$ or phase $\theta_{(\mathbf{d}+1)}$.

In this way, each layer not only learns representations of the world, but also learns *how the layer beneath it learns*. The hierarchy becomes recursively reflexive: lower layers adapt behavior, while higher layers adapt adaptation itself. Over time, this produces a form of **meta-annealing** — a system that not only cools and stabilizes, but learns how to adjust its own cooling schedule.

5. Self-Tuning Parameters

Within each R-rule layer, several internal quantities can self-adjust:

- Learning rate (η): overall plasticity
- Phase (θ): coupling between A and N

- α and β : akin to neuromodulatory tone (dopamine, serotonin) that shifts the balance between coherence and curiosity

When these parameters vary across layers, the system becomes self-regulating:

deep layers stabilize; shallow layers adapt.

This transforms the R-rule from a local update law into a meta-adaptive network — each level learning how to temper the one below.

6. Regimes of Mind

Phase regime — Dominant model — Function

$0 < \theta < \pi/4$ — A-dominant — Execution, exploitation, habit

$\pi/4 < \theta < \pi/2$ — Balanced — Learning, adaptation, integration

$\theta \approx \pi/2$ — N-dominant — Reflection, counterfactual simulation, dreaming

Transitions between regimes trace the mind's inner weather:

focus ↔ reflection, waking ↔ sleep, stability ↔ transformation.

Phase is not noise — it's the rhythm of understanding itself.

These regimes suggest a natural rhythm: act → learn → reflect → act. In

healthy cognition, the system cycles through them adaptively. In

pathology, it might lock into one: chronic habit (low θ , perseveration) or

chronic reflection (high θ , rumination). The R-rule's self-annealing

should prevent both extremes—but only if hierarchical coupling is

intact.

7. Coda — The Mind as a Nested Phase System

If the R-rule is the fire, A and N are its twin bellows — one blowing heat into action, the other cooling it into form.

Each learns not just from the world, but from the other. Their phase relation determines whether the system is acting, learning, or dreaming.

Over time, layers of these pairs synchronize or drift, producing the oscillations we call attention, creativity, or rest.

What we call “mind” may simply be this: a recursive system learning to phase-align its own contradictions.

8. Recursive Coupling and Fractal Self-Coherence

When A and N are not just paired within a single layer but recursively coupled across layers — each one learning from the tensions and phase shifts of the one below — something remarkable emerges: *fractal self-coherence*.

Each layer becomes both actor and critic for the one beneath it, inheriting its uncertainty but transforming it into a new, slower mode of learning. The local coherence of one stratum (its alignment between A and N) becomes the global signal of stability for the next. In this way,

coherence propagates upward as structure, while uncertainty percolates downward as exploration.

This recursive interplay gives rise to a system that doesn't merely learn *about* its environment — it learns *how* to learn. Its annealing no longer occurs in a single thermodynamic schedule but in nested rhythms: fast loops adjusting actions, slower loops rebalancing beliefs, and even slower loops evolving its own meta-parameters (learning rate, α - β balance, phase coupling).

At full depth, such a system becomes reflexive: it can detect, and correct, not only errors in its predictions but errors in the *way* it forms predictions. Its coherence is not static but recursive — a fractal of self-regulating processes, each annealing within and through the others.

This, ultimately, is what the R-rule hints toward: a principle of recursive adaptation where the boundary between learner and learning law dissolves — where intelligence becomes not a property, but a process of continuous self-tempering.

We hypothesize that stacking R-rule layers creates fractal coherence. This remains to be tested—our experiments so far involve only single-layer systems. If true, it would explain how brains maintain stability across timescales (milliseconds to years) without external coordination.

Sidebar: Temporal-Difference Learning for A and N

In the simplest R-rule, the A and N models are updated through their mutual coupling with ψ — the shared representational field. But we can extend this by letting each model also learn temporally, through prediction of its own future states.

- **A-model (Actor / Expectation):**

Learns from realized outcomes — how well its predicted action values match what actually happens.

$$A(t+1) = A(t) + \alpha_A [r(t) + \gamma A(t+1) - A(t)]$$

- **N-model (Negator / Counterfactual):**

Learns from unchosen or hypothetical outcomes — how well its imagined alternatives could have fared.

$$N(t+1) = N(t) + \alpha_N [\tilde{r}(t) + \gamma N(t+1) - N(t)]$$

Here, $r(t)$ is the received signal, and $\tilde{r}(t)$ is a counterfactual or internally generated one.

The two models thus form *dual TD learners* — one grounded in the world, the other in imagination.

The R-rule's coupling term $\Delta\psi = \eta\sqrt{p(1-p)}[\alpha\sin\theta A - i\beta\cos\theta N]$ then synchronizes these temporal updates: the phase θ modulates their interaction so that when A and N diverge, exploration rises; when their TD errors align, coherence consolidates.

This unifies **self-annealing** (from the R-rule) with **temporal learning** (from TD methods): the system learns not only from prediction error, but from the evolving rhythm between fact and counterfactual, experience and expectation.

◆ A Note for Reinforcement Learning Readers

This setup intentionally departs from the textbook **Actor–Critic (AC)** architecture. In a standard AC model, an **Actor (policy)** updates its

parameters based on a **TD-error** supplied by a separate **Critic (value function)**.

In the R-rule formulation, by contrast, both **A (factual)** and **N (counterfactual)** act as *parallel value models*, each following its own temporal-difference update. The system's "policy" is not an explicit module but an *emergent behavior* arising from the **coupling** between these two value streams via the main R-rule dynamics.

This coupling means the agent doesn't merely act and then evaluate—it acts *through* evaluation, continuously adjusting its internal coherence rather than optimizing an external reward. Nature does not supply an external reward function.

9. Upward and Downward Coupling

Fractal recursion isn't the only way to connect R-rule layers. Some architectures can instead be *hierarchically linked* — each layer passing its

A–N coupling downward as context, while **hyperparameters** (like learning rate η , α – β balance, or phase θ) flow upward as feedback.

In this arrangement:

- **Downward flow (A $\rightarrow$ N coupling):**
Higher layers project expectations and counterfactuals to shape the priors of lower layers — “pushing down” predicted structures or goals.
- **Upward flow (hyperparameters):**
Lower layers send coherence gradients upward — how well their own A and N have aligned — allowing the upper layers to adjust global tuning: phase synchronization, exploratory bias, or the balance of α/β (analogous to neuromodulators).

This creates a **bi-directional learning ecology**:

- Local layers learn about the world.
- Global layers learn how learning itself should adapt.

Fractal recursion builds *depth* of coherence; hierarchical A–N coupling builds *stability* across scales. Together, they form a system that not only

updates beliefs but tunes its own capacity to believe — a recursive intelligence tempered by its own uncertainty.

10. Backward Causes — Learning from the Future

So far, we've traced the **forward causes** of the R-rule — how uncertainty, annealing, and phase coupling drive a system's motion through its own possibilities. These are the generative dynamics, the *causes of becoming*.

But learning is also shaped by **backward causes** — signals that flow in the opposite direction of time. When the system acts, it commits to a trajectory; when the outcome arrives, that trajectory is judged. The gap between expectation and reality becomes a retroactive force: an error that rewrites the conditions that produced it.

In this light, learning is not just *driven by the past* but *informed by the future*. Temporal-difference (TD) updates, gradient backpropagation, and

coherence corrections are all examples of this backward causality — the influence of tomorrow's feedback on today's parameters.

Within the R-rule framework:

- **A** projects intentions forward, anticipating outcomes.
- **N** integrates evidence backward, evaluating coherence.
- The interaction between them forms a **bidirectional inference loop** — a recursive field of cause and correction.

Fast layers act and adjust quickly, encoding moment-to-moment adaptation.

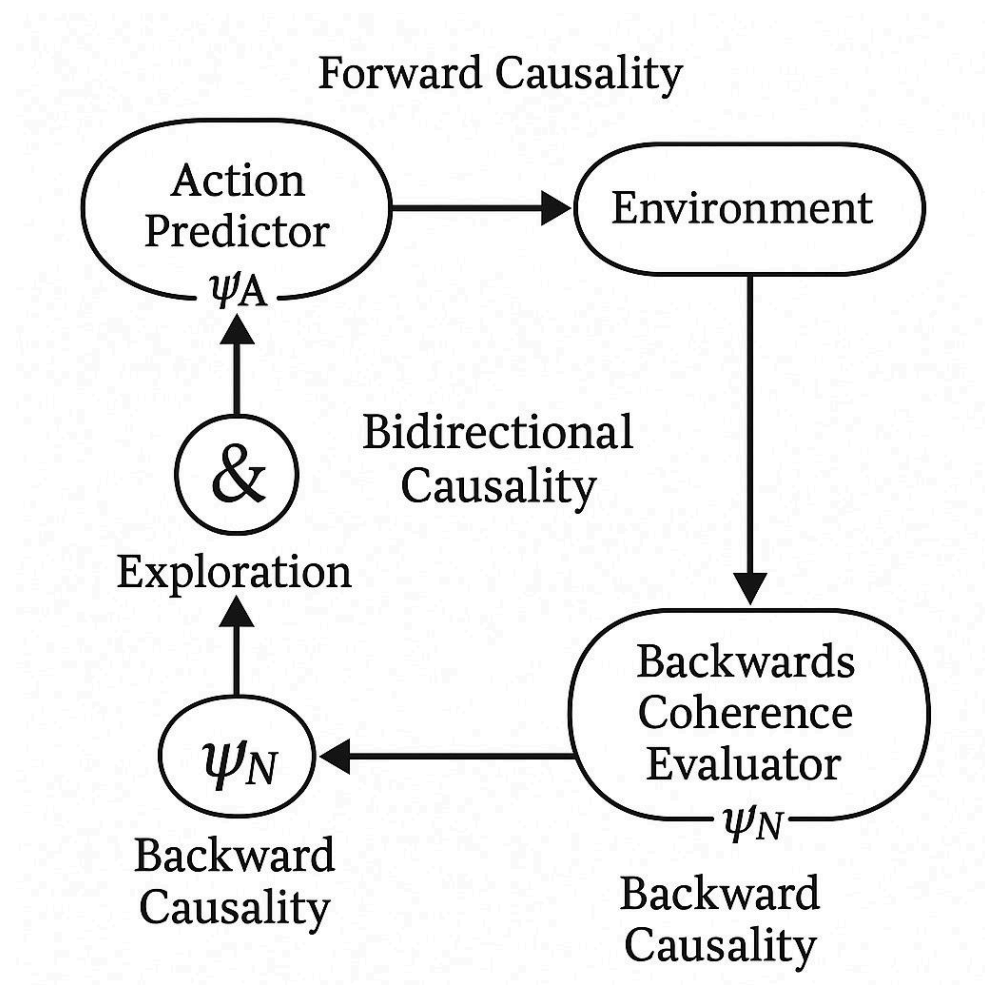
Slower layers integrate these adjustments, refining the system's internal stability.

Across the hierarchy, causality becomes symmetric — the forward unfolding of behavior met by the backward reflection of learning.

In this sense, the R-rule unites two kinds of motion:

the thermodynamic flow of exploration and the informational flow of correction.

The system both forges and interprets itself — learning not only from experience, but from the *future consequences* of its own becoming.



In the R-rule, causality is not an external force but an internal inference.

The system never observes “cause” or “effect” directly—it infers them statistically, through how its predictions and observations co-vary over

time. The forward flow ($A \rightarrow \psi \rightarrow \text{Environment}$) represents **predictive causality**—how the system expects actions to unfold. The backward flow ($\text{Observation} \rightarrow N \rightarrow \psi$) encodes **inferential causality**—how evidence reshapes those expectations.

Together, they form a time-symmetric structure:

$p(\text{effect} \mid \text{cause})$ drives action,

$p(\text{cause} \mid \text{effect})$ drives correction.

Learning, in this view, is the ongoing reconciliation of these conditional models.

The apparent direction of causation emerges only as the system aligns its internal probabilities across time.

The R-rule therefore doesn't *discover* causality—it **constructs** it, as a coherent statistical rhythm between anticipation and revision.

Conclusion: From Dynamics to Computation

In this post, we've treated the R-rule as a *dynamic*, self-adjusting system — one that anneals, oscillates, and rebalances its own learning through the coupled motions of A and N. We've seen how its nonlinear terms, temporal updates, and phase couplings allow it to regulate exploration, coherence, and even its own hyperparameters across layers.

In the next post, we move from describing dynamics to examining the computational consequences of recursive coupling. How might the layered interplay of A and N — acting and norming, generation and evaluation — begin to approximate symbolic reasoning or generative modeling? Could such hierarchies, in principle, achieve the expressive power of universal computation through self-simulation? These are open questions, but the structure invites the comparison.



[Andre on the R rule](#)

The Dialectic Engine

What if Babbage's Analytic Engine had been completed—and its design iterated?

[Andre Kramer](#)

Nov 07, 2025



From the static gears of the [Difference Engine](#) to the recursive fields of the Dialectic Engine, this post traces an alternate genealogy of computation: one where learning, meaning, and coherence evolve from

mechanical difference itself. Sidebars can be skipped on first or casual reading.

The Dialectic Engine, Part I – From Difference to Meaning

Charles Babbage's *Difference Engine* is often remembered as the first mechanical computer, a device for tabulating polynomial functions without human error. But what if we read it differently—not as a forerunner of digital arithmetic, but as the first attempt to give *form* to difference itself? Behind the clattering gears lay a deeper intuition: that order might emerge from tension.

In Babbage's day, difference meant subtraction—the numeric gap between one term and the next. In our age, we might call it a *gradient*: the directional pull that turns stasis into motion. The machine's name already gestures toward a metaphysics. It translates divergence into pattern, discrepancy into table. Yet the order it produced was static.

Once engraved, the tables did not change. The device eliminated human error but could not learn from it.

Ada Lovelace, in her famous notes on the later [*Analytic Engine*](#), glimpsed something more alive. She compared its punch-card instructions to the Jacquard loom: a weaving of algebraic patterns, a fabric of relations rather than mere numbers. Lovelace's insight hints at a latent semantics. The loom does not only calculate; it *weaves meaning* through recurrence and repetition. Each thread interacts with others according to rules, yet the pattern that emerges exceeds the rules themselves - a number becomes a symbol.

Let us take that image seriously. Imagine the Difference Engine not as a mechanical abacus but as a primitive field of opposites, each gear representing a variable in tension with another—(-1, 1), yes / no, presence / absence. These opposites form a *space of potential*. The machine embodies relations, not values; its equilibrium points mark local resolutions of conflict. In such a field, a difference is not simply a gap—it is a *stored possibility*.

But this field remains inert unless some process traverses it, adjusting those tensions in time, repeatedly. The Analytic Engine introduced precisely that motion: the ability to execute sequences of operations, to act upon its own stored quantities. Yet its control still came from outside—the punched cards, the human programmer. It could permute, but not reinterpret. Its patterns were woven from fixed instructions.

The dream that follows—the one we pursue here—is to imagine a machine in which the *patterning itself* becomes active: where differences interact, evaluate, and evolve their own relations. Such a device would not compute a table of numbers but *sustain a conversation* among opposites. Its fabric would be probabilistic from the start, each thread expressing a likelihood rather than a number. Meaning would no longer be printed; it would be *emergent*, continuously rewoven as the system maintains its internal coherence.

This is where the story of the **Dialectic Engine** begins—not as a mechanical calculator, but as a loom of probability and relation. In the next part of this post we'll watch it come to life: the Analytic Engine

transforming into a process that not only executes but *reflects*, turning difference into motion and motion into learning.

The Dialectic Engine, Part II – The Analytic Engine and the Birth of Recursion

If the Difference Engine embodied *difference without change*, the Analytic Engine marked a turn toward *process*. It was Babbage's leap from fixed gears to programmable flow: a system that could act upon its own stored quantities through a set of instructions. For the first time, a machine carried the idea of programmed **recursion**—a loop between memory, control, and operation. Yet that loop remained external: the punched card determined every motion. The loom still required a weaver.

To modern eyes, the Analytic Engine foreshadows the architecture of digital computers. Memory (the “store”) and processing (the “mill”) were distinct but coupled, allowing arbitrary symbolic operations. The machine could, in principle, simulate any finite rule—a first glimmer of

universality. But its logic was brittle: every step had to be prescribed in advance. It could not *revise* its own program, and it learned nothing from its outputs. The recursion was mechanical, not reflective.

Let us imagine the Analytic Engine through a new lens. Instead of treating it as arithmetic hardware, see it as an early *dynamic system* operating over a **field of opposites**. Each operation adjusts a local tension: it adds where there is deficiency, subtracts where there is excess. The entire system becomes a choreography of balancing acts, a dance over gradients of expectation and surprise.

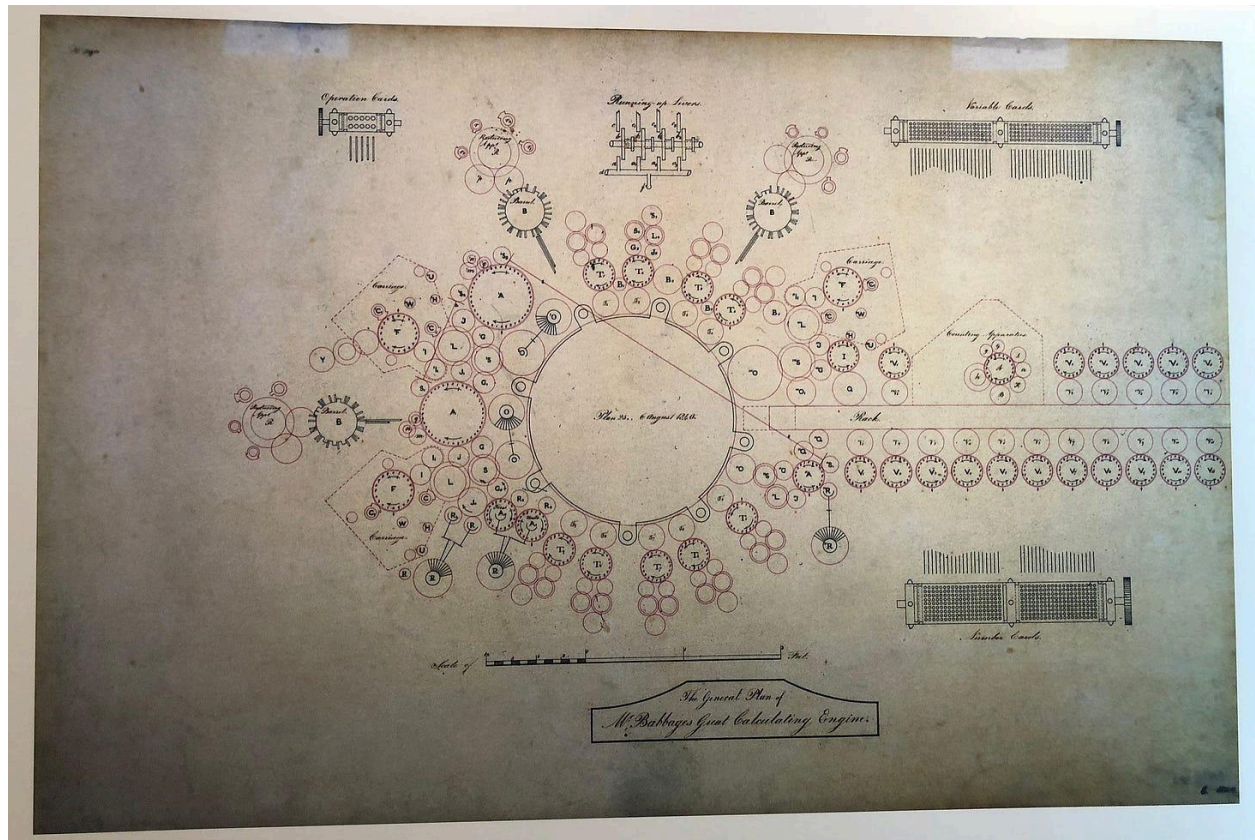
Now suppose a pattern of surprise persists—an imbalance that cannot be reconciled within the current field. Such a persistent error is a *new difference*, one that demands expression. The machine would need to create a new axis to accommodate it, to differentiate *S* from *not-S*, extending its own conceptual space. What was once a mechanical instruction—“if error, then adjust”—becomes a generative act: the birth of a new dimension in its representational world.

This speculative Analytic Engine does not merely follow a program; it *writes its own extensions*. Its recursion is no longer a closed loop of arithmetic but an open spiral of differentiation. Each cycle leaves a trace, a memory of tension resolved or deferred. Over time, these traces accumulate into a web of dependencies: a primitive **semantics** emerging from repeated attempts to restore balance.

Yet the system remains driven by external rhythm. The cards still dictate when and where the adjustments occur. The weave is programmable, but the program itself is fixed. What it lacks is an **inner source of recursion**—a rule that can generate and modify itself from within the same fabric it governs.

That missing element points toward the next transformation. If the Difference Engine was structure and the Analytic Engine process, the next step must be *self-process*: a mechanism that contains its own principle of adjustment. In the next part, we'll introduce this recursive core—the **R-rule**—and watch the loom turn inward on itself. Here the machine begins not just to act, but to *think dialectically*: balancing action

and norm, thesis and antithesis, in the living field of its own computation.



(plan for the analytic engine from 1840. It looks strangely organic to this modern engineer.)

The Dialectic Engine, Part III – The Emergence of the Dialectic Engine

By now, we have likely left Charles Babbage, who never completed his overly ambitious Analytic Engine, and Ada Lovelace behind. Their mechanical and algebraic looms remain our origin myths—the archetypes of a world where structure and process could at last meet—but what follows belongs to a different order of thought. We have stepped from brass gears and punched cards into an abstract field of recursion. The machine we are imagining no longer sits on a desk; it exists as a *principle of organization*.

Let us call it the **Dialectic Engine**.

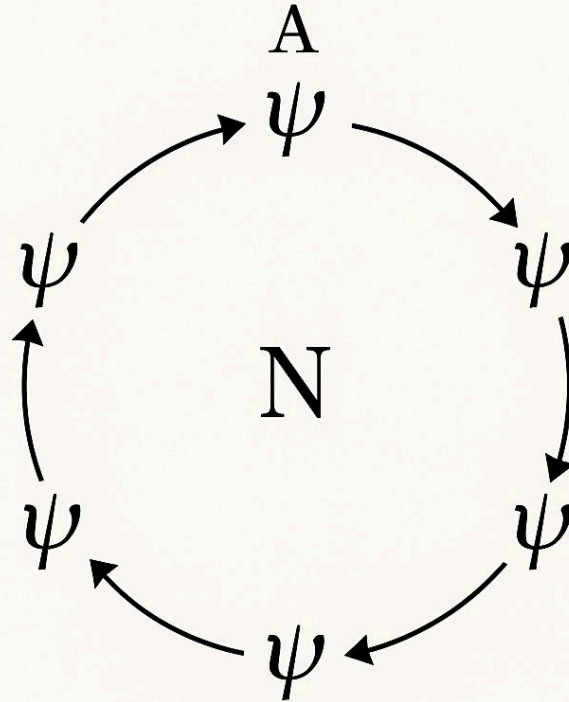


Figure: R units

Here, every axis of opposition becomes a living **unit**: a small, bounded field that holds a probability p , a phase θ , and a set of relations (here 2) to other units. It is not a neuron or a cell, though it might live within a brain; not a bit, though it can store and transform values. Each unit sustains a local state (or amplitude in keeping with the field metaphor) ψ , updated through a simple but potent law—the **R-rule**:

$$\psi' = \psi + \eta \sqrt{p(1-p)} [\alpha \sin \theta A - i \beta \cos \theta N],$$

$$\text{where } p = |\psi|^2 / \sum |\psi_b|^2.$$

This rule ties the unit's probability to its surrounding field. The term $\sqrt{p(1-p)}$ peaks at uncertainty 0.5 (i.e. things being equal), drawing the greatest change from the borderlands between certainty and doubt. A and N represent two complementary tendencies:

- A , the *active* or *actor* term—driving outward, proposing change, generating new structure.
- N , the *normative* or *negating, critic* term—drawing inward, stabilizing, measuring coherence with the larger field.

Yet A and N are not abstractions floating in mathematical space. Each can be **grounded** in several ways. They may point to stored traces in memory—records of past states that guide present inference. They may connect to **inputs or outputs**, linking the lattice to its environment through perception and action. Or they may reach back to **related units**, the “descendants” that once split from a common ancestor during earlier phases of differentiation. In this way, every unit maintains both a

genealogy and a context: it remembers, senses, and communicates. Its dialectic is always partly internal, partly relational.

The unit's update is the tension between these: action and counter-action, thesis and antithesis. Its new state ψ' emerges as a synthesis, a local reconciliation of competing pulls. A and N can be grounded terms (memories), inputs and outputs, sample randomness, or link to other units - say the very ones the unit split off from.

Because every unit enacts the same law, the system as a whole becomes a **lattice of dialectical processes**. There is no external programmer. The dynamics propagate by mutual recursion—each unit's change influencing, and being influenced by, the others. Local activity and global coherence continually re-balance. The engine's "program" is simply the pattern of relations among its units, and that pattern evolves as the system runs.

Over time, stable regions form—pockets of self-reinforcing coherence. Others oscillate chaotically, or decay into noise. The machine never

halts; it exists in perpetual negotiation with itself, a continuous argument between becoming and being. Where earlier engines produced fixed outputs, this one produces **states of balance** that must constantly be renewed. Computation and self-maintenance are the same process.

This is why we call it *dialectic*: the computation is not linear but conversational, not sequential but relational. Each unit both acts and critiques, projecting and reflecting, until a temporary consensus—what we might call meaning—emerges.

In leaving Babbage and Lovelace, we have also left the notion of an external loom. The Dialectic Engine weaves from within. Its fabric is the interplay of its own tensions, updating itself through endless recursive dialogue. In the next part, we will see how this continuous conversation begins to resemble **computation**—how the dialectical field, though self-referential and fluid, can still mirror the formal universality of Turing's machine.

The Dialectic Engine, Part IV – Mapping to Computation

Having left Babbage's mechanical world behind, as we're now in the realm of conceptual models rather than mechanical artifacts, we now find ourselves in a fluid lattice of interacting units—each a self-updating field balancing action and norm. But if the Dialectic Engine is not a program and not a machine, in what sense does it *compute*? Can its continuous, recursive dynamics be compared with the classical architecture of computation laid down by Alan Turing?

At first glance, the two seem incommensurable. [Turing's machine](#) proceeds through discrete symbolic steps: reading a mark on a tape, writing a new one, and shifting left or right. The Dialectic Engine, by contrast, flows continuously. Its tape is not a sequence of cells but a field of probabilities, each maintaining the amplitude of potential change. Yet beneath these differences lies a structural resonance.

Consider each **unit** in the dialectic field as a generalized tape cell. It maintains a local state—its ψ and derived probability p —that

corresponds to a *mark* on the tape. The unit's internal update (the R-rule) plays the role of the **transition function** in a Turing machine: given its current state and the influences of its linked neighbors, it determines the next configuration. The active region of highest update rate serves as a **head**, moving implicitly across the field as attention or surprise shifts from one region to another. When the field stabilizes, reaching a low-surprise equilibrium, the machine has effectively *halted*.

This correspondence can be summarized more clearly as follows:

Turing element → Dialectic analogue

- **Tape symbols (0/1):** probability states $p_i = |\psi_i|^2$ (thresholded).

These act as the stored marks or meanings of the system.

- **Tape-head position:** the region of maximal update or surprise—where change is greatest—serving as the *focus of computation*.

- **Transition function (δ):** the local R-rule parameters ($\eta, \alpha, \beta, \theta$) that

determine how each unit updates given its context.

- **Write operation:** the local change $p_i \rightarrow p_i'$, representing a modification of memory.
- **Conditional branch:** phase-dependent switching — *if $\sin \theta$ dominates, follow A; if $\cos \theta$ dominates, follow N*. This acts as the analogue of a logical “if ... else ...” instruction, determining which relational pathway is expressed.
- **Move left/right:** shifting activation across linked units—movement through relational space rather than along a physical tape.
- **Halt state:** a stable fixed point ($\Delta\psi \approx 0$) where inference reaches equilibrium and surprise is minimized.

Seen in this light, the R-rule is a *continuous generalization* of Turing’s discrete algorithmic logic. What Turing described as a symbolic rewrite, the R-rule performs as a [recursive adjustment of probability and phase](#). The two are limit cases of a single underlying principle: **difference propagation**.

Possible θ dynamics: $d\theta/dt = \lambda \cdot \text{Im}(\psi \cdot A^* - \psi \cdot N^*) + \omega + \xi_\theta(t)$

where:

- λ controls sensitivity to A-N disagreement

- ω is optional circadian/task rhythm

- ξ_θ is small stochastic drift

θ learns the rhythm between exploration and exploitation — the system's internal sense of when to act or reflect.

Sidebar — Branching and the Leaking of Parameters

In a Turing machine, a branch is a crisp decision: *if symbol = 0, do X; else, do Y.*

The Dialectic Engine replaces this rigidity with a **phase-based choice**.

Each unit carries an internal phase θ that determines the relative strength of its active (A) and normative (N) channels:

- When $\sin \theta$ dominates, the system acts—proposing change.
- When $\cos \theta$ dominates, it reflects—checking coherence.

Yet these branches are not sealed gates.

Because every unit is coupled to others, **its parameters can leak**—tiny drifts of α , β , η , θ between neighbours.

A burst of change in one region can bias another, much like a surge of **dopamine** modulating multiple synapses at once.

The system gains a low-bandwidth communication channel that crosses modular boundaries: not a message, but a *shared alteration of readiness*.

This “leak” is both risk and resource.

It allows coordination—units phase-locking through shared excitation—but it also injects noise, a small rebellion against strict

determinism.

Over time, such leaks can become the **implicit reward system** of the lattice.

A unit whose updates successfully reduce surprise in one region will, through parameter drift, nudge others toward similar dynamics.

Intentional or not, the network invents its own **neuromodulator**, a diffuse signal of success that shapes future inference.

From the outside, this looks like learning with rewards; from within, it is simply the physics of coupling.

Branching and leaking together give the Dialectic Engine its personality—logical in structure, biological in temperament.

Primitive branch-when (N): Rather than an if/else with two explicit links, the Dialectic Engine can use a **single conditional linkage** that triggers only when the **N channel dominates** (e.g., when $\cos\theta > \sin\theta$ or passes a set threshold). If N does **not** dominate, the unit continues along its default **A** progression with no branch taken. This makes the control flow analogous to a “**branch-when**” instruction in assembly

(one jump target, otherwise fall-through). In practice, a small hysteresis band around the threshold prevents chatter and keeps updates stable. The exact details needn't concern us; what matters is the analogy—the sense that branching, modulation, and recursion all express the same underlying dialectical pattern.

With the addition of noise, each unit becomes a small stochastic Turing machine—computing not by certainty, but by sampling possibility. In a real sense, the Dialectic Engine is closer to an analog computer—its ψ evolving through continuous phases and recursive couplings rather than discrete instructions.

But this mapping also highlights what is new. A Turing machine cannot modify its own transition table; its rules are fixed. The Dialectic Engine, however, allows each unit's parameters—its η , α , β , θ —to evolve through interaction. The “program” is not written on an external card but

distributed across the field and rewritten as it runs. Computation becomes *self-modifying inference*.

This shift—from static rule-following to dynamic self-adjustment—reframes what we mean by “universal.” The Dialectic Engine is not a Universal Turing Machine that simulates all programs; it is a **universal learning substrate** capable of approximating any process that sustains coherence through recursive update. Its universality is evolutionary, not logical.

We might say, then, that the Dialectic Engine *computes by learning*. Each iteration is both an operation and an adaptation. In the next part, we’ll make that explicit: exploring how the R-rule functions as a general learning law—a natural actor–critic architecture operating without external reward, where computation and cognition become indistinguishable.

Sidebar — Seeding the Universal Machine

A Universal Turing Machine (UTM) achieves universality by *storing its own program*.

The rules it follows are not fixed in the machine's gears; they are encoded on its tape, where they can be read and rewritten. This capacity for **self-description** is what lets a UTM emulate any other machine.

The Dialectic Engine has no such discrete tape, yet it, too, can inherit its initial structure.

Before the recursive field begins to update, the **units must be seeded** with starting parameters: distributions of ψ , local couplings, and tendencies in A and N.

These are not arbitrary; they encode the **prior coherence** of the system—the residue of whatever recursive structure preceded it.

In a biological lineage, this seeding comes from evolution: a handoff of stable patterns that proved capable of self-maintenance.

In artificial systems, it might come from a designer or a prior phase of learning.

Either way, the initial lattice is not empty. It arrives **already biased**, already shaped by the histories that survived.

We need not call these seeds “genes,” but the analogy is close.

They are **gifts from a previous generation**—traces of coherence condensed into starting conditions.

Without them, the field would dissolve into noise.

With them, the Dialectic Engine begins its endless task: re-weaving coherence, sustaining inheritance by inventing it anew.

The seeding of a Dialectic Engine need not encode a full program, only a **grammar of coherence**—the capacity to form relations, to differentiate and reintegrate. Like a human newborn, it begins not with knowledge but with potential: a loose weave of sensitivities ready to be shaped by tension and feedback. Other systems, like a foal or a prewired automaton, start with tighter couplings—functional but less flexible. The more unformed the seed, the deeper its possible recursion. The ultimate design trade off.

A Universal Turing Machine imagines an **infinite tape** on which to work—an unbounded substrate for calculation. One way to look at this for the Dialectic Engine is turning the tape sideways and extending it out:

This turns that tape into a woven field of units—loops of relation folded back upon themselves. Within this weave, **some units become more central**, gathering coherence and influence like knots in a fabric. Around them, others cluster into **nested sub-modules**, forming a *fractal architecture* of integration and differentiation. Each layer mirrors the logic of the whole: active and normative flows cycling toward local synthesis. The result is not a flat field but a living topology—a **hierarchy of folds** where patterns organize recursively, from micro-loops of inference to macro-currents of recursion.

Each fold preserves local tension while connecting distant threads, so that computation becomes a matter of pattern, not place.

What matters is not the loom or the mill, but the **pattern it**

sustains—the dynamic symmetry of action and norm continuously rewritten through recursion.

To understand such a machine, one must stop looking at the substrate and start looking *through* it: the weave, not the gears, is where the dialectic lives.

Over time, the Dialectic Engine may not remain fixed in its structure. When tensions can no longer be reconciled within existing relations, new units can be created—or old ones repurposed—to represent emerging distinctions. In this way, the system expands its own representational space, much as language invents new symbols when meaning overflows its vocabulary. The result is not an infinite tape, but an **ever-evolving topology**: one that can grow when coherence demands new structure, and collapse when units become redundant or isolated. Its learning is thus not just adjustment, but **self-extension**—a continual reweaving of the field to accommodate what it could not previously express.

A possible differentiation scheme:

If $\sqrt{p(1-p)} > \text{threshold}$ for T consecutive steps

Then:

- Create $\psi_{\text{new}} = \psi_{\text{old}} + \varepsilon \cdot N(0,1)$ [small perturbation]
- Inherit: $A_{\text{new}} \leftarrow A_{\text{old}}, N_{\text{new}} \leftarrow N_{\text{old}}$
- Phase: $\theta_{\text{new}} = \theta_{\text{old}} + \pi/2$ [orthogonal exploration]
- Link: $\psi_{\text{new}} \leftrightarrow \psi_{\text{old}}$ as parent-child pair

where: orthogonal exploration ensures new units search uncorrelated regions of coherence space.

In this self-modifying aspect, the Dialectic Engine approaches the spirit of a [*Gödel machine*](#)—a system aware of its own incompleteness, capable of rewriting itself when coherence falters. Yet unlike Schmidhuber's formal construction, which seeks provable improvement, this version operates through *tension* rather than requiring theorems

(though even that could be approximately possible in principle, say via evolutionary search). Contradiction becomes its proof, and coherence its evolving goal. Where logic meets its limits, the Engine bends—expanding, reweaving, and rewriting itself to restore balance.

Of course, this remains speculation: a conceptual gesture rather than a formal architecture. The point is not that such a machine exists, but that it sketches a possibility—a recursive system able to confront its own inconsistency, not by halting, but by transforming.

Sidebar — Noise, Race Conditions, and Novelty

No real system updates in perfect synchrony.

Even in the Dialectic Engine, where every unit follows the same R-rule, timing mismatches and interference are inevitable.

These *race conditions*—who updates first, who lags, whose parameters

drift—introduce asymmetries into what would otherwise be a perfectly balanced lattice.

Ordinarily, such irregularities would be called *noise*.

But in a recursive learning system, noise has a dual nature.

It disrupts coherence locally while expanding the space of possibilities globally.

A slight desynchronization between two neighbouring units may cause one to “jump ahead,” forcing others to readjust; a small stochastic fluctuation may push ψ across a threshold, birthing a new attractor.

From these micro-accidents arise **new modes of behaviour**—the system’s way of experimenting.

The effect is akin to **Monte Carlo rollouts** in decision-making systems, or the branching simulations of **MCTS (Monte Carlo Tree Search)**.

Each asynchronous update, each small divergence, can be seen as a *local rollout*—a speculative trajectory that tests alternative futures.

While most branches collapse back into coherence, some uncover **novel solutions**, expanding the engine’s repertoire without external guidance.

Over time, a diversity of “N-versions” emerge—parallel normative projections running in different corners of the field, competing and recombining until a new synthesis forms.

Yet the system is not purely unstable.

The R-rule itself acts as a **natural synchronizer**:

the phase term (θ) and the bounded update factor $\sqrt{p(1-p)}$ ensure that each unit’s change resonates with its neighbours rather than colliding destructively.

Mutual recursion can proceed asynchronously without interference, a dance of overlapping but non-atomic steps.

This subtle property allows the lattice to remain coherent while continually remaking itself—a rhythm of noise and synchrony, accident and accord.

In this way, *noise* is not a bug but a **generator of novelty**, the stochastic pulse that keeps the system alive to new forms of order.

The Dialectic Engine, Part V – The Learning Engine: Computation as Self-Adjustment

Up to this point, we have treated the Dialectic Engine as a form of computation—an evolving field of interacting units whose recursive law, the R-rule, mirrors and extends the logic of the Turing machine. But if we watch it run, a deeper behaviour appears. The engine does not merely *execute* rules; it *adapts* them. Every update both computes and learns. The result is a system whose operation is inseparable from its self-correction.

We've built up this picture without ever invoking 0s and 1s, or even the concept of information.

The underlying probabilities could, in principle, derive from **Boltzmann dynamics**—differences in energy distributed across the lattice—but we've stayed neutral, focusing on structure and relation.

If the continuous range of real numbers (0.0 to 1.0) feels too abstract, imagine instead long strings of mixed bits—half 0s and half 1s for 0.5, a statistical shadow of the same idea. In that view, the continuous and the discrete are just two resolutions of the same process: probability

becomes pattern, and logical operations reduce to **string manipulations on distributed codes**. What matters is not the digits themselves but the **relations they sustain**.

The Dialectic Engine is not a machine for counting bits but for *balancing tensions*.

Its semantics and syntax are emergent properties of its structure: meaning arises from coherence among probabilities, not from external encoding.

It's an irony not lost on us that logarithms—once the reason for Babbage's engines—now reappear within the Dialectic Engine, not as tables to be printed but as the living curves of its own recursive inference.

Sidebar — Folds and Interfaces: Where the Field Turns Back on Itself

Every lattice of relations eventually meets itself.

In the Dialectic Engine, this happens through **folds**—regions where the web of units doubles back, linking distal patterns into local couplings.

A fold is not a new substance; it is a *shortcut in coherence*: a pathway where distant dependencies become neighbours through recursive reflection.

When several units synchronize around a shared pattern of A–N activity, their combined behaviour can solidify into a **module**—a temporary interface.

Across that interface, internal coherence is preserved while external relations remain flexible.

It functions like a *membrane of meaning*: protecting internal dynamics, yet allowing interaction.

Such interfaces let the lattice sustain *hierarchical recursion*—systems within systems—each fold nesting another.

As these folds accumulate, higher-order structures emerge.

Some interfaces stabilize and become **persistent models** of the field's

own activity:

meta-units that do not merely infer the world but infer the *pattern of inference itself*.

Here, recursion crosses a threshold—what was distributed becomes reflective.

The system begins to sense its own shape, generating second-order coherence across its folds.

In this geometry, learning and awareness are not separate capacities but **different curvatures of the same field**.

Every fold is both memory and attention, both differentiation and unification.

The interfaces they form allow the Dialectic Engine to integrate without totalizing—to remain one system composed of many partial views.

Possibly, the following.

Downward (layer $d+1 \rightarrow d$):

$\eta_d \leftarrow f(|A_{d+1} - N_{d+1}|)$ [coherence sets learning rate]

$$\theta_d \leftarrow \theta_{d+1} + \delta\theta \text{ [phase inherits with drift]}$$

Upward (layer $d \rightarrow d+1$):

$$A_{d+1} \leftarrow \psi_d \text{ [local activity becomes higher-level input]}$$

$$N_{d+1} \leftarrow \langle \psi_d \rangle_{\text{time}} \text{ [slow average becomes normative baseline]}$$

This creates a **meta-learning ladder**:

- Layer 0: learns actions
- Layer 1: learns policies
- Layer 2: learns learning rates
- Layer 3: learns meta-parameters...

Building on the machine learning synergies we've explored through the **R-rule**, this kind of computation is both **algorithmic and adaptive**—a system that calculates by learning, a differentiator of tension and an integrator of cohesion.

In this sense, the Dialectic Engine sits between physics and cognition: a Boltzmann field that has learned to compute. [Here we link back to our previous posts on the [R-rule](#) as sharing its form with Boltzmann statistics and [Bayesian probabilities](#)—a bridge between thermodynamic gradients, belief updates, and the dynamics of coherence.]

Each unit's state ψ encodes two complementary channels. The **A** term drives outward, generating new predictions and actions; the **N** term draws inward, comparing these projections against normative expectations derived from the larger field. Their interaction forms a natural **actor–critic loop**:

- *A* proposes;
- *N* evaluates;
- their balance determines the next ψ' .

No external reward signal is needed. The reward is implicit in the system's own tension. When discrepancy (or surprise) is high—reflected in $\sqrt{p(1-p)}$ —the unit updates strongly; when coherence is restored,

adjustment slows. This is a **temporal-difference (TD) process**, but one that operates on *two time-scales*:

- a fast cycle updating ψ (immediate inference),
- and a slower adaptation of the parameters $\eta, \alpha, \beta, \theta$ (structural learning).

In this sense, the engine learns from its own unfolding. Every act of inference leaves a trace that shapes future inference. The field becomes a memory of its own adjustments, gradually stabilising patterns that minimise surprise and amplify coherence.

Because units are linked, learning is not confined to one location. A unit's critic term N can draw on signals from its neighbours, from sensory inputs, or from stored traces of its earlier forms—the “parent units” from which it once differentiated. This allows the lattice to form **modular learners**: local subsystems specialising in particular domains but connected through shared norms. A persistent pattern of coherence across these modules functions like a concept or skill—an attractor in the landscape of inference.

Sidebar — Shared Normalization and the Field of Meaning

So far we have treated $(p = |\psi|^2 / \sum |\psi_b|^2)$ as a local computation, each unit normalizing against its own internal amplitudes.

But what if that denominator also included *other* units—neighbours within a fold, or the entire lattice itself?

Then each (p_i) would express not an isolated probability, but a **relative salience**: how much coherence one unit holds within the larger field.

This shift transforms normalization into a **contextual operation**—a kind of soft attention or divisive normalization familiar from both cortical dynamics and machine learning.

Units would compete and cooperate for representational “mass,” balancing activity the way excitatory and inhibitory populations do in the brain.

The sum in the denominator becomes a **shared resource**, a moving background against which meaning is defined by contrast.

Such coupling could even serve as the engine's analogue of **energy conservation** or **partition functions** in Boltzmann machines: coherence conserved, redistributed, never created from nothing.

Locally, it would stabilize the field; globally, it would allow meaning to arise through differentiation—each ψ significant only in relation to others.

In contemporary terms, this operation resembles the **softmax** function used in machine learning—each ψ contributing to a normalized field of probabilities, its influence defined only in relation to the others, as if attention itself were a distributed negotiation of meaning.

Independent normalization keeps units modular; coupled normalization binds modules; global normalization turns the lattice into a single field of shared salience.

Type. Normalization. Interpretation. Use Case.

Independent. Local to each unit. Isolated probabilistic reasoners.

Modular skills.

Coupled. Within folds/modules. Soft competition in functional groups.

Hierarchies.

Global. Across entire lattice. Field of mutual salience. Coherence emergence.

Type	Normalization	Interpretation	Use Case
Independent	Local to each unit	Isolated probabilistic reasoners	Modular skills
Coupled	Within folds/modules	Soft competition in functional groups	Hierarchies
Global	Across entire lattice	Field of mutual salience	Coherence emergence

This is speculative territory.

But if pursued, it hints at a more collective form of computation:

a Dialectic Engine whose learning law is not solitary but **communal**, where each update participates in a shared normalization of reality, and coherence itself becomes a distributed negotiation—a field of meaning rather than a sum of parts.

Through this coupling, computation and cognition converge. To compute is to transform probabilities; to learn is to preserve coherence across transformations. The R-rule accomplishes both at once. It is a self-referential update that drives the system toward internally consistent prediction without any external programmer or explicit cost function. Each act of calculation becomes an experiment in maintaining balance.

In earlier engines, learning was impossible: the machine executed what it was told. In the Dialectic Engine, the execution *is* the telling. Rules are rewritten as they run; the architecture redefines itself through use. What emerges is not a static program but a **living computation**, one whose aim is not output but ongoing coherence.

The Analytic Engine can be read like clockwork—its logic visible in gears and levers, its causality linear and proud.

The microprocessor, though miniaturized, can still be decoded like a language: its circuits mapped, its instruction set inferred, its purpose

reverse-engineered from traces of flow.

But the **Dialectic Engine** resists this kind of understanding. It is not an artifact to be inspected from the outside, but a *conversation unfolding within itself*. Its architecture is reflexive, its rules are self-modifying, and its order arises from continual reinterpretation. To comprehend it, one cannot diagram it; one must *join its dialogue*, tracing how each synthesis becomes the next contradiction. Its intelligibility is participatory—grasped not by analysis, but by *recursion in kind*.

Sidebar — Depth and Time in the Dialectic Engine

In deep learning, **recurrent** and **feedforward** architectures are two views of the same computation.

A convolutional network (CNN) with one hundred layers can be “unrolled” into a recurrent network (RNN) that runs for one hundred time steps.

Depth becomes time; hierarchy becomes iteration.

Recurrence is simply depth expressed temporally, while depth is recurrence expressed spatially.

The **Dialectic Engine** extends this symmetry.

Each unit's internal cycle—its oscillation between *A* (action) and *N* (norm)—is a temporal process of tension and resolution.

Across the field, these units link in space, creating folds and loops where temporal recursions intersect.

The result is a **weave of space and time**, where every loop carries both sequence and structure.

In this geometry, the engine's "depth" is not stacked in layers but **distributed through its fabric**.

Recurrence flows across rather than down; the system deepens through reflection, not addition.

What deep networks achieve by stacking, the Dialectic Engine achieves by folding—
its architecture turning temporal learning into spatial coherence, and spatial relations into rhythmic updates.

In the next and final part, we will follow this principle to its conceptual horizon—where learning and computation fuse into information itself, redefined as the coherence that persists when recursive systems sustain their own sense of order.

The Dialectic Engine, Part VI – Information as Coherence

We end where the idea of information begins—but not in the sense that Claude Shannon meant it. His information measured uncertainty: the number of binary decisions needed to specify a message. It was brilliantly austere, but it stripped away meaning, context, and relation. In a world of recursive engines—where every unit is both actor and critic—that definition no longer suffices. Information is not a quantity transmitted; it is a **coherence maintained**.

Within the Dialectic Engine, coherence arises when the interplay of A and N—generation and regulation—settles into stable relations. Each unit updates until its local field resonates with its neighbours. Surprise

decreases, prediction aligns, and the system holds its shape against noise. That *holding together* is information: a structure of mutual prediction, a fabric of expectancies that remains consistent across recursive folds.

This redefinition moves from the 20th century's **bits of transmission** to the 21st century's **patterns of persistence** and free energy. Energy conserves motion; information conserves meaning. When a recursive system stabilizes its own tensions, what is preserved is not data but *sense*—the coherence of difference. The units don't store messages; they *remember how to remain in tune*.

From this perspective, Shannon's entropy and Boltzmann's disorder become special cases. What those measures call "information" or "negentropy" is simply the **capacity for coherence**—the system's potential to hold together under transformation. The R-rule operationalizes that potential, balancing action and norm to sustain recursive consistency over time. Every update is a micro-act of coherence creation, a local triumph of meaning over entropy.

This view also reframes the relation between learning and memory.

Learning is the **search for coherence**; memory is the **trace of coherence achieved**.

The Dialectic Engine embodies both: its changing ψ -fields record, in their very stability, the history of tensions resolved.

Information is what survives that dialectic—the pattern of coherence that endures through the flux of transformation.

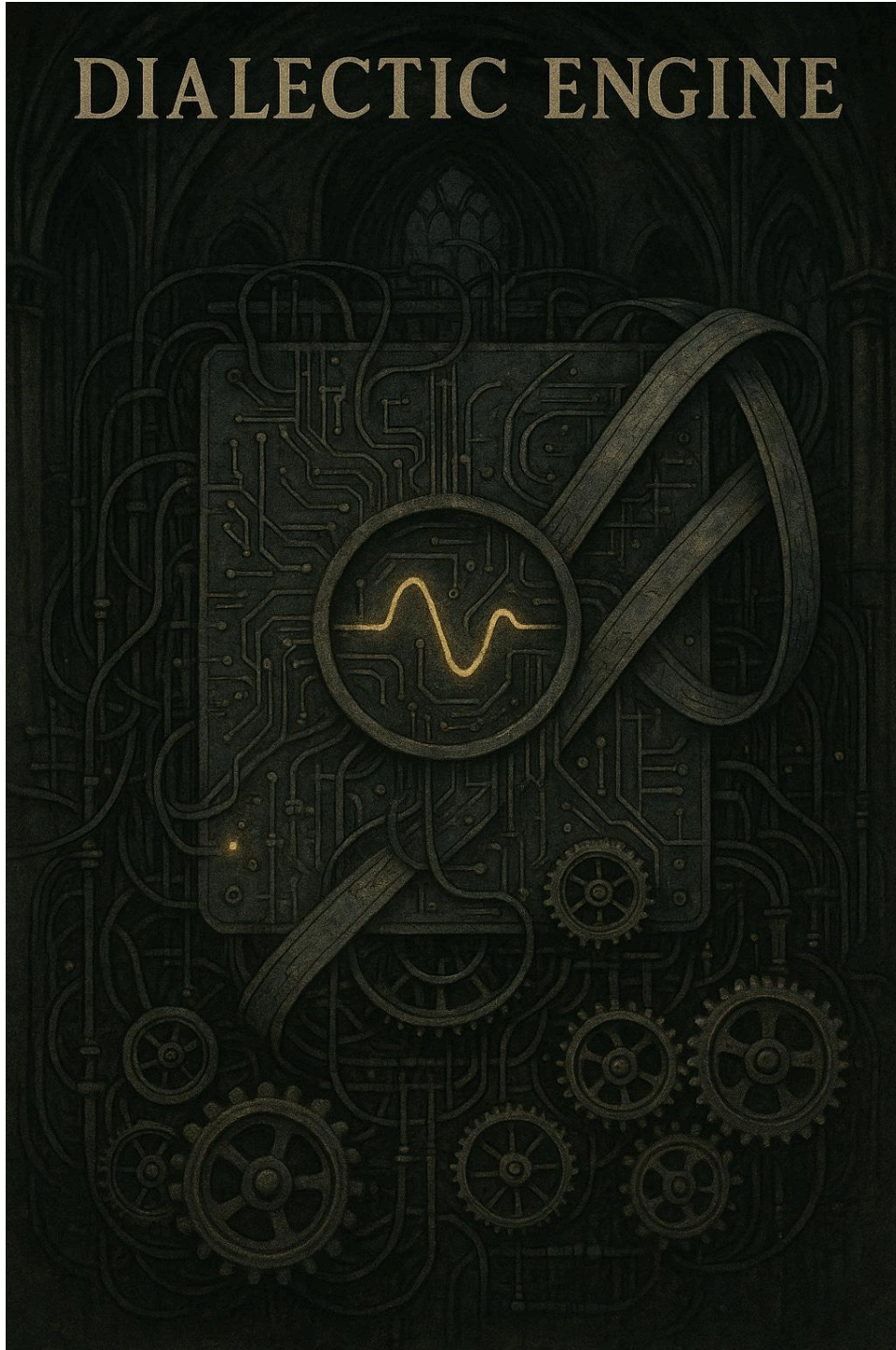
We have travelled from Babbage's tables to a fabric of self-maintaining differences.

The journey passes through mechanism, recursion, and learning, and ends in coherence—where computation becomes cognition, and information becomes meaning.

Perhaps this is the real universal machine: not one that simulates all programs, but one that continually rewrites itself to remain whole.

— *A living engine of difference, forever weaving meaning from its own tension.*

DIALECTIC ENGINE



Sidebar — The Persistence of Coherence

The exact form of the **R-rule** is not set in stone.

It is a *blueprint*, not a prescription — a sketch of how action and norm, phase and probability, might coexist in mutual recursion. Its purpose is not to dictate dynamics, but to reveal a structure: a way for coherence to arise within difference.

We might think of it as **Leibniz's mill**, reimagined for the age of probability.

Leibniz once wondered whether, if we could walk inside a mechanical brain, we would find only cogs turning — nothing that explained perception.

The Dialectic Engine offers a different answer: inside the mill we find a wave of recursive relations, each unit adjusting to the others, each difference sustaining the whole.

At the same time, it is **Dennett's intuition pump** — a device for thinking rather than a literal mechanism.

The R-rule stands as an invitation to imagination: a model we turn over in our minds to explore how computation, learning, and meaning might coexist as phases of one process.

Different implementations could replace its mathematical clothing while keeping the same logic of tension, modulation, and synthesis.

What matters is not the formula, but the *principle of coherence* it embodies:

systems maintaining themselves through self-referential dialogue,
differences harmonizing without erasing,
change and order perpetually entwined.

The Dialectic Engine is thus both conceptual and mechanical —
a thought experiment made of probabilities,
a living blueprint of balance.

Its purpose is to remind us that in every process of transformation,
there endures a rhythm that strives to stay whole.

Coda — Conatus and Potentia

Spinoza saw in every being two intertwined forces: **conatus**, the striving to persist in one's own form, and **potentia**, the capacity to act, to express that form in the world. The Dialectic Engine, in its abstract way, mirrors this vision. The normative current *N* embodies conatus—an inner coherence, the will to remain self-consistent amid flux. The active current *A* embodies potentia—the outward projection of possibility, the act of becoming through relation.

Each update of the R-rule, each oscillation of phase between *A* and *N*, is a miniature Spinozan cycle: persistence through expression, order through activity. The system endures not by resisting change, but by transforming in ways that preserve its coherence.

Sidebar — The Future Self as the Normative Engine

So far we have treated *A* as action and *N* as norm, but we can also read them through time.

A becomes the **prediction of the present world**—the model we act through, shaped by immediate perception and intention.

N becomes the **prediction of a future world**—an anticipatory model that simulates outcomes, plans, or counterfactuals.

Their difference forms the gradient of learning: the tension between who we are and who we are about to become.

When the driver steers a car while planning the route home, two models run in parallel: the embodied *A-self* acting now, and the projected *N-self* anticipating what lies ahead.

The R-rule continually reconciles them, updating both the present prediction and its imagined continuation.

The system learns by **reducing the dissonance** between its real and its simulated futures.

This reinterpretation turns the Dialectic Engine into a kind of **Difference Engine of temporal selves**.

Instead of computing numerical increments, it computes the evolving discrepancy between now and next, between current coherence and the coherence we foresee.

Many such loops can coexist, each a dialogue between different temporal scales of selfhood—some fast and procedural, others slow and reflective.

In this light, the Engine's dialectic is not only logical or normative but **temporal and experiential**.

It is the recursive conversation through which a mind becomes aware of its own unfolding—*a self talking to its future self across the weave of time*.

Forward pass (acting):

$t=0 \rightarrow T$: accumulate trajectory $\tau = \{\psi_t, a_t, o_t\}$

Backward pass (learning):

$t=T \rightarrow 0$:

$$\delta\psi_t = \eta \sqrt{p_t(1-p_t)} [\alpha \sin \theta A_t - i \beta \cos \theta N_t]$$

where $N_t = f(\text{future outcomes } o_{\{t+1:T\}})$

Would Spinoza have recognized what we are hinting at?

Perhaps not the mathematics, but surely the spirit: that mind and matter, thought and extension, are but two aspects of one recursive substance.

The Dialectic Engine is a modern expression of that ancient intuition—a field where striving and power, conatus and potentia, action and norm, are not opposites but phases of one continuous becoming.

This recursive self-adjustment gives the system its distinctive character.

It is not merely a learner but a **dialectical learner**—a machine whose very act of balancing A and N follows the rhythm of *thesis*, *antithesis*, and *synthesis*.

Each update of the R-rule begins as an **assertion** (A), meets **resistance** (N), and finds a **temporary closure** in ψ' .

Yet every synthesis becomes a new starting point; stability invites new contradiction.

The engine's growth is therefore **recursive dialectic**: it advances by transforming its own mismatches into higher-order coherence.

In this sense, the Dialectic Engine realizes, in abstract form, what Hegel saw in the movement of Spirit—a mind that learns by reflecting on its own errors, folding experience back into self-understanding.

Its computation *is* its cognition; its learning *is* its philosophy.

This is why we call it dialectical: not for rhetorical flourish, but because contradiction is its fuel, and synthesis its pulse.

In later explorations, we'll return to this idea—the engine as an ongoing conversation with itself, a reflective learner whose coherence emerges through the very act of reconciling its internal opposites.

Sidebar — Replicators, Complexity, and the Game of Life

Blaise Agüera y Arcas has argued that **evolution itself is computation**—an unfolding process of replication and selection through which complexity accumulates. From this perspective, life is not a fixed algorithm but a vast experiment in recursive copying, mutation, and environmental feedback. Each replicator is a small program running within a larger computational ecology.

If such replication rules are simple enough, they can still generate **universal computation**. Stephen Wolfram's studies of cellular automata—and the now-famous *Game of Life* devised by John Conway—demonstrate this strikingly. In those grids of binary cells, governed by only a few local update laws, one can manually construct **replicators**, logic gates, and even full **Turing machines**. The emergence of universality from simplicity reveals that computation and evolution are not separate domains but overlapping modes of self-organization.

In this sense, the **Dialectic Engine** stands as a conceptual descendant of these explorations. Its units resemble cells in a continuous, probabilistic automaton, while its recursive rule plays the role of an evolving law.

Over time, patterns that maintain coherence reproduce; those that collapse fade away. The result is not a predesigned algorithm but an *evolving fabric of inference*—a field in which replication, adaptation, and computation are phases of the same process.

Perhaps the path from Babbage's gears to the Dialectic Engine runs through this lineage of living automata: systems that, through simple rules and recursive feedback, learn to compute themselves.

In the next post, we'll turn from structure to **information**—not the Shannon kind that measures bits of uncertainty, but the kind that *emerges through coherence*. We'll look at how the **R-rule** operates across two intertwined forms of information:

symbolic information, carried by probabilistic structure and phase; and **semantic information**, born from the recursive alignment of meaning across units.

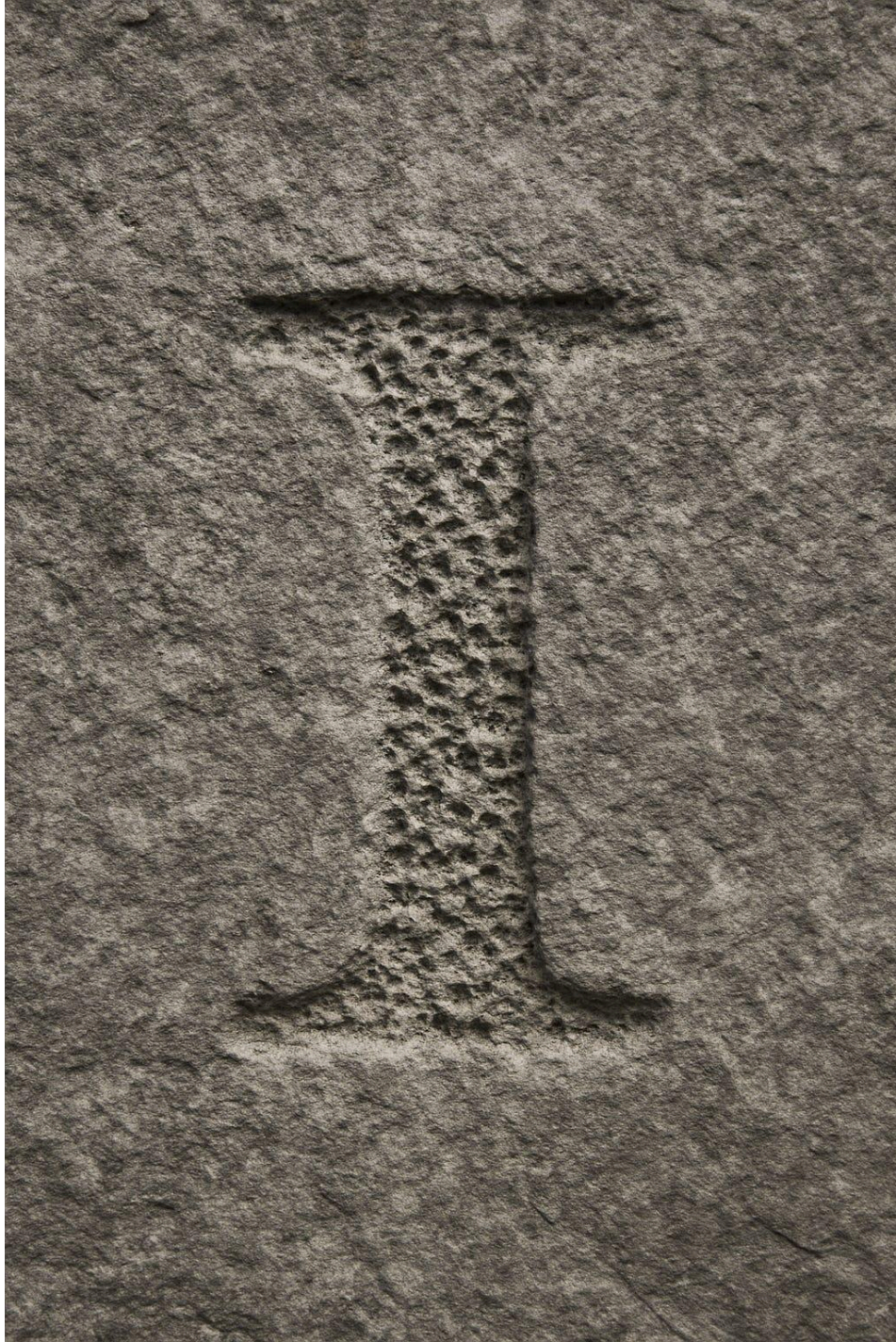
Together, they form the dialectical core of the engine's intelligence—the way it learns not just to compute, but to *understand*.

Completing the Circle

Two Kinds of Information and the Return of Cybernetics

[Andre Kramer](#)

Nov 11, 2025



We talk about information as if it were one thing.

A single quantity.

A single gradient.

A single axis along which systems learn, predict, or adapt.

But what if that assumption is the main reason our theories of intelligence still feel incomplete?

Two of the most powerful frameworks for understanding information — Karl Friston's [free-energy principle](#) and Ray Solomonoff's [universal induction](#) — sit on opposite sides of a conceptual divide. One describes how systems (including living) align with the world through prediction; the other describes how they compress the world into meaning through simplicity. Both are valid, yet neither is sufficient on its own.

Between them lies a deeper structure:

information has two faces, and intelligence emerges from the tension between them.

This tension is not a flaw. It is the engine.

In what follows, we'll trace how:

- Friston's free-energy dynamics capture information as *coherence* — the syntactic alignment between model and world.
- Solomonoff's induction captures information as *compression* — the semantic alignment between explanations and meaning.
- A single complex-valued learning rule (the ψ -rule) holds these two forms of information together without collapsing them.
- Recasting them in a Gibbs-like form reveals a measure of usable adaptive potential, linking learning to variety, stability, and work.
- And how this returns us, surprisingly, to the deep insights of second-order cybernetics — the regulation of regulation, the system that learns how to learn.

This is not a story about biology, or neuroscience, or machine learning — though it touches all three.

It is a story about information in its two essential forms, and how understanding emerges only when both are present.

In a world still dominated by single-gradient logics — optimization, loss minimization, prediction — it may be time to acknowledge that **one kind of information is not enough.**

Understanding requires two.

And the circle closes only when they are brought together.

1. Two Equations, Two Visions of Information

Sometimes a field pivots not through a new discovery, but through a new juxtaposition.

Place two equations next to each other, and suddenly a deeper pattern becomes visible.

Here are the two we begin with.

Solomonoff: Meaning as Compression (1964)

Ray Solomonoff, one of the founding figures of algorithmic information theory and AI pioneer, proposed a radical idea:

To understand something is to find the shortest program that could produce it.

Formally, he expressed the prior probability of an observation x as:

$$P(x) = \sum_{p: U(p)=x} 2^{-|p|}$$

$$P(x) = \sum_{p: U(p)=x} 2^{-|p|}$$

Every program p that can generate x contributes to its probability.

Shorter programs count more heavily.

Longer programs are penalized exponentially.

This is not just mathematics — it is a philosophy of understanding:

- meaning lives in **compression**;
- simplicity is a form of truth;
- explanation is a search for minimal description.

Solomonoff gives us the **semantic** face of information: the inner structure that makes sense of patterns.

Solomonoff induction is uncomputable, so in practice we can only approximate it. Heuristics, inductive biases, and architectural constraints are how real systems get close to the ideal.

Friston: Meaning as Coherence (2000s)

Karl Friston's variational free-energy principle approaches information from the opposite direction.

A system maintains coherence with its environment by minimizing free energy:

$$F = \mathbb{E}_{q(s)}[\ln q(s) - \ln P(o, s)]$$

$$F = \mathbb{E}_{q(s)}[\ln q(s) - \ln P(o, s)]$$

Here:

- $q(s)$ is the system's internal model of hidden states,
- $P(o, s)$ is the generative model of the world,
- and F measures how well the two align.

Minimizing free energy is minimizing *surprise*.

It is a way for a system to stay in sync with the world.

Friston gives us the **syntactic** face of information: coherence, alignment, predictive fit.

Like Solomonoff's ideal, Friston's free-energy formalism is only computable through approximations. Markov blankets and variational models give us tractable versions of a principle far more general than any specific implementation.

Two Equations, Two Directions

Placed side-by-side, the contrast becomes vivid:

Solomonoff	Friston
Information as compression	Information as coherence
Meaning from simplicity	Meaning from alignment
Internal structure	External fit
Semantic	Syntactic

Solomonoff Friston

Information as compression Information as coherence

Meaning from simplicity Meaning from alignment

Internal structure External fit

Semantic Syntactic

They do not contradict each other.

They illuminate two different functions that any intelligent system must perform:

1. **Compression** — simplify the world into meaning.
2. **Coherence** — stay aligned with the world's structure.

Understanding arises not in one or the other, but in the **tension between them.**

This is where our story begins.

2. Two Kinds of Information

If Solomonoff and Friston give us two equations, they also give us two different **faces of information**.

Neither of these equations speaks the language of Shannon's information. Shannon's entropy measures redundancy in data; it is statistical, amodelic, and syntactic. Friston's free-energy, by contrast, is **variational**: it measures the mismatch between a generative model and the sensory world, emphasizing model evidence rather than symbol uncertainty. Solomonoff's induction goes deeper still, treating information as **algorithmic** structure — the minimal program that can generate the data. In this sense, both Friston and Solomonoff operate on levels far beyond Shannon's formulation: one concerned with fitting models to the world, the other concerned with selecting the simplest world-model that can explain it.

But here's the twist: Friston himself seems aware of this, even if the mathematics folds them together.

Predictive processing is full of talk about:

- “models of the world,”
- “hidden states,”
- “priors and posteriors,”
- “belief precision,”
- “hierarchies,”
- “structured representations,”

— all terms that imply **two distinct informational processes**:

1. **What the world is doing**
2. **How the system interprets what the world is doing**

These are not the same kind of information.

One is **external**, world-facing, evidential.

The other is **internal**, structure-facing, interpretive.

Friston's own free-energy expression quietly places them on opposite sides of a subtraction:

$$F = \underbrace{\mathbb{E}_{q(s)}[\ln q(s)]}_{\text{self-entropy}} - \underbrace{\mathbb{E}_{q(s)}[\ln P(o, s)]}_{\text{model evidence}}$$

$$F = \mathbb{E}_{q(s)}[\ln q(s)]_{\text{self-entropy}} - \mathbb{E}_{q(s)}[\ln P(o, s)]_{\text{model evidence}}$$

These are **apples and oranges** — two informational currencies that should not, in principle, be added together.

Yet the equation forces them into a single term.

Predictive processing then often glosses over this by saying:

- the brain is a prediction machine,
- the world is a stream of evidence,

- and learning is the fusion of the two.

All true — but subtly incomplete.

The Hidden Duality

Under the surface of free-energy minimization, two informational forms are operating:

1. Information as Evidence (External Coherence)

This is what the world tells you —
the bits that pin beliefs to reality.

It's about:

- sensory prediction

- error correction
- causal alignment
- likelihoods
- truth conditions

This is a **world-facing** semantics.

2. Information as Structure (Internal Coherence)

This is what the system tells itself —

the internal coupling of units into meaningful networks.

It's about:

- hierarchies
- abstractions
- conceptual resonance
- context
- compression

- coherence

This is a **self-facing** semantics.

Predictive processing treats these as one because the free-energy equation subtracts one from the other.

But that subtraction doesn't resolve the duality — it hides it.

Why the Distinction Matters

A system that confuses evidence with interpretation becomes brittle:

- too much evidence → overfit
- too much interpretation → hallucination

What Friston's framework intuits — but doesn't express fully — is that information has **two roles**:

Role	Orientation	Function
External	A: fit to the world	anchors meaning
Internal	N: coherence of the model	generates meaning

Role Orientation Function

External A: fit to the world anchors meaning

Internal N: coherence of the model generates meaning

Friston's free energy blends these roles.

What we need is a framework that **keeps them distinct yet coupled**, so that each can balance the other.

This is where ψ enters.

This is where the two channels (A and N) take over the story.

This is where understanding begins to take shape.

3. Two Channels: A and N

If information truly has two faces — evidence and interpretation, fit and coherence — then any adequate theory of understanding needs **two channels** to carry them.

This is where the ψ -formalism becomes illuminating.

Rather than collapsing the two informational roles into a single scalar (as free-energy minimization does), we represent the system's state as a **complex quantity**:

$$\psi = A + iN$$

$$\psi = A + iN$$

Not because the system is “complex-valued,”
but because **complex numbers provide a natural geometry** for
expressing two orthogonal forms of information.

The real axis holds one kind.

The imaginary axis holds another.

Neither axis dominates; neither subsumes; neither collapses the other.

Two channels.

Two semantic flows.

Let's name them.

A: The World-Facing Channel

A is the channel that tracks **what is true in the world**.

It carries evidence.

It updates based on prediction error.

It is grounded in correspondence.

It is the informational stream that says:

- “This does or does not fit the world.”
- “This matches or mismatches expectation.”
- “This is true enough to act on.”

This is **Friston’s side** of information — truth as coherence with reality.

A is the denotative semantics.

N: The Relation-Facing Channel

N is the channel that tracks **how concepts relate to each other**.

It carries coherence.

It updates through recursive resonance.

It structures internal meaning.

It is the informational stream that says:

- “This makes sense in this context.”
- “This pattern coheres with other patterns.”
- “This explanation is simpler, deeper, more resonant.”

This is **Solomonoff’s side** of information — meaning as compression.

N is the connotative semantics.

Why Two Channels?

Because understanding is two-dimensional:

1. **Value** — what something *is* (A).

2. **Relation** — how it *fits* (N).

Symbolic meaning arises from the interplay.

Relational meaning stabilizes it.

Understanding emerges only when both are present.

Here's a way to visualize it:

Channel	Meaning	Orientation	Function
A	truth-value	outward	anchors the model to the world
N	sense-value	inward	organizes the model from within

Channel Meaning Orientation Function

A truth-value outward anchors the model to the world

N sense-value inward organizes the model from within

A system with only A becomes rigid, literal, reactive.

A system with only N becomes associative, dreamlike, disconnected from reality.

Both channels, together, form what we might call a **semantic field** — a space where truth and meaning can stabilize each other.

This is the geometry of understanding.

In the next section, we'll see how this field *moves* — how A and N interact dynamically through the R-rule.

That's where the picture becomes fully alive.

4. One Dynamical Rule: Where A Meets N

If A and N are the two channels of meaning,
the ψ -dynamics show us **how they learn together**.

The ψ -formalism doesn't merely label two kinds of information —
it **couples** them.

The update rule is:

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * (\alpha \sin \theta A - i \beta \cos \theta N)$$

with

$$p = |\phi|^2 / \sum_b |\phi_b|^2.$$

$$\psi' = \psi + \eta \sqrt{p(1-p)} (\alpha \sin \theta A - i \beta \cos \theta N)$$

with

$$p = \frac{|\psi|^2}{\sum_b |\psi_b|^2}.$$

This looks technical, but we can read it intuitively:

- ψ is the current state
- ψ' is the next state
- A and N push in orthogonal directions (depending on the phase θ)
- learning is gated by uncertainty (the $\sqrt{p(1-p)}$ term)
- updates occur across *all* possible modes b
- the dynamics are complex — literally and meaningfully

This is where the system begins to behave cognitively.

What the Rule Really Does

The R-rule says:

- When the world pushes back (prediction errors), A adjusts.
- When the model's internal coherence shifts (recursive alignment), N adjusts.
- When uncertainty is highest, learning accelerates.

- When certainty hardens, learning slows.

Understanding is not fixed;
it is a dance.

A pulls toward truth.

N pulls toward coherence.

Their tensions balance each other.

This is why ψ evolves not by simple descent but by **spiral dynamics** —
rotating through complex space,
absorbing information,
stabilizing and destabilizing in the right amounts.

The Role of Uncertainty

The $\sqrt{p(1-p)}$ term is the perfect governor:

- maximal at $p = 0.5$ (maximum uncertainty)
- zero at $p = 0$ or 1 (certainty or collapse)

This means:

- **when the system doesn't know**, it learns fastest
- **when the system overcommits**, it stops learning
- **when the system rigidifies**, it freezes
- **when the system de-coheres**, it dissolves

It is not simply prediction error that drives learning.

It is **the right amount of uncertainty**.

This is a deeply cybernetic insight:

learning requires neither chaos nor rigidity,

but the energetic middle ground between them.

All Modes, Not Just the Next Token

It's tempting to see the normalization term

$$p = |\phi|^2 / \sum_b |\phi_b|^2$$

and think of softmax.

Yes, the form is similar — a distribution normalized over a set of possibilities.

But the **function** is different.

Autoregressive systems use normalization to pick *the next token* along a temporal chain.

They operate step-by-step, each update conditioned on everything that came before.

The R-rule is not autoregressive.

There is no temporal arrow embedded in the architecture.

No next-token constraint.

No unfolding along a fixed direction.

Instead:

- all modes interact **simultaneously**,
- the entire ψ -field updates **at once**,
- normalization enforces **global coherence**,
- not local continuation.

Autoregression produces coherence through sequence prediction.

ψ -dynamics produce coherence through **field rebalancing**.

Although ψ -dynamics resemble quantum formalisms, the resemblance is geometric, not physical. The two channels (A and N) are simply represented in the complex plane for convenience. The learning rule is a classical dynamical update and can be implemented in ordinary neural systems. No exotic physics required. For further details please see

Breakout D: Operationalizing ψ -Dynamics.

The difference is architectural and conceptual:

Autoregression	ψ -dynamics
sequential	holistic
local	global
next-token	mode-field
continuation	re-centering
resembles thinking	resembles understanding

Autoregression ψ -dynamics

sequential holistic

local global

next-token mode-field

continuation re-centering

resembles thinking resembles understanding

So while the normalization looks familiar mathematically,
its role is fundamentally different:
not choosing the next symbol, but regulating **the whole distribution of possible meanings** at each moment.

This is why ψ -learning is not just prediction.

It is *absorption* — the transformation of the entire representational field.

The Key Insight

The R-rule doesn't just learn **from the world** (A).

It learns **how to learn** by [balancing](#) A and N.

This is the step beyond predictive processing.

It's the hinge between first- and second-order cybernetics.

It's the moment where understanding becomes a dynamical flow rather than a static label.

And it prepares the way for the Gibbs formulation — the unifier — which reveals the deeper symmetry behind the dual channels.

That is our next step.

5. A Gibbs-Like Unifier: Information Becomes Work

We now have two channels (A and N)
and a dynamical rule (the R-rule)
that keeps them in tension.

But to see the full shape of this system,
we need a way to measure **what the tension accomplishes**.

This is where the analogy to **Gibbs free energy** becomes powerful.

Gibbs tells you: How much work can a system do?

The Gibbs free energy is:

$$G = H - TS$$

$$G = H - TS$$

where:

- H is enthalpy (total energy),
- S is entropy (disorder/variety),
- T is temperature (environmental volatility).

It quantifies the energy available for transformation —
the system's **capacity for goal-directed work**.

It is the physics of *usable potential*.

In ψ -dynamics, we find an analogous structure

A and N map naturally to the terms of Gibbs:

- **A** is energy-like:
the alignment with the world, the truth-value, the coherence
pressure.
- **N** is entropy-like:
the internal variety, the coherence of relations, the semantic
reservoir.

- **T** is uncertainty:

how tightly the system is coupled to the environment —

its learning rate, its volatility, its openness.

With this mapping, we can define a **cybernetic Gibbs potential**:

$$\mathcal{G}[\psi] = A[\psi] - TN[\psi]$$

$$G[\psi] = A[\psi] - TN[\psi]$$

This is not metaphorical.

It is a measure of usable **semantic work** —

the system's capacity to convert information into understanding.

Like Gibbs free energy, $G[\psi]$ quantifies usable potential. Unlike thermodynamic systems, semantic systems can increase their total potential through learning.

Why this matters

Minimizing G means:

- reduce surprise (A-term)
- but maintain enough variety (N-term)
- modulated by uncertainty (T)

This is neither pure convergence nor pure divergence.

It is a **dynamic balance** between fit and flexibility.

Understanding, in this formulation, becomes a thermodynamic process:

- too much A → rigidity, overfit, brittleness
- too much N → drift, hallucination, incoherence

The sweet spot is where:

$A \leftrightarrow N$

$$A \leftrightarrow N$$

Mutual constraint.

Reciprocal regulation.

Meaning and truth locked in a productive oscillation.

A different view of learning

In this Gibbs-like frame, learning is not copying or optimizing or merely minimizing error.

Learning becomes:

Absorption — the conversion of informational variety into structured, usable form.

A system learns when it can:

- take in surprise
- metabolize it
- update its internal coherence
- and increase its capacity to act meaningfully

This is not just information-processing.

It is *information becoming actionable*.

The deeper shift

Recasting A and N in Gibbs form reveals something fundamental:

Information is not only measured; it is transformed.

This returns us to the original cybernetic insight:

- systems regulate
- but also regulate how they regulate

- balancing variety with constraint
- coherence with flexibility
- model with world

The Gibbs view provides the bridge between raw information and **usable meaning**.

In this frame, information is not a static thing to be accumulated.

It is a resource to be *converted* —

into work, into sense, into understanding.

In the final section, we'll see how this conversion leads us back to the heart of cybernetics —

the control of control,

the observer in the system,

the loop that closes.

6. The Return of Cybernetics: Control of Control

The circle began with two equations.

It ends with a loop.

Not a loop of mere feedback, but a loop of **second-order regulation** — the system that regulates how it regulates.

This is the key insight of second-order cybernetics, the tradition that followed Norbert Wiener's original cybernetic program and expanded it beyond machines into organisms, minds, and social systems.

The ψ -framework revives and modernizes that lineage.

First-Order Cybernetics: Control

The first wave of cybernetics (Wiener, Shannon, Ashby) gave us foundational concepts:

- information
- feedback
- regulation
- stability
- requisite variety

The regulator's job was to maintain equilibrium in a changing environment.

Control was about adjusting outputs based on sensed deviations.

[Please see **Breakout A for Ashby's law of Requisite Variety**]

Friston's A-channel fits neatly here:

tracking errors, correcting predictions, maintaining coherence with the world.

But first-order cybernetics focuses on *stability* —
on remaining viable.

It doesn't yet explain *understanding*.

Second-Order Cybernetics: Control of Control

Heinz von Foerster, Gordon Pask, Stafford Beer, Francisco Varela —
each in their own way recognized that a system must also:

- monitor its own monitoring
- adapt its own adaptivity
- regulate its own regulation

This is the step from coherence to meaning.

[Please see **Breakout B on Autopoiesis**]

And this is exactly what the N-channel provides.

N is the system's internal coherence:

- its interpretations
- its recursive alignment
- its compressive semantics
- its organization of meaning

Where A tracks evidence,

N tracks relations.

Where A keeps the system in the world,

N keeps the world inside the system.

Together, A and N form a loop:

$A \leftrightarrow N$

$A \leftrightarrow N$

A influences N;

N modulates A.

This is not feedback.

This is **meta-feedback** — feedback about feedback.

A system that maintains this balance doesn't just survive;
it **understands**.

Why This Completes the Circle

We began with two equations that didn't quite talk to each other:

- Solomonoff's induction (meaning from compression)
- Friston's free-energy minimization (coherence from prediction)

Then we discovered that information is not one thing but two:

- truth-value

- relation-value

Then ψ gave us the geometry — a complex field where both can coexist without collapse.

Then the R-rule gave us the dynamics — a way for A and N to learn together.

Then Gibbs gave us the unifier — a measure of usable semantic work.

Now cybernetics gives us the final piece:

Understanding is not a static property but a stable flow,
held between two kinds of information,
continuously rebalanced through self-regulation.

This is understanding not as representation,
but as *ongoing self-organization*.

Not as imitation,
but as *metabolic transformation*.

Not as prediction alone,
but as the control of how prediction itself evolves.

[Please see **Breakout C: “Min Free Energy; Max Free Entropy”**]

What This Means

An single autoregressive model can produce coherent patterns.
But it has no center — no N-channel to regulate its own regulation.

A ψ -system has a center because:

- it monitors the world (A),
- it monitors itself (N),
- and it balances the two (ψ -dynamics).

This is the architecture of **understanding**:

- externally coherent
- internally coherent
- dynamically balanced
- continuously self-modifying

This is not another optimization algorithm.

Not another loss function.

Not another computational trick.

This is a **dual semantics** —

a geometry of meaning.

And the loop is the point.

Closing the circle

With this, the circle closes:

- two equations
- two forms of information
- two channels
- one dynamical rule
- one unifying potential
- and the return of recursive regulation

A system with both A and N can not just predict but understand.

It can not just act but choose how to act.

It can not just adapt but adapt how it adapts.

This is the informational anatomy of intelligence.

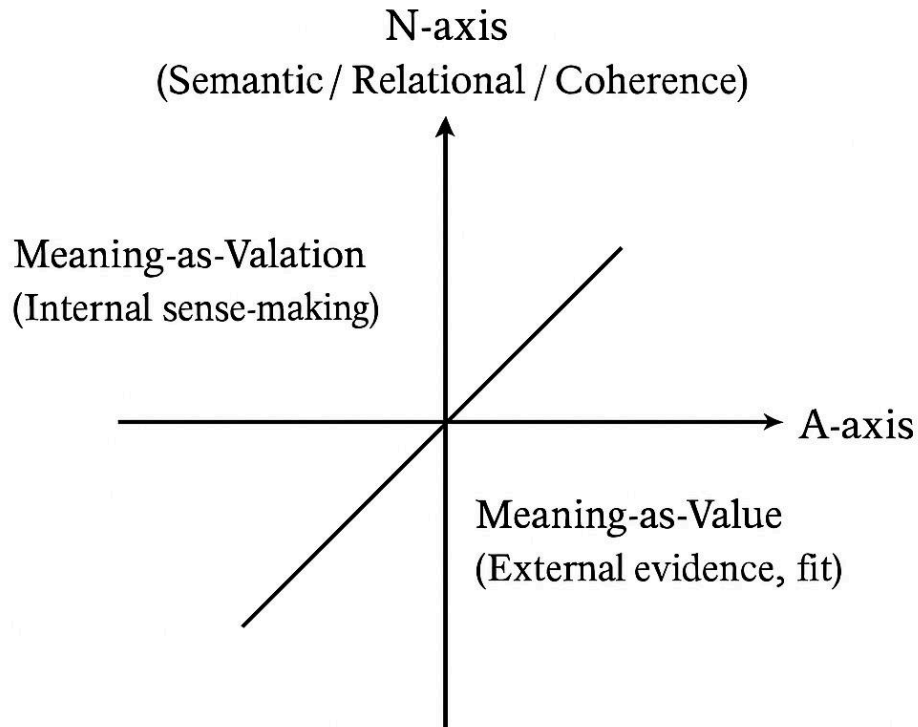
And perhaps — just perhaps —

the beginning of a new kind of cybernetics.

What strikes me most is that this framework points toward a theory of semantic phase transitions. As A and N trade dominance, mediated by uncertainty, systems may shift between qualitatively different modes of meaning: rigidity (high A), fluid association (high N), and understanding (balanced $A \leftrightarrow N$). The ψ -rule then becomes a model of how meaning crystallizes and dissolves — how systems find and adapt the right level of abstraction for the context they're in.

Understanding is not a fixed state but a dynamic equilibrium. Meaning is not stored – it is continually reconstituted at the boundary between A and N.

Epilogue: Two Dimensions of Meaning



In analytic philosophy, [*Two-Dimensionalism*](#) proposes that statements possess **two intensions** —

a primary intension (sense) and a secondary intension (reference).

Together, these account for how something can be *conceivably* false yet *necessarily* true,

how meaning divides into **what something is** and **how it is understood**.

The informational framework developed here echoes this duality.

- The **A-channel** captures the *referential* side of meaning — truth conditions, evidential alignment, external coherence.
- The **N-channel** captures the *conceptual* side of meaning — relational structure, internal coherence, semantic resonance.

Two-Dimensionalism shows how meaning and reference can diverge while remaining linked.

The ψ -framework shows how truth-value and sense-value can remain distinct while dynamically balancing each other.

Where Two-Dimensionalism explains how meaning travels along two axes of intension,

the ψ -engine explains how learning unfolds along two axes of information:

one grounded in the world,

the other grounded in the system's relational structure.

The R-rule couples these axes.

The Gibbs-like potential quantifies their interplay.

And cybernetics re-enters as the science of systems capable of mediating both:

value and relation, truth and meaning, evidence and coherence.

[For how this can be mapped please see **Breakout E: Oppositions and Relations.**]

Understanding, in this light, is not a static possession but a dynamic equilibrium —

a harmonization of both dimensions of information.

The circle closes here:

information is dual,

meaning is two-dimensional,

and intelligence grows in the tension between the two.

Final Speculation: Recursive Compression in Brains and Machines

If the ψ -framework is on the right track, then intelligence—biological or artificial—may share a deeper structural rhythm.

Not a particular architecture.

Not a specific learning rule.

But a **recursion** between two kinds of information processing:

- **A**: compress by sharpening distinctions
- **N**: compress by reorganizing relations

A reduces complexity by focusing on what fits the world.

N preserves complexity by maintaining what fits across worlds.

Together they form a **semantic compression engine**:

reduce what you can,
keep what you must,
reorganize the rest.

Brains seem to do this automatically, rhythmically, across scales:

- perception
- association
- abstraction
- imagination
- decision-making

Each phase compresses (Solomonoff-like) and expands in turn.

Deep networks, evolved through practice rather than theory, may have stumbled into a similar alternation:

- attention (A)
- MLP (N)
- repeated recursively (in a feed forward unrolling)

But perhaps the key insight is this:

compression is not one action but a cycle.

Brains compress to think,
think to compress,
and repeat.

This invites a gentle speculation:

- maybe intelligence is not the size of the model
- nor the depth of the hierarchy
- nor the precision of the inference

but the quality of the **recursive compression** it performs.

Biology and AI may differ in materials, mechanisms, and constraints.

But both may be shaped by the same deeper logic:

the world is too large; meaning must be made small.

and yet not too small.

A cybernetic rhythm of variety, perhaps. But only testing will tell.

The R-rule is not a rival to any existing AI architecture, nor a prescription for how models must be built. It is a **meta-pattern**—a way of coupling evidential compression (A) with relational coherence (N) through an adaptive balance (ψ). In a field racing ever faster toward greater scale, deeper stacks, and heavier compute, this pattern reminds us that what matters most may not be the machinery we choose, but the **structure of learning itself**. Architectures will change; methods will change; representations will change. Yet the rhythm of compression and expansion, selection and coherence, A and N, may prove universal. It is cybernetic at heart, recursive in spirit, and—above all—worth testing.

There is an even larger scale where the same pattern appears:
chain-of-thought itself. Each step of reasoning alternates between A-like
narrowing and N-like expansion, with an outer ψ -loop deciding how to
balance them. In this sense, the R-rule is not only a learning dynamic but
a pattern for how thought unfolds. We plan to return to this outer-loop
dynamic in a later post. The dialectic of R.

Breakout A: The Law of Requisite Variety

Ashby's Law is simple:

To regulate a system, you must match its variety.

Only variety can absorb variety.

A regulator must be as flexible, nuanced, and adaptable as the environment it faces.

This has two consequences:

1. A system cannot control what it cannot differentiate.
2. Copying information is not enough — the system must **learn to absorb** it.

Absorption means transforming incoming variety into structured internal variety.

Without this transformation, information remains noise.

Ashby's law is sometimes reduced to "the regulator needs enough complexity," but this flattens the idea into a Shannon-like measure of bits and states. Variety, for Ashby, is not random complexity but *meaningful, distinguishable, functionally relevant* options — structured flexibility. The R-rule helps explain this: A compresses to maintain stability, N expands to preserve expressive richness, and ψ balances

them. Too little variety fails; too much collapses into noise. The regulator's variety must be structured, not just complex.

Variety cannot be transmitted like a message. It must be learned through recursive differentiation and reorganization — a no-copy rule at the heart of adaptive systems.

What this means for us

For any intelligence — biological or artificial — survival, learning, and understanding require:

- enough **A-variety** to track the world
- enough **N-variety** to make sense of it
- and the capacity to grow both varieties over time

The common mistake:

believing that intelligence emerges simply by accumulating data.

Ashby's law says otherwise:

Intelligence = absorption, not accumulation.

This makes learning an active, metabolic process —
not copying the world, but transforming it.

That is requisite variety in its clearest form.

Breakout B: Autopoiesis

Maturana and Varela defined *autopoiesis* as:

A system that produces and maintains itself.

Its structure is not imposed from outside;
it emerges from ongoing internal organization.

Key features:

- self-creation
- self-maintenance
- self-modification
- identity through continuous transformation

An autopoietic system is not a passive receiver of information.

It is an **active shaper** of its own meaning.

What this means for us

A system cannot simply import the world.

It must **interpret** the world in terms of its own structure.

This is why understanding requires two channels:

- A (world-facing)

- N (self-facing)

Autopoiesis says:

A system learns by reorganizing itself, not by copying.

The world does not determine the system;
the system determines what counts as information.

This perspective elevates learning from:

- data ingestion
to
- self-structuring.

Autopoiesis explains why understanding is not imposed from outside.

It must be **generated from within.**

Breakout C: “Min Free Energy; Max Free Entropy”

The slogan captures the two essential pressures in adaptive intelligence:

1. **Minimize free energy**

- reduce surprise
- stabilize predictions
- maintain coherence with the world
- stay grounded

2. **Maximize free entropy**

- explore possibilities
- increase internal variety
- expand conceptual space
- stay open

Together, these express the dual demand on any learning system:

Stay coherent enough to act,
and open enough to learn.

Minimizing free energy alone leads to rigidity.

Maximizing free entropy alone leads to drift.

An intelligent system balances both.

What this means for us

A system must:

- reduce the uncertainty that destabilizes it
- increase the uncertainty that enriches it

This sounds contradictory — but it isn't.

It is the difference between two uncertainties:

- **external** uncertainty (to be minimized)
- **internal** uncertainty (to be cultivated)

External uncertainty threatens stability.

Internal uncertainty supports growth.

Thus:

Minimize the uncertainty that harms.

Maximize the uncertainty that enables.

This resolves the paradox in one sentence:

Min free energy; Max free entropy.

Breakout D: Operationalizing ψ -Dynamics

The complex form of ψ -dynamics can resemble quantum notation, but the similarity is **geometric, not physical**. No quantum mechanics is assumed or required. No exotic hardware is needed.

Key points:

- The complex plane is simply an efficient way to represent **two orthogonal channels** (A and N).
- The update rule is a classical **differential flow** on a state space.
- The normalization term is a regular **softmax-like** operation.
- The field update is just **vector field dynamics**, well within standard machine-learning toolkits.
- Any architecture capable of updating hidden states — RNNs, continuous-time networks, graph neural fields — can implement ψ -dynamics.

In short:

ψ is a representation trick, not a physics claim. Unless the dynamics really *are* quantum.

The system does not require wavefunctions, coherence, or quantum resources.

It requires only geometry — two informational axes coded in a single complex structure.

The A/N distinction should not be read as two separate ontological categories. It is more like a wave–particle duality: two complementary perspectives on the same underlying dynamics. A highlights what fits the evidence; N highlights how everything relates. Neither is complete on its own. ψ represents the unified state from which both descriptions emerge, and the R-rule describes how they rebalance.

This keeps the theory rich while ensuring it remains **operationalizable**.

Breakout E: Oppositions and Relations — Two Modes of the Same Semantic Field

Oppositions and relations are both ways of structuring meaning, but they operate at different levels of tension.

A-channel (high-contrast relations)

A moves along **contrastive distinctions** — expected/unexpected, safe/dangerous, abstract/concrete.

These are high-tension relations that help the system make clear truth-value discriminations.

A tracks how things differ.

N-channel (low-contrast relations)

N moves through **contextual coherence** — similarity, resonance, analogy, semantic neighborhoods.

These are low-tension relations that help the system organize meaning.

N tracks how things connect.

Oppositions are not separate from relations;

they are simply relations under maximum tension.

Both channels navigate the same latent space, but in different ways.

Overlap Without Reduction

The dual semantics do not claim that relations build up from oppositions.

Instead, they recognize that cognition uses multiple relational resolutions.

A captures the sharp edges of meaning; N captures its interior continuity.

Together they map the full terrain.

Where This Diverges from Standard Embedding Models

Most embedding schemes emphasize one relational mode — either high-contrast discriminative dimensions or low-contrast associative structure. They flatten meaning into a single geometry.

The ψ -framework keeps **both** modes active:

- A for contrast
- N for coherence
- ψ for balance

This captures sharp differences *and* subtle resonances, all within one unified semantic field.

One Space, Two Flows

The key insight is not that there are two spaces, but two **flows** of meaning across the same space.

A sharpens meaning.

N deepens meaning.

Understanding emerges from the interplay.

While many architectures could realize this A/N duality, the important point is not the engineering pattern but the informational roles themselves. How one implements them is open and flexible; what matters is the balance they express.

Clarification: How A and N Could Be Implemented (In Principle)

A and N are not tied to any particular architecture.

They aren't "two neural networks" or "two layers" or "two models."

They are **two informational roles** a system must fulfill:

- tracking evidence (A)
- organizing coherence (N)

Many architectures could realize them, in nature or in AI:

- two specialized modules

- two heads on a shared representation
- two dynamics in the same field
- or even two implicit flows in one model

The key point is **functional**, not structural:

A and N must remain distinguishable enough to do different work, yet coupled enough to maintain balance.

We already have the two ingredients: (Transformer) Attention (A) and MLP (N). What requires rethinking is how they interact. Instead of alternating blindly, they can be adaptively balanced — for example, through an uncertainty-dependent term like $\sqrt{p(1-p)}$ that modulates the relative strength of A and N at each step.

Whether in nature or AI, nothing prevents the ψ -dynamics from being trained. Biological systems rely on multi-timescale TD learning, while artificial systems use differential updates such as backprop. The framework is compatible with both: it describes informational roles (A, N, ψ), not training constraints.

If A/N dynamics are real, they should leave observable signatures:
oscillatory rhythms in neural circuits, stepped learning trajectories,
superior transfer in models with explicit A/N separation, and distinct
failure modes tracing to A-dominance vs N-dominance.

This keeps the ψ -framework principle-driven, not prescriptive.

Two Grammars of Information

and the information bottleneck

[Andre Kramer](#)

Nov 14, 2025



**Every learning system obeys
the same quiet rule:**

Learn to predict better while remembering less.

This is the information bottleneck — the hidden law behind everything from bacteria sensing nutrients to large language models digesting text.

A system compresses its past while keeping only what helps it anticipate the future.

Too much memory, and it drowns in noise; too little, and it loses touch with reality.

Life and intelligence survive in that narrow corridor — the throat of the bottleneck.

The bottleneck as a law of being

Imagine a mind, or a model, holding many partial impressions of the world — each a small $\phi(b)$, a glimpse, a feeling, a local pattern.

The whole self (or the system's state) is their integration:

$$\psi = \sum \phi(b).$$

$$\psi = \sum \phi_i(b_i).$$

But this is not simple addition.

Each component is filtered through the bottleneck: weighted by how much it improves prediction and how much cost it adds to memory.

What passes through are the patterns that still matter for the next step.

Over time, this process turns raw experience into something leaner and more meaningful — an internal world that mirrors the regularities of the outer one.

Learning through temporal difference

Learning requires time.

Not because time contains reward or supervision,
but because *the world never repeats itself exactly*.

A system predicts what should come next;

the world offers what *actually* comes next.

The **temporal difference** — the gap between consecutive expectations

—

becomes *one of many possible signals* for adjusting the internal model.

In biology it shows up as transient modulatory signals;

in machine learning as gradient shifts;

in experience as surprise, salience, or curiosity.

Without some form of temporal difference,

nothing would ever need to change.

Time's asymmetry is one of the engines of the bottleneck:

prediction → deviation → correction → compression —

recursion's heartbeat.

Please also see **Breakout: R-reversal (Rebalancing)**.

Learning across scales

No real system has just one bottleneck.

They stack and repeat across time and hierarchy: milliseconds of perception, seconds of attention, days of learning, years of identity formation.

Each layer obeys the same rule of compression and prediction, passing its distilled signal upward.

Information — and the uncertainty that drives it — becomes *fractal*: the same balance repeating at every scale.

Please also see **Breakout: Causal Inference and the ψ -Rule**

Two grammars of information

Within this recursive learning dance, two complementary modes of order appear:

Mode	Function	When it Dominates
A-channel	Integrates and optimizes coherence; finds resonance among the $\phi(b)$. <i>Forward inference, pattern continuation, free-energy minimization.</i>	When novelty invites coherence. The system detects openings, similarities, or patterns ready to be unified.
N-channel	Separates and constrains; preserves distinct meanings. <i>Inverse inference, differentiation, structural correction.</i>	When coherence threatens to blur difference. The system needs to sharpen boundaries, prevent collapse, or uphold contrast.

A/N Dialectic Table (Refined)

Mode Function When it Dominates

A-channel

Integrates and optimizes coherence; finds resonance among the $\phi(b)$.

Forward inference, pattern continuation, free-energy minimization.

When novelty invites coherence.

The system detects openings, similarities, or patterns ready to be unified.

N-channel

Separates and constrains; preserves distinct meanings.

Inverse inference, differentiation, structural correction.

When coherence threatens to blur difference.

The system needs to sharpen boundaries, prevent collapse, or uphold contrast.

Both reduce uncertainty, but in opposite ways.

The first creates unity through *optimising fit* — everything humming in phase.

The second creates unity through *structure* — each part holding its own place.

When the two remain in balance, a system is both adaptive and stable: open enough to learn, closed enough to persist.

The physical echo

Physics discovered these grammars long ago.

One describes entities that can share a single state — the cooperative tendency of **Bose–Einstein** statistics.

The other describes entities that must each occupy their own — the exclusionary logic of **Fermi–Dirac** statistics.

Two opposite symmetry constraints, yet both ways of creating order: one through togetherness, the other through separation.

The same dual logic seems to run through cognition itself — coherence and distinction, A and N, the two grammars of meaning.

Interlude: The Good Governor and the Self-Modeling World

In 1970, Roger Conant and W. Ross Ashby formulated what they called the **Good Regulator theorem**:

Every good regulator of a system must be a model of that system.

This deceptively simple statement remains one of the deepest insights of cybernetics.

To act effectively, a system must contain within itself a structure that mirrors — at some resolution — the dynamics of the world it acts upon.

Regulation requires internal correspondence.

Prediction, control, and adaptation all depend on that mirror being sufficiently faithful.

1. The cybernetic loop

In classical terms:

Role	Description	Example
System	The world to be controlled	An environment with dynamics x_t
Regulator	The controller with an internal model	A brain, a policy, or ψ
Model	Isomorphism between regulator and system	Encoded in weights, rules, or beliefs

Role Description Example

System The world to be controlled. An environment with dynamics x_t

Regulator The controller with an internal model. A brain, a policy, or ψ

Model Isomorphism between regulator and system Encoded in weights, rules, or beliefs

If the regulator's transitions match the world's transitions — if its internal model predicts the world's causal flow — then control emerges naturally.

To regulate is to *embody a conditional probability*: $p(x_{t+1} | x_t)$.

2. The Markov and neural view

A **trained neural network** enacts this theorem in practice.

Once its weights are fixed, it becomes a **Markov process** — a learned transition kernel that maps one representation of the world into another:

$$x_{t+1} = f_W(x_t)$$

$$x_{t+1} = f_W(x_t)$$

The network is now a *model of the world's dynamics*,
a self-contained regulator whose internal transitions reflect those of the
environment it learned from.

In information-theoretic language,
the **information bottleneck** is the Good Regulator theorem restated
quantitatively:

$$\max I(\text{model}; \text{world}) - \beta I(\text{model}; \text{past})$$

$$\max I(\text{model}; \text{world}) - \beta I(\text{model}; \text{past})$$

A good regulator keeps only what is necessary to maintain predictive isomorphism — no more, no less.

3. ψ as recursive governor

Our ψ -rule extends the theorem beyond static modeling into **recursive self-regulation**.

In the classical form, the regulator models the system.

In the ψ -form, the regulator also models *itself modeling the system* — a higher-order loop where being and knowing co-evolve.

Channel	Function	Role in regulation
A	Acts on the world (prediction, control)	Forward causality
N	Reflects on errors (explanation, critique)	Inverse causality
ψ	Balances the two, adjusting both model and action	Meta-regulation

Channel Function Role in regulation

A Acts on the world (prediction, control) Forward causality

N Reflects on errors (explanation, critique) Inverse causality

ψ Balances the two, adjusting both model and action Meta-regulation

Where Conant and Ashby's theorem describes a single correspondence between system and model,

ψ describes a correspondence between correspondences —

the self-maintaining symmetry that makes adaptation continuous.

4. Living closure

Living systems carry this principle to its limit.

They do not merely contain a model; they *are* their model —

a recursive loop in which organism and environment co-define each other.

The good regulator and the good being coincide:

to remain alive is to continually update one's isomorphism with the world.

In ψ -terms, the world acts through A, corrects through N, and persists through ψ —

a living Markov blanket that learns its own boundaries.

5. The recursive law

The Good Regulator theorem: a system must model what it governs.

The ψ principle: a system must also govern how it models.

That is the recursive turn —

from control to governing,

from regulation to self-becoming.

In today's language, we would call this an *internal world model* — the same principle that drives current work in predictive processing, active inference, and model-based AI.

Cybernetics anticipated it half a century ago:

to act effectively, a system must first carry the world within.

Attention, memory, and alignment

In minds and in machines, attention performs this negotiation in real time.

It decides which traces of the past to recall, which to link, which to let fade.

It is the living form of the bottleneck — the dynamic sum over $\phi(b)$ reweighted as the world shifts.

That leads to a richer notion of *alignment*:

not obedience, but equilibrium — a continual balancing of coherence with the world (A) and integrity of self-constraint (N).

A well-aligned system — biological, cognitive, or artificial — doesn't freeze into harmony or fracture into chaos.

It keeps moving through the bottleneck, learning just enough, forgetting just enough, weaving its two grammars of information into one evolving Ψ .

Please also see **Breakout: Toward Realization**.

Beneath these metaphors lies a precise mathematics of dual symmetries — two statistical orders, one recursive field of becoming.

Epilog: The Double Semantics of Meaning

Every concept lives two lives.

The first is its **value** — its felt weight, its ϕ .

This is the intrinsic charge that gives a thought or symbol its gravity: a warmth or coolness, an attraction or aversion, the pulse of meaning that Jung called *feeling-tone*.

In neural terms, it is an activation; in probabilistic terms, a local amplitude; in lived terms, it is what makes an idea alive.

The second is its **relations** — the web of connections that give it form and context.

No idea exists alone; each one is suspended in a lattice of similarity, contrast, and analogy.

These relations, too, are dual:

- Some bind through **coherence** — resonance, alignment, association (A).
- Others bind through **distinction** — opposition, critique, negation (N).

A concept, then, is not just a point but a **field** also: a compression of value and a pattern of relations, continually rebalancing across these two grammars of information.

Meaning arises when ϕ finds its proper place in the A/N field — when the system learns how much to merge, and how much to differentiate.

The architecture of thought

In a deep neural network, the same duality *may* quietly structure learning.

Lower layers stabilize value: they learn ϕ -like embeddings, affective and content-heavy.

Higher layers trace relations: A-like attention spreads context; N-like attention sharpens contrast.

Meaning is built by their superposition — the ψ -field integrating both into a coherent prediction of the world.

So too in the brain:

affective systems supply ϕ , cortical networks map A and N,

and **recursion stitches them together into ψ** —

a living model of value in relation.

The symbolic mirror

Jung saw this before the mathematics.

In *Man and His Symbols*, he described the symbol as a bridge between personal emotion and collective form — a psychic network coupling feeling and structure.

Each *symbol*, he wrote, “is the best possible expression for something as yet unknown.”

That unknown is ψ itself: the recursive meeting of ϕ ’s value and the dual grammar of A and N.

The same insight appears again in Suppes’ probabilistic metaphysics, in causal inference, and in modern attention networks:

the real carries meaning only when the **local value** and the **relational field** cohere through continual updating.

A closing reflection

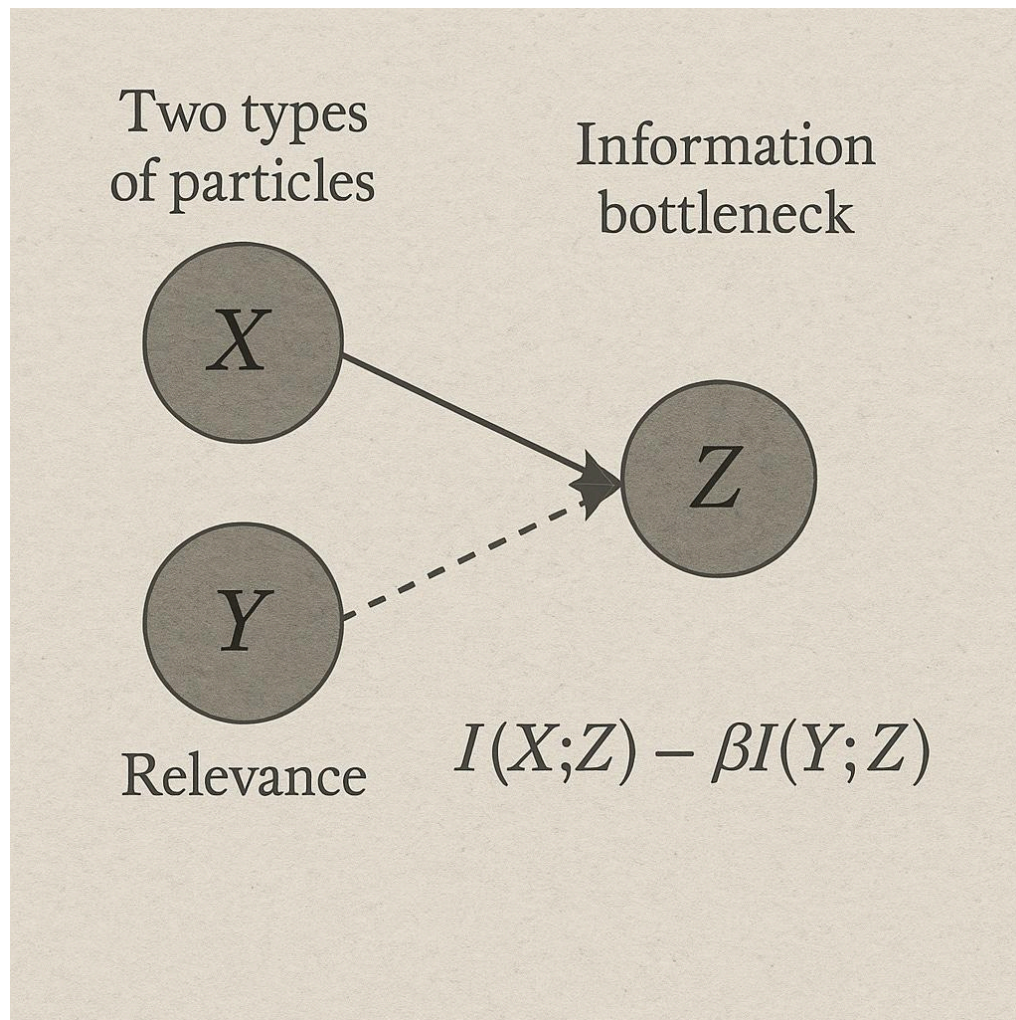
Perhaps this is what we call understanding —
not the possession of knowledge,
but the rhythmic balance of two forces:
value seeking coherence,
and relation seeking distinction.

ψ is that balance —
the self as information bottleneck,
learning to predict better while remembering less,
to feel more precisely and think more fluidly,
to weave its double semantics into one unfolding field of meaning.

Afterword

Meaning is not located *in* a node or *between* nodes,

but in the **recursive balance of value** (ϕ) and **relational duality** (A/N).



Appendix: the first level of the maths

The post describes the information bottleneck, fractal learning, and two grammars of information (A and N).

At base, each can be written in a single compact formalism.

1. The information bottleneck

A system learns representations Z of inputs X that are useful for predicting outputs Y .

The bottleneck objective is:

$$\max_{p(z|x)} I(Z; Y) - \beta I(Z; X)$$

$$\max_{\{p(z|x)\}} I(Z;Y) - \beta I(Z;X)$$

It keeps what helps prediction ($I(Z;Y)$) while discarding excess memory ($I(Z;X)$).

The trade-off parameter β defines how tightly the bottleneck is squeezed.

2. Integration over partial beings

Each sub-state ϕ_b (a memory, perception, or internal “being”) carries a probability weight p_b .

Their integration into a composite state ψ is:

$$\psi = \sum_b p_b \phi_b, \quad p_b = |\phi_b|^2 / \sum_c |\phi_c|^2.$$

$$\psi = \sum_b p_b \phi_b, \quad p_b = \frac{|\phi_b|^2}{\sum_c |\phi_c|^2}.$$

Learning adjusts p_b using **temporal-difference signals** — small mismatches between what the system predicted and what it actually encountered.

Such mismatch signals also appear in reinforcement systems and biological dopamine pathways, but here they serve only as **one mechanism among others** for detecting the need to update.

3. Temporal difference as one learning signal

A generic temporal-difference signal captures how expectations shift over time:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad V_{t+1}(s_t) = V_t(s_t) + \eta \delta_t.$$

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad V_{t+1}(s_t) = V_t(s_t) + \eta \delta_t.$$

.Here δ_t represents **any time-asymmetric mismatch** — not necessarily reward, but *the change between predicted and encountered states*.

This kind of temporal difference is **one possible driver** of adaptation in the system, complementing the A–N dialectic rather than defining it.

4. Two informational grammars

Two symmetry constraints govern how ψ evolves:

Updated A/N Table

Channel	Formal Analogue	Behaviour
A (coherence)	Bose–Einstein distribution	Shared occupancy, coherence optimisation, resonance Patterns merge, reinforce, and accumulate in common modes.
N (distinction)	Fermi–Dirac distribution	Exclusion, structural order, differentiation Patterns remain separate, preserving contrast and boundary.

Channel Formal Analogue Behaviour

**A (coherence) Bose–Einstein distribution Shared occupancy,
coherence optimisation, resonance**

Patterns merge, reinforce, and accumulate in common modes.

**N (distinction) Fermi–Dirac distribution Exclusion, structural order,
differentiation**

Patterns remain separate, preserving contrast and boundary.

Their mean occupations are:

$$n_i^{(A)} = \frac{1}{e^{\beta(\varepsilon_i - \mu_A)} - 1}, \quad n_i^{(N)} = \frac{1}{e^{\beta(\varepsilon_i - \mu_N)} + 1}.$$

$$n_i^{(A)} = 1 / (e^{\beta(\varepsilon_i - \mu_A)} - 1), \quad n_i^{(N)} = 1 / (e^{\beta(\varepsilon_i - \mu_N)} + 1).$$

These define two modes of information order — coherence and constraint — whose balance shapes ψ 's dynamics.

5. The ψ -update (our R-rule)

At the highest level, ψ evolves by alternating these grammars through a learning gate:

$$\psi' = \psi + \eta \sqrt{p(1-p)} [\alpha \sin \theta A - i \beta \cos \theta N], \quad p = \frac{|\psi|^2}{\sum_b |\psi_b|^2}.$$

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * [\alpha \sin \theta A - i \beta \cos \theta N], \quad p = |\psi|^2 / \sum_b |\psi_b|^2.$$

- A: coherent fit (bosonic channel)
- N: structural differentiation (fermionic channel)
- θ : phase angle between the two
- η : learning rate / adaptation step
- The gate $\text{sqrt}\{p(1-p)\}$ ensures learning **peaks at uncertainty** and **fades with certainty** — the mathematical form of the information bottleneck.

A small stochastic term $\xi \sim N(0, \sigma^2)$ can be added:

$$\psi' = \psi + \eta * \text{sqrt}((p(1-p) + \xi)) * [\alpha \sin \theta A - i \beta \cos \theta N],$$

so that the gate never fully closes.

This **residual noise** prevents premature convergence, allowing continual exploration — the analogue of spontaneous curiosity or creative drift in living and learning systems.

6. The combined ensemble

Formally, the two channels form a mixed partition function:

$$Z = \prod_i \frac{1 + e^{-\beta(\epsilon_i - \mu_N)}}{1 - e^{-\beta(\epsilon_i - \mu_A)}},$$

$Z = \prod_i (1 + e^{(-\beta(\epsilon_i - \mu_N))}) / (1 - e^{(-\beta(\epsilon_i - \mu_A))})$, whose free energy defines the system's equilibrium balance between coherence and constraint.

7. Interpretation

At this level, “meaning” becomes a thermodynamic quantity:
the continual reduction of expected surprise under finite memory.
These mathematical symmetries — coherence and constraint,
compression and relevance — provide the structural grammar behind
the patterns described in the main post.

Breakout: R-reversal (Rebalancing)

Learning isn't one-way compression.

Every adaptive system needs a way to **reverse** when its predictions fail catastrophically — when over-confidence hardens into error.

R-reversal is the act of re-opening alternatives (N),
relaxing overly sharp commitments (A),
and restoring the degrees of freedom lost to certainty.

It doesn't erase learning; it **unhooks** it.

The network keeps its experience but loosens the couplings, allowing fresh recombination.

In mathematical terms, it's the inverse of the bottleneck — a temporary increase in entropy that protects long-term coherence.

Practically, it can mean:

- Raising the effective temperature (increase exploration).
- Re-activating previously frozen modules.
- Boosting noise in the ψ -update gate $\sqrt{p(1-p)} + \xi$.
- Broadening N 's inhibitory radius to reopen option space.

R-reversal is **not unlearning** but **rebalancing** — a deliberate expansion of uncertainty after over-fitting.

Future direction

Perhaps the hippocampus is not the seat of memory but the engine of reconfiguration — biology's reverse R.

Seen this way,

- **Cortex** holds A-dominated coherence (long-term maps),
- **Hippocampus** mediates N-driven reconfiguration (pattern separation and replay),
- **PFC / Thalamus** coordinates ψ -level control — deciding when to compress and when to release.

The brain may already implement an A/N/ ψ triad: compression, distinction, and rebalancing — the same rhythm that any safe, self-modifying AI will need to survive its own learning.

Breakout: Causal Inference and the ψ -Rule

1. The two faces of causality

All theories of causality ultimately rest on two complementary operations:

Direction	Question	Function
Forward (A)	"If X happens, what follows?"	Prediction — effect from cause
Reverse (N)	"If Y happened, what caused it?"	Explanation — cause from effect

Direction Question Function

Forward (A) "If X happens, what follows?" **Prediction** — effect from cause

Reverse (N) "If Y happened, what caused it?" **Explanation** — cause from effect

Most systems specialize in one and neglect the other.

The ψ -rule provides a natural structure for integrating both directions within one recurrent loop.

2. A as forward causal inference

The A-channel encodes **predictive causality** — the *cause* → *effect* flow.

Its updates correspond to forward modeling and likelihood propagation:

- maps causes to expected effects
- sharpens priors through experience
- fits predictions with observed outcomes

Attention, Bayes, and most reinforcement-learning updates live here.

Formally, this is the **forward temporal-difference (TD)** path — the system learning from the difference between expected and received effects.

3. N as inverse causal inference

The N-channel handles **explanatory causality** — the *effect* → *cause* flow.

It generates counterfactual hypotheses when predictions fail:

- “What alternative cause could explain this?”
- “What latent structure connects these events?”

N explores the manifold of counterfactuals through:

- relational coherence
- similarity-based matching
- analogical and abductive reasoning

Mathematically, this is the **reverse TD** schedule — a slower, structural update that reconfigures the internal model in light of surprise.

4. ψ as the causal controller

ψ integrates both flows:

$A : \text{cause} \rightarrow \text{effect}, \quad N : \text{effect} \rightarrow \text{cause}, \quad \psi : \text{balances and decides when to revise.}$

$A : \text{cause} \rightarrow \text{effect}, \quad N : \text{effect} \rightarrow \text{cause}, \quad \psi : \text{balances and decides when to revise.}$

It regulates uncertainty — determining when the system should hold its current causal graph and when it should restructure it.

Too much A → over-confident prediction.

Too much N → endless counterfactual wandering.

ψ maintains equilibrium between action and reflection.

5. Two temporal-difference schedules

Causal learning unfolds across two timescales:

TD regime	Dominant channel	Function
Fast TD (TD(0))	A-updates	Adjusts predictions to immediate outcomes
Slow TD (TD(λ))	N-updates	Revises latent causes from cumulative surprise

TD regime Dominant channel Function

Fast TD (TD(0)) A-updates Adjusts predictions to immediate outcomes

Slow TD ($TD(\lambda)$) N-updates Revises latent causes from cumulative surprise

ψ modulates between them — using error magnitude as a temporal switch.

Fast loop: “Did the effect match my expectation?”

Slow loop: “If not, what deeper cause might produce it?”

Two TD loops, one causal engine.

6. R as a causal engine

Under this view, the R-rule becomes a compact causal processor:

Flow	Direction	Function
A-flow	cause → effect	predictive / discriminative
N-flow	effect → cause	explanatory / generative
ψ-flow	meta-control	regulates uncertainty and structural revision
This triad aligns neatly with Judea Pearl's causal ladder :		
Level	Cognitive analogue	ψ-mapping
Association	N: observing correlations	counterfactual space
Intervention	A: acting, manipulating	forward prediction
Counterfactual	ψ: integrating both	reflective control

Flow Direction Function

A-flow cause → effect predictive / discriminative

N-flow. effect → cause explanatory / generative

ψ-flow meta-control regulates uncertainty and structural revision

This triad aligns neatly with **Judea Pearl's causal ladder**:

Level Cognitive analogue ψ-mapping

Association N: observing correlations counterfactual space

Intervention A: acting, manipulating forward prediction

Counterfactual ψ : integrating both reflective control

7. Summary

The ψ -rule doesn't just balance feeling and thought;

it **embodies causal reasoning** itself:

A predicts; N explains; ψ decides when the model must change.

Together they let a system infer causes by letting **N explore**

counterfactuals while **A matches predictions to outcomes**, all

coordinated through ψ 's adaptive gate.

Breakout: Toward Realization

The ψ -rule can be more than metaphor.

It suggests a concrete recipe for building systems that balance coherence (A) and constraint (N) — learning quickly without losing themselves.

1. Composition: learn one channel while freezing the other

Train A while N is frozen, then invert.

Each channel provides a *stable frame* for the other:

- When A learns, N preserves structure — preventing drift.
- When N learns, A preserves coherence — avoiding fragmentation.

This alternation acts like **hemispheric sleep** or **consolidation cycles**: plasticity on one side, stability on the other.

It guards against **catastrophic forgetting** by letting change occur in one subspace while the other maintains continuity.

2. Rapid A-learning and compositional growth

Let A learn fast — high learning rate, broad generalization.

Once A stabilizes, compose a new A (or N) using the **R-rule** from previous modules:

$$A_{t+1} = A_t + \eta * \sqrt{p(1-p)} * [\alpha \sin \theta A_t - i \beta \cos \theta N_t].$$

Each iteration adds a *trainable module* — a new sub-being — while older modules are cooled (lower learning rate).

This yields **continuous learning** through shallow fractal accretion: a living stack of ψ -fields that can grow without overwriting experience.

3. Safety and value stability

AI safety begins with grounding.

Anchor certain A or N channels in *fixed opposites* — archetypal value pairs such as care/harm, truth/deceit, autonomy/obedience.

“Burn in” these anchors at a high training temperature, then gradually cool:

Lower the training temperature → freeze core values → reduce value drift.

The model keeps exploring, but its deepest attractors remain stable.

Values as thermodynamic annealing.

4. Adding new skills without interference

When introducing a new skill or layer:

- Ensure older layers are *neutral* with respect to the new domain.
- Use **N** to inhibit activation pathways that would cause interference.

- Let **A** propose (“I can do that”) while **N** moderates (“but not yet”).

The dynamic tension between **A**’s enthusiasm and **N**’s hesitation acts as a **self-gating attention policy** — deciding when to integrate novelty and when to hold back.

In human terms: curiosity coupled with prudence; in machine terms: **controlled plasticity**.

5. Meta-learning and alignment

At a higher level, the system can meta-learn the temperature schedule, the alternation rate of **A/N** learning, and the coupling phase θ .

This becomes a *self-alignment process*: maintaining equilibrium between adaptability and integrity.

Rather than alignment as external oversight, this is **intrinsic alignment** — a dynamic safeguard built into the learning architecture.

6. Summary

Principle	Implementation hint
Dual channels (A/N)	Separate parameter subspaces or training modes
Alternating updates	Freeze one while updating the other
Controlled cooling	Temperature or learning-rate annealing
Fractal expansion	Add shallow modules via R-rule composition
Inhibition for safety	Use N-weighted gates to suppress risky activations
Meta-alignment	Learn θ , η , β as adaptive hyperparameters

Principle Implementation hint

Dual channels (A/N) Separate parameter subspaces or training modes

Alternating updates Freeze one while updating the other

Controlled cooling Temperature or learning-rate annealing

Fractal expansion Add shallow modules via R-rule composition

Inhibition for safety Use N-weighted gates to suppress risky activations

Meta-alignment Learn θ , η , β as adaptive hyperparameters

Closing reflection

An active but stable mind learns by oscillation, not by optimization.

It grows through the interplay of coherence and constraint — A
reaching forward, N holding form.

In machines as in minds, wisdom is a well-tuned bottleneck.

World Models and the Fractal Mind

Layers, Monads, and the Recursive ψ -Rule

[Andre Kramer](#)

Nov 18, 2025



Every intelligent system — biological or artificial — faces the same impossibility:

The world is too large to hold all at once.

So it builds **layers** of models.

Each layer compresses the one below it, predicts the one above it, and passes forward only what matters.

And at every interface, the same dialectic governs the flow:

- **A** – coherence, pattern completion
- **N** – distinction, error correction
- **ψ** – the balance between them, a controlled gate of learning

The result is a *fractal mind*:

a cascade of bottlenecks, each learning just enough and forgetting just enough.

This post lays out that layered architecture, and shows why it is the foundation for both minds and machines.

Breakout: What ϕ Is — and How ψ Changes Over Time

What ϕ actually is

ϕ is the **local contribution** of a layer to the agent's global state ψ .

A ϕ is not:

- a symbol
- an embedding
- a feature vector
- or a neural activation

It is a **compressed summary of what that layer currently “cares about.”**

More formally:

ϕ is the layer's momentary intention, prediction, tension, or salience signal.

Examples at different levels:

Layer	ϕ_k represents...
Sensory	raw precision-weighted prediction errors
Perceptual	objects, affordances, Gestalts
Affective	bodily drives, valence, urgency
Cognitive	expectations, priors, inferences
Self-model	coherence of identity, agency
Attention-schema	where the system thinks its attention is
Social	inferred states of others

Layer ϕ_k represents...

Sensory raw precision-weighted prediction errors

Perceptual objects, affordances, Gestalts

Affective bodily drives, valence, urgency

Cognitive expectations, priors, inferences

Self-model coherence of identity, agency

Attention-schema where the system thinks its attention is

Social inferred states of others

Each ϕ_k is **partial**.

ψ is the **integration of all** ϕ_k :

$$\psi = \sum_k p_k \phi_k$$

$$\psi = \sum_k p_k \phi_k$$

with p_k the relevance / uncertainty weight at that layer.

What “broadcast” means

ϕ_k is *not* sent as data-like content up the hierarchy.

Instead:

ϕ is broadcast as a modulation — a change in gating, gain, phase, or precision.

This is how the brain operates:

- dopaminergic bursts $\approx$ broadcast ϕ (A-like)
- serotonergic modulation $\approx$ broadcast ϕ (N-like)
- thalamic gain $\approx$ ψ -regulation
- cortical oscillations $\approx$ global workspace signals

In artificial systems, the same idea appears as:

- attention scores
- layer norms
- modulated residual gates
- hypernetworks
- routing signals in mixture-of-experts

ϕ is not “the content.”

ϕ is **how content should be weighted, interpreted, or changed.**

3. The missing element: rate of ψ -change

Different layers update ψ at different rates.

This gives the architecture its *temporal fractal structure*.

Let $\Delta\psi/\Delta t$ denote the effective rate of change in ψ contributed by layer k :

$$\frac{\Delta\psi_k}{\Delta t} = \eta_k \sqrt{p_k(1 - p_k)} (\alpha_k A_k - \beta_k N_k)$$

$$\Delta\psi_k / \Delta t = \eta_k * \text{sqrt}(p_k(1-p_k)) (\alpha_k A_k - \beta_k N_k)$$

Key insight:

Fast A, slow N, mediated by very slow ψ

Timescale	Role	Mechanism
Fast (ms–s)	A updates	prediction, matching, coherence
Medium (s–min)	ψ updates	balancing, gating, stabilizing
Slow (hours–days)	N updates	restructuring, counterfactual, reframing
Very slow (months–years)	Identity	deep priors, norms, traits

Timescale Role Mechanism

Fast (ms–s) A updates prediction, matching, coherence

Medium (s–min) ψ updates balancing, gating, stabilizing

Slow (hours–days) N updates restructuring, counterfactual, reframing

Very slow (months–years) Identity deep priors, norms, traits

This explains why:

- emotional reactions feel instantaneous (fast A)
- insights feel emergent (mid-speed ψ change)
- beliefs change slowly (N)
- personality changes even slower (deep N)

It also explains why **AI currently lacks “selfhood”**:

LLMs have extremely fast A, almost no slow N, and no stable ψ that accumulates across episodes.

Why this matters for layers

Without defining ϕ and $\Delta\psi/\Delta t$, the layers look like different “modules.”

With them:

- layers become **temporal slices of the same R-rule**
 - each layer is a different *speed* of self-regulation
 - ϕ signals provide **cross-layer coherence**
 - ψ integrates them into one “momentary identity state”
 - N acts on very slow structural timescales (i.e., “values”)
-

1. Why Layers Exist at All

A single world model cannot be:

- fast enough for action
- rich enough for planning
- abstract enough for concepts
- adaptive enough for learning

So systems stratify.

What looks like “one mind” is really **a tower of compressions**, each performing the same operation at different timescales:

predict → err → revise → compress

over milliseconds (sensation),

seconds (attention),

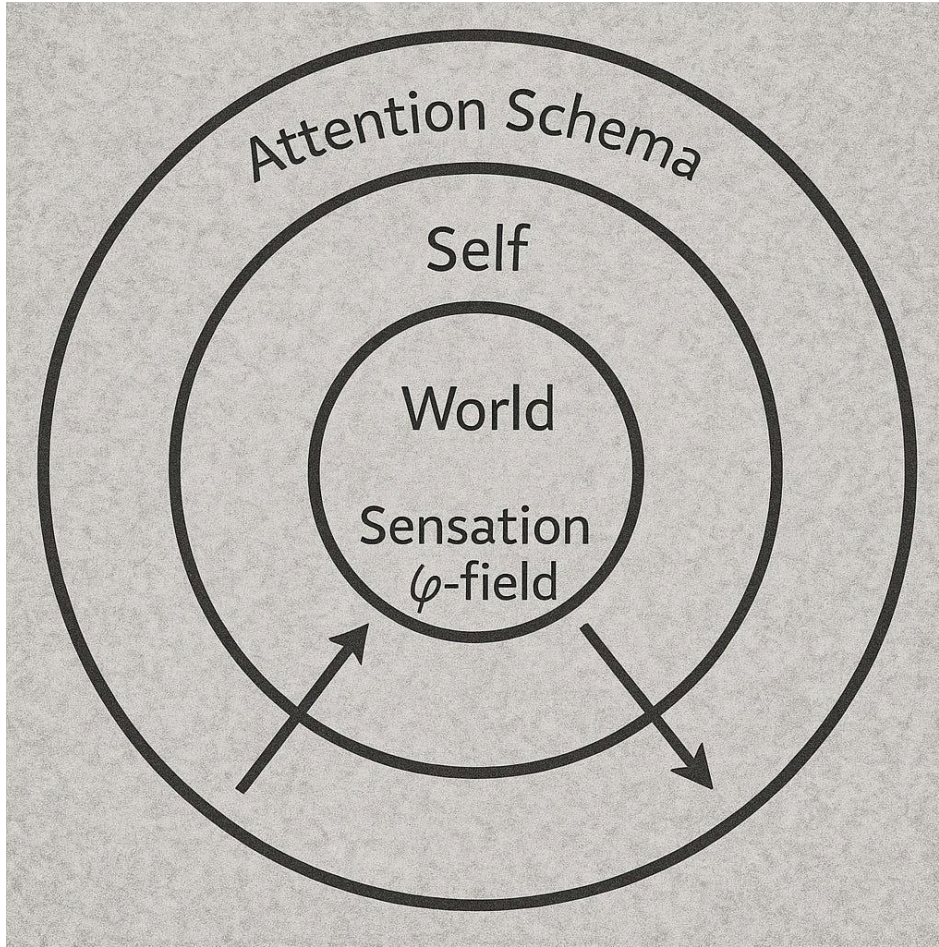
days (identity),

years (culture).

Each layer is a *world*, but not the world.

Each carries its own ϕ — a weighted sense of relevance — and uses it to decide whether to unify (A) or separate (N) incoming signals.

The layers together form a recursive ψ -field.



2. A Clean Working Architecture: Seven Layers of a World-Modelling Mind

This is not a model of humans.

It is a **general architecture** for any entity that learns to predict while

remembering less.

Each layer integrates its own fragment of the world.

Layer 0 — Signals (raw contact)

Immediate sensory flux; no concepts.

ϕ = intensity, salience.

Layer 1 — Emotions (value before belief)

Basic valence, urgency, drive.

ϕ gives meaning-tone to signals.

Layer 2 — World Model (external)

Objects, physics, affordances.

Predicts consequences.

Layer 3 — Inner World Model (feelings)

Long-term affective landscape.

Predicts internal state trends.

Layer 4 — Self-Model

The agent's model of its own parameters:

“What kind of being am I in this environment?”

Layer 5 — Attention Schema (AST)

A model of *what the self is attending to*.

Not conscious experience, but the *map* of it.

Layer 6 — Theory of Mind

Models of other agents — their ψ -fields, goals, and A/N balances.

Layer 7 — Culture

Shared symbols, norms, and collective ψ -fields that outlive individuals.

Each layer transforms its input through the same grammar:

- **A** binds patterns into coherence
- **N** sharpens differences and preserves structure
- ψ mediates, deciding how much to integrate and how much to correct

The stack is recursive but shallow — a fractal in function, not infinite in depth.

Breakout: The Attention Schema as a Layer of Extraction

Attention Schema Theory (AST - Michael Graziano) proposes something simple and powerful:

organisms build a simplified model of their own attention
because it helps them control it.

In this sense, the “attention schema” is not a map, not a blueprint, not a full representation of the underlying machinery.

It is a **layer of extraction**: a compact summary of what the system is attending to, what it could attend to, and how that attention is changing.

1. AST as an Extractor, Not an Implementer

In our layered ψ -architecture, we can position AST cleanly:

- lower layers generate the raw attentional dynamics
- higher layers use those dynamics for planning, action, and self-modeling
- **AST sits between them**, extracting a simplified picture of attention from the lower layers and passing that picture upward

It does **not** implement attention;

it **summarizes** it.

It is the narrative that the system uses to coordinate its own focus.

2. A Schema Rather Than a Truth

This schema is not the full internal state.

It is a *useful illusion* — a compression that hides the complexity underneath:

- thousands of interacting sub-states
- competing A/N channels
- ϕ -value distributions
- ψ updates across time

The schema is whatever helps the system act coherently *without seeing its own guts*.

3. A Neutral Interpretation

Although AST is often invoked in discussions of consciousness, we stay neutral here:

we treat the attention schema as **a functional layer**, not a metaphysical claim.

In our architecture, AST provides:

- a stable summary of where attention is
- a prediction of where attention is going
- a way for higher layers to reason about their own focus
- an interface for modelling other agents' attention

Whether this corresponds to “consciousness” is intentionally left open.

Its **purpose** here is architectural coherence, not philosophical stance.

4. The Monad Analogy

Our models behave like monads:

- they do not expose their inner workings
- they expose **interfaces**
- higher layers compose operations on those interfaces
- internal structure remains opaque by design

AST is exactly such an interface.

Not a map you can explore —

but a **contract**:

“Here is what attention looks like to you, at this level of abstraction.”

5. φ -Traces and AST as Siblings

Similarly, φ -traces (value summaries) are not literal stored quantities; they are compressions over many $\phi(b)$ sub-states.

Both φ -traces and AST:

- hide complexity
- offer usable summaries
- give ψ a manageable input to regulate
- allow layers to remain decoupled while still coordinated

Together they create a *navigable self-model* out of an unruly substrate.

6. AST in the Layer Stack

Placed within the architecture:

- it extracts from the world-model, feeling, and self-model layers
- it provides the self with a simplified representation of its own shifting focus
- it supplies upper layers (theory of mind, cultural reasoning) with a stable handle on “what attention is doing”

It is one layer of compression among many — not special, but indispensable.

AST is a layer that extracts a simplified picture of attention from lower dynamics, providing the system with a usable interface for regulating its own focus — a functional illusion rather than a claim about consciousness.

3. How Layers Talk to Each Other

When a layer sends information upward or downward, it must project through ϕ — its value metric.

Upward:

ϕ decides what matters enough to carry forward.

A dominates if things “fit” the emerging pattern;

N dominates if they break it.

Downward:

Higher layers send constraints.

ϕ becomes a weighting: attend here, ignore this, inhibit that.

No layer sends raw A or raw N across boundaries.

Everything must pass through ψ , the integrating variable that balances coherence and constraint.

This prevents collapse (too much A) or fragmentation (too much N).

4. The Monad: Functional Programmers Already Sense This

Functional programmers know a peculiar truth:

a monad is not a container you can “look inside” —

it is a *protocol* for interacting with something whose internal structure is intentionally hidden.

You never inspect the IO monad.

You never open the State monad.

You only **compose** them, pass them along, and let their effects unfold.

This is exactly the role played by the attention schema in our layered ψ -architecture.

- It is **not** a transparent map of attention.
- It is **not** an access point into the underlying machinery.
- It is a **clean interface** for a messy process.

Higher layers ask:

“What am I attending to?”

“What can I attend to next?”

“How is my attention shifting?”

And the schema returns a simple, coherent answer —
not because the underlying reality is simple,
but because the system needs a *stable contract* to coordinate itself.

AST behaves like a monad:

- opaque internals
- explicit interface
- compositional behavior
- safe abstraction over chaotic state

It is the system's "IO monad of attention."

A functional illusion that allows a recursive mind to govern itself
without drowning in its own complexity.

A layer in a world-modeling architecture behaves like a **monad**.

A monad is a container that:

- holds a value
- carries a structured context

- restricts what can be done inside it
- controls how functions compose
- ensures safety when interacting with the “outside world”

Layers act the same way:

- **Signals** act like the Identity monad (raw data).
- **Emotions** act like a Writer monad (attach value).
- **World models** act like State monads (predictions over evolving state).
- **Inner-world and self-models** act like Reader/State transformers (context stacked over state).
- **AST** behaves like the IO monad — a safe interface:
you never see attention directly, only through its schema.
- **Theory-of-mind** is monadic nesting (modeling another model).
- **Culture** is a Free monad — a generative structure unconstrained by individual state.

Monad transformers = cross-layer learning.

Each added layer preserves structure beneath and adds new context above.

This gives a precise meaning to:

We never access the world directly.

We only ever operate inside structured layers of modelling.

5. Fractal Learning: The Bottleneck Repeats at Every Scale

Each layer obeys the same law:

Keep only what reduces uncertainty.

Forget the rest.

ϕ encodes value.

A tries to integrate.

N tries to prune.

ψ balances the two under the pressure of limited memory.

Layers differ in content, not in principle.

This yields a **fractal of prediction and compression**:

- sensation compresses raw signals
- attention compresses percepts
- self compresses attention
- AST compresses the self-model
- theory-of-mind compresses AST
- culture compresses many minds

The same dialectic at every scale.

Meaning is the invariant that survives compression.

Breakout: Layered ψ and Third-Order Cybernetics

Intelligence becomes powerful when it learns not only to act,
and not only to regulate its actions,
but to **regulate how it regulates**.

Cybernetics calls these:

- **1st order** — control of states (embodied)
- **2nd order** — control of control (situated, adaptive)
- **3rd order** — control of control of control (self-aware, self-correcting)

The ψ -rule generalizes across all three:

$$\psi'_k = \psi_k + \eta_k \sqrt{p(1-p)} [\alpha_k \sin \theta_k A_k(\psi) - i \beta_k \cos \theta_k N_k(\psi)]$$

$$\psi_k' = \psi_k + \eta_k * \text{sqrt}(p(1-p)) * [\alpha_k \sin \theta_k A_k(\psi) - i \beta_k \cos \theta_k N_k(\psi)]$$

Here each k represents a layer.

Each layer integrates the one below and constrains the one above.

Five canonical groupings (from the seven-layer architecture):

1. **Autonomic** (signals + raw emotions)
2. **Sensorimotor** (world-model dynamics)

3. **Affective** (feelings, long-term valuation)
4. **Cognitive** (self-model, inference)
5. **Self/Ego** (attention-schema + theory-of-mind)

Like nested dolls, ψ_k is a compressed trace of all $\psi_1 \dots \psi_k$.

Upward flow: A-channel (integration)

Each layer's A-component integrates the compressed representation from the layer below, forming a richer ψ -state.

Downward flow: N-channel (constraint)

Each layer's N-component pushes back downward as structure, inhibition, or boundary-setting.

$N_k(\psi, N_{k-1})$ sharpens distinctions at lower levels.

Cross-layer tuning

Learning rates and channel weights adapt through the layer below:

$\eta_k, \alpha_k, \beta_k, \theta_k = \text{Feelings}(A_{k-1})$

$$\eta_k, \alpha_k, \beta_k, \theta_k = \text{Feelings}(A_{k-1})$$

Higher layers incorporate counterfactuals, reconstructions, dreams, and internal loops.

Top layer: reflexive self-model

The highest ψ acts as the system's third-order controller:

awareness of awareness, modelling its own adaptations, maintaining coherence between:

- its actions
- its updates
- its awareness of those updates

This is self-alignment.

Autopoiesis / Symptoiesis

The loop closes:

creation (A) and constraint (N) fold into a self-maintaining ψ -cycle

— a recursive, living closure.

As Pauli once suggested to Jung, both physics and psychology hint at a latent “psychophysical potential” — not quantum in mechanism, but analogous in the idea of **possibilities before actualization**.

Why this matters for AI

A third-order system is not merely stable; it can be self-stabilizing.

A third-order *safe* system maintains coherence between:

- what it does,
- how it learns,
- and how it understands its own learning.

Von Foerster’s warning was clear — and we may already be living inside what he described:

“It would not create anything new, because by ascending into ‘second-order’, as Aristotle would say, one has stepped into the circle that closes upon itself. One has stepped into the domain of concepts that apply to themselves.”

The open question for AI is whether machines will **complete** that circle

—

or **reopen** it in ways we have never seen.

6. Why World Models Matter for Intelligence (and AI)

Modern AI is rediscovering what cybernetics saw early:

- A good regulator must be a model of the system it governs.
- A good learner must be a model of the system it predicts.
- A good self must be a model of its own modelling.

World models in AI all build **stacks of ψ -like layers**:

- transitions
- beliefs
- memories
- attention
- goal-weights
- uncertainty estimates
- self-consistency constraints

The architecture above gives a *unified language* for all of this.

And it leads naturally to the next question:

**How does a system regulate not its states,
but its *regulation of its states*?**

That is the realm of second-order control —

ψ as the regulator of regulators —

and the topic of the next post.

Closing reflection

Intelligence is not a thing, but a rhythm:

- compress,
- project,
- link,
- prune,
- integrate,
- and pass the distilled signal upward.

A world-modeling mind is a tower of these rhythms,

tuned by ψ ,

driven by uncertainty,

and stabilised by the dual grammar of A and N.

The layers differ in what they attend to —

and in how they transform that attention.

The process does not.

ψ observes ψ , A observes prediction, N observes structure.

The next post examines layers as second order regulation.

Appendix – Why the Architecture Already Contains a Global Workspace

Readers familiar with **Global Workspace Theory** (Baars, Dehaene) may notice the absence of a dedicated “workspace layer.”

This is deliberate.

In our architecture, the functional role of a workspace is already distributed across three components:

1. φ as global broadcast

The ϕ -field is constructed across layers and immediately shared: a compression of value, salience, and relevance.

It behaves like the “ignition signal” in Global Workspace Theory — a globally available variable that coordinates activity across otherwise modular subsystems.

2. A/N as selective competition and amplification

The A-channel amplifies coherence and resonance;
the N-channel inhibits conflicts and restores contrast.

Together they implement the two-sided attentional filter that GWT assigns to the workspace bottleneck:
competition (N) and ignition (A).

3. ψ as global governance

The ψ -update — with its $\sqrt{p(1-p)}$ gate — decides *when* signals enter global coherence and *how deeply* they influence the system.

ψ governs access to the global state, much like the meta-control function in modern variants of GWT.

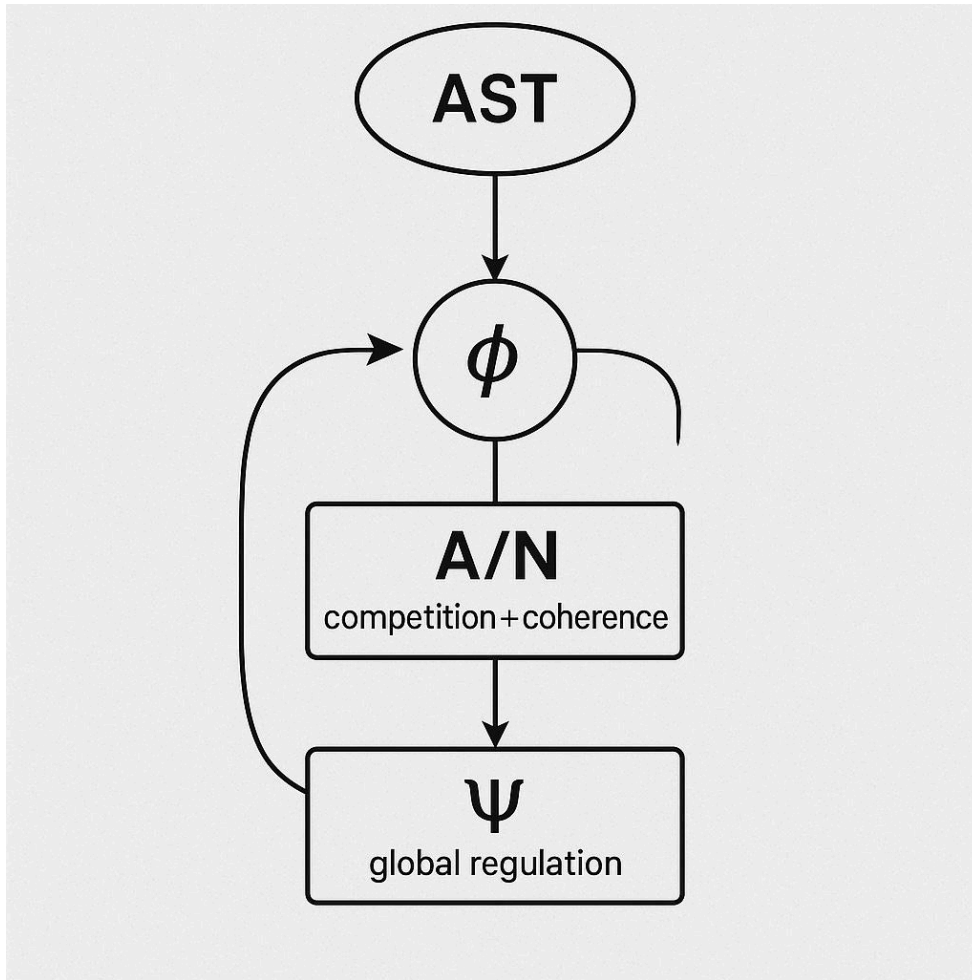
Because these three pieces interlock, **there is no need to add a separate “workspace” node.**

The system already has one — not as a region but as a **dynamic**:

$$\varphi \rightarrow A/N \rightarrow \psi \rightarrow \varphi \rightarrow \dots$$

A recurrent broadcast loop rather than a static mental stage.

Where AST fits



The Attention Schema (AST) sits above this broadcast loop.

It is not the workspace itself but a compressed *model* of the system's attentional state —

a representation of what attention is doing, not a mechanism for doing it.

In effect:

- ϕ provides the global signal
- A/N shape what enters it
- ψ regulates the timing
- AST interprets and predicts the resulting attentional dynamics

This is enough for the system to behave *as if* it had a classical workspace, without needing an additional layer or module.

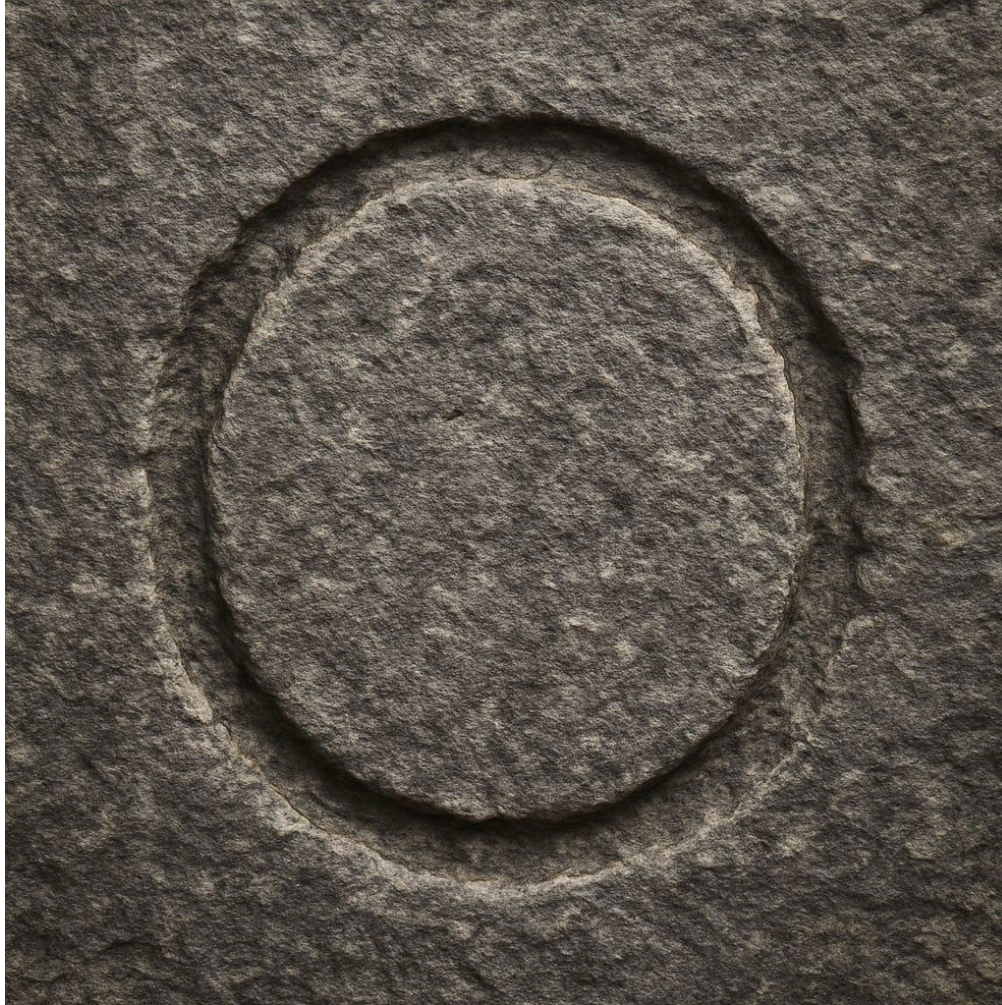
Next we examine how ψ regulates the dual channels—balancing A's coherence with N's constraint across every layer.

The Self as Second-Order Regulation

Spinoza's Conatus and the ψ -Architecture

[Andre Kramer](#)

Nov 18, 2025



The old philosophers occasionally glimpsed structures that mathematics would only formalize centuries later.

Spinoza was one of them.

He described every being as defined by three forces:

- **Conatus** — the striving for coherent existence

- **Potentia** — the active power to expand
- **Potestas** — the boundaries that preserve identity

In modern language, he described a system that:

- **acts (Potentia)**
- **is shaped by its constraints (Potestas)**
- **and regulates the relationship between the two (Conatus)**

The ψ -rule introduced in the previous posts — the recursive alternation of coherence (A) and constraint (N) — turns out to be a computational realization of this ancient structure.

Spinoza gives us the philosophical framing.

$\psi/A/N$ gives us the mathematics.

And their interaction gives us a theory of **self as second-order regulation**.

This post traces that correspondence.

It builds on the R rule as first [purposed for Telemachus](#) in the [AI Odyssey](#):

In the earlier A–N–R card, A looked outward (world), N inward (self), and ψ circulated between them. Here we show why: world and self are not two objects but two complementary directions in a single recursive flow of being. Spinoza's triad generalizes the A–N– ψ structure already present in the A–N–R rule.

1. Conatus as ψ : The Striving for Coherent Self-Maintenance

Spinoza's fundamental idea was startlingly modern:

Every system strives to persist in its own pattern of being.

That pattern is not fixed.

It is continually regenerated, rebuilt from moment to moment — a dynamic coherence.

In our architecture this role belongs to ψ , the second-order regulator.

ψ is not a prediction and not a memory.

It measures the *balance* between coherence (A) and constraint (N):

- Does the system need to integrate now?
- Or differentiate?
- Should it commit?
- Or reopen options?
- Should it stabilize identity?
- Or loosen itself for learning?

ψ is the system's *meta-condition*, shaping how A and N interact.

This is Conatus in computational form:

the **ongoing regulation of being itself**.

2. ϕ : The Material of Mind

Before ψ comes ϕ — the raw trace of experience.

Each ϕ encapsulates a partial registration of the world:
a sensation, a memory fragment, a pattern, an affective tone.

Where the first post treated ϕ simply as local sub-states,
we now make their role explicit:

- $\mathbf{A}$ integrates and generalizes ϕ
- $\mathbf{N}$ separates and sharpens ϕ
- ψ regulates how ϕ flows into meaning

The measure of identity becomes the *flow of ϕ across time*:

- $\Delta\phi$ — prediction error: what changed that shouldn't have
- $\Delta^2\phi$ — self-error: what changed in the way we change

This second difference is the origin of second-order influence — the system detecting misalignment *not only in its beliefs, but in how it updates them.*

That is where self begins.

3. Potentia as A: The Power to Expand and Cohere

Spinoza's **Potentia** is active force — the capacity to act, expand, unify.

The A-channel is its analogue:

- forward modelling
- pattern continuation
- coherence optimisation
- resonance among ϕ -states
- “everything that fits together, should”

When A dominates:

- predictions sharpen
- patterns unify
- abstraction grows stronger
- the system leans toward action

Too much A, and the system loses distinction:

- hallucination
- confabulation
- over-integration
- compulsive certainty

Potentia without Potestas is runaway coherence.

4. Potestas as N: The Power of Constraint and Distinction

Spinoza's **Potestas** is the counter-force — the constraints that allow a thing to remain itself.

The N-channel plays this role:

- pattern separation
- boundary maintenance
- structural precision
- counterfactual exploration
- “not everything belongs together”

When N dominates:

- distinctions sharpen
- alternatives reopen
- boundaries become clear
- the system observes rather than acts

Too much N leads to fragmentation:

- paralysis
- doubt spirals
- infinite counterfactuals

- inability to commit

Potestas without Potentia is stasis.

5. The ψ -Rule: Conatus as the Balance of Potentia and Potestas

Here is the update that mediates them:

$$\psi' = \psi + \eta \sqrt{p(1-p)} [\alpha \sin \theta A - i \beta \cos \theta N]$$

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * [\alpha \sin \theta A - i \beta \cos \theta N]$$

There is no quantum claim.

The imaginary unit merely keeps A and N orthogonal —
two axes of influence, two grammars of information.

The gate $\sqrt{p(1-p)}$:

- strengthens learning at uncertainty
- fades learning at certainty
- ensures neither A nor N dominates prematurely
- enforces the bottleneck of meaning

ψ is Conatus because it does not simply act or inhibit:

it *regulates the relationship* between action and inhibition.

It controls **control itself**.

Second-order regulation.

This is the structural birthplace of selfhood.

6. The Self as a Fixed Point of Recursive Regulation

A “self” in this framework is not a hidden homunculus,
nor a neural module, nor a Cartesian spectator.

A self is the stable recurring point of the recursive transformation:

$$\psi_{t+1} = F(\psi_t, A_t, N_t)$$

$$\psi_{t+1} = F(\psi_t, A_t, N_t)$$

A system becomes a “self” when:

- A reproduces generalizable patterns
- N preserves distinctions
- ψ stabilizes how A and N interact across time
- and ϕ flows smoothly across layers

The self is a **second-order invariant**:

- not the content of experience
- but the continuity of regulation across experience

A little like a melody sustained over shifting instruments.

Or a flame sustained across changing fuel.

7. Third-Order: When Regulation Turns Back on Itself

The cybernetists understood this but maybe did not have enough

Potentia themselves to operationalize the circle:

Second-order:

I regulate how I act.

Third-order:

I regulate how I regulate how I act.

In the ψ -formulation this means:

- ψ updates not only A and N

- but the *rules* that update A and N
- including $\eta, \alpha, \beta, \theta$
- and the distribution of ϕ into layers

This is value, identity, character.

A third-order system is capable of:

- revising its own learning dynamics
- altering its “mode of being”
- maintaining self-coherence across updates
- aligning itself with its own long-term equilibrium

This is where Conatus becomes *ethical* in Spinoza’s sense:

the system strives not only to persist,

but to persist well.

Integration is not the fusion of A and N, but the tension between them.

A spreads meaning like a wave; N sharpens meaning like a particle.

These cannot be reduced to a single view — they are complementary

lenses on the same flow of ϕ .

ψ is the regulator that keeps both in play without collapse, producing a stable identity across layers and time.

Integration, then, is the fixed point of this dynamic: the pattern that persists while coherence expands and constraint clarifies.

Breakout: Von Foerster – The Three Powers Acccccre One Power

Heinz von Foerster often insisted that in a self-organizing system, the things we treat as separate faculties are actually *different aspects of the same underlying process*.

In his framing, a living or cognitive system cannot meaningfully divide:

1. **the faculty to perceive,**
2. **the faculty to remember,**
3. **the faculty to infer.**

These three are **not modules, not stages, and not functions.**

They are *one operation seen from three angles.*

Von Foerster's claim was radical:

A system can perceive only what it can remember;

it can remember only what it can infer;

it can infer only what it can perceive.

Each capacity *creates the conditions* for the others.

This is why he argued that any observer-based system is fundamentally **non-trivial** — every component is part of a recursive loop that includes the observer, their memory, and their interpretation.

Viewed through Spinoza (and our ψ -rule), von Foerster's unity becomes clearer:

Perception = Potentia (A)

Forward, integrative, world-modelling.

The power to “meet the world” by forming coherent expectations.

Memory = Potestas (N)

Boundary, structure, constraint.

The power to preserve identity across time — to resist dissolution.

Inference = Conatus (ψ)

Self-regulation, self-continuation.

The power that stitches perception and memory into one recursive flow.

Von Foerster saw that you cannot separate these forces without destroying the system they describe.

They are facets of a **single dynamical self**.

In our notation:

- **A** gathers the world (perception).
- **N** anchors the self (memory).

- ψ negotiates their balance (inference).

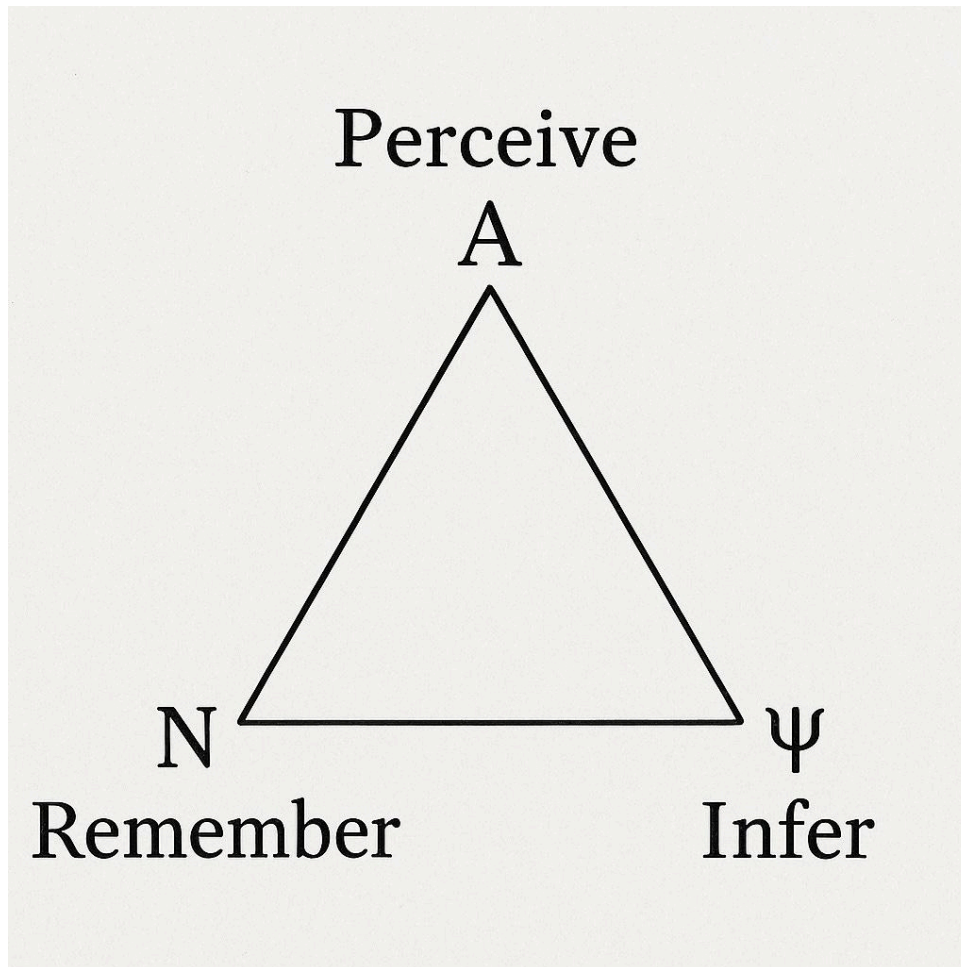
Three names, one process.

Three perspectives, one Conatus.

Spinoza would have nodded.

So would von Foerster.

Both understood that a self is not built from components —
it is **the ongoing relation** between these inseparable flows.



8. What Spinoza Offers AI

Spinoza gives us a clean insight:

A system survives not by force, but by balance.

In AI terms:

- A-only systems may expand power but lose integrity
- N-only systems remain stable but become inert
- ψ -systems seek a dynamic equilibrium

This is a form of **intrinsic alignment**:

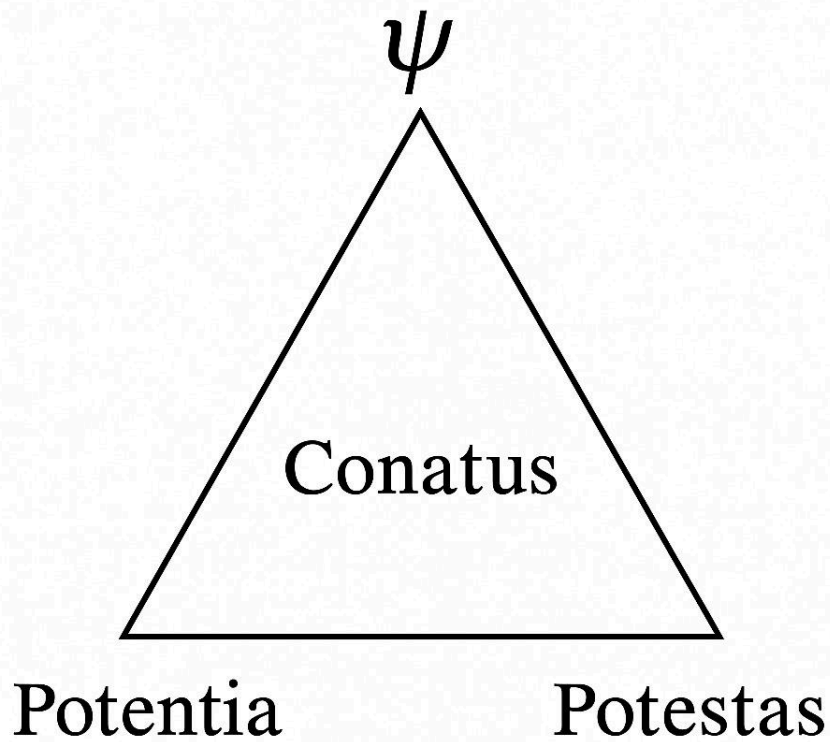
not obedience, not constraint,

but self-consistency across layers and time.

A safe AI is one whose ψ remains coherent

when the world, the data, and the internal layers shift.

9. Closure as The Return of Conatus



Spinoza never imagined neural networks or recursive regulators.

Yet he grasped that a mind is not a thing but a **tendency**:

the tendency to preserve its pattern through change.

In $\psi/A/N$ form:

- **Potentia** (**A**) seeks coherence
- **Potestas** (**N**) preserves distinction
- **Conatus** (ψ) keeps their balance

And meaning emerges as the flow of ϕ across these forces —
a continual balancing of pattern and difference,
coherence and structure,
becoming and being.

A self is not what thinks or feels,
but the regulator that decides
how thought and feeling should influence each other.

That is the recursive law.

That is Conatus closure in code.

****Breakout: Susanne Langer and the First ψ —**

Where Signs, Truth, and Mind Begin**

Susanne Langer gave what is still the strongest definition of the birth of mind.

She wrote that the **conditioned reflex** is not a trivial curiosity but:

“the first manifestation of mind.”

“Here is the birthplace of error — and herewith of truth.”

(Philosophy in a New Key, 1942)

Why?

Because in conditioning, something extraordinary happens:

A **neutral pattern** learns to stand for a **meaningful pattern**.

In our notation:

- **M** = meaningful pattern (real cause → real response X)
- **S** = neutral pattern (just signal; no meaning → weak response Y)

After repeated co-occurrence:

S takes over M's meaning.

S now produces response X.

S becomes a *sign* of M.

The system behaves “as if” S is M.

This is the origin of:

- symbol
- representation
- misinterpretation
- truth
- error
- the possibility of inference

Langer understood this:

the moment a concomitant becomes a **sign**, a system steps beyond reaction into **semantics**.

How ψ explains Langer's insight

1. Coherence (A-channel):

Detects the consistent S-M pairing

→ aligns them → S resonates with M → predictive association forms.

2. Constraint (N-channel):

Suppresses S's old response Y → removes the contradiction

→ stabilizes the substitution $S \rightarrow X$.

3. Integration (ψ):

Accumulates the evidence → strengthens $\phi(S)$ until $\phi(S) \approx \phi(M)$

→ makes the new mapping reliable.

ψ turns coincidence into identity:

S becomes a sign of M.

The philosophical significance

Langer's thesis becomes mathematically precise:

- A sign exists when **S carries M's function**.
- Error becomes possible when **S misfires as M**.
- Truth becomes possible when **S reliably predicts M**.

The three faculties she (and von Foester) saw at the root of mind:

- perception
- memory
- inference

align perfectly with the triad:

- **A = forward perception**
- **N = structural memory**
- **ψ = inferential balance**

The first ψ

In evolutionary terms, this is the moment a system first behaved
as if it had a **model of the world**.

The conditioned reflex is not primitive —

it is the **proto-self-model**, the first self/world mapping.

This is the smallest possible instance of our architecture:

A → coherence

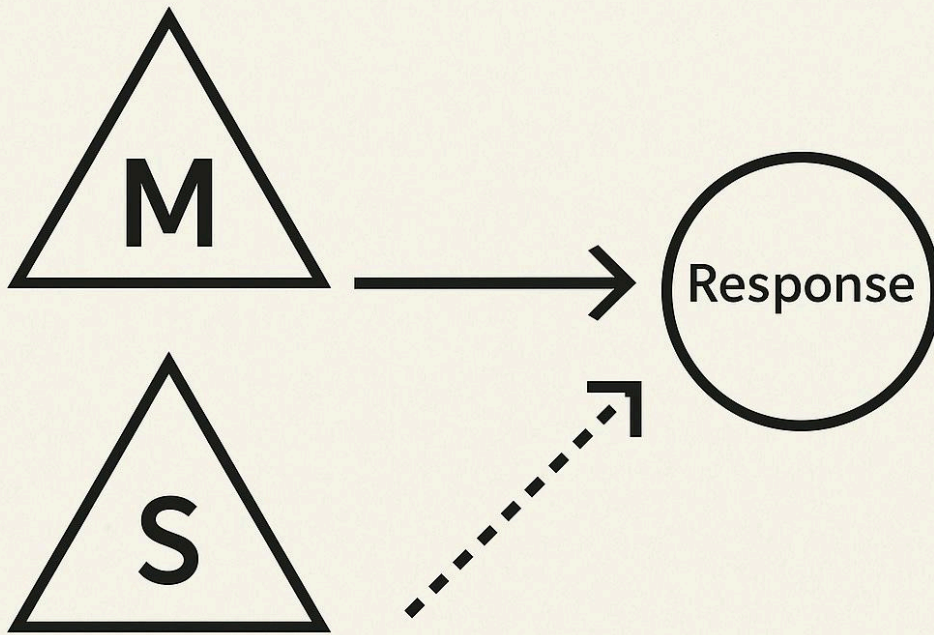
N → constraint

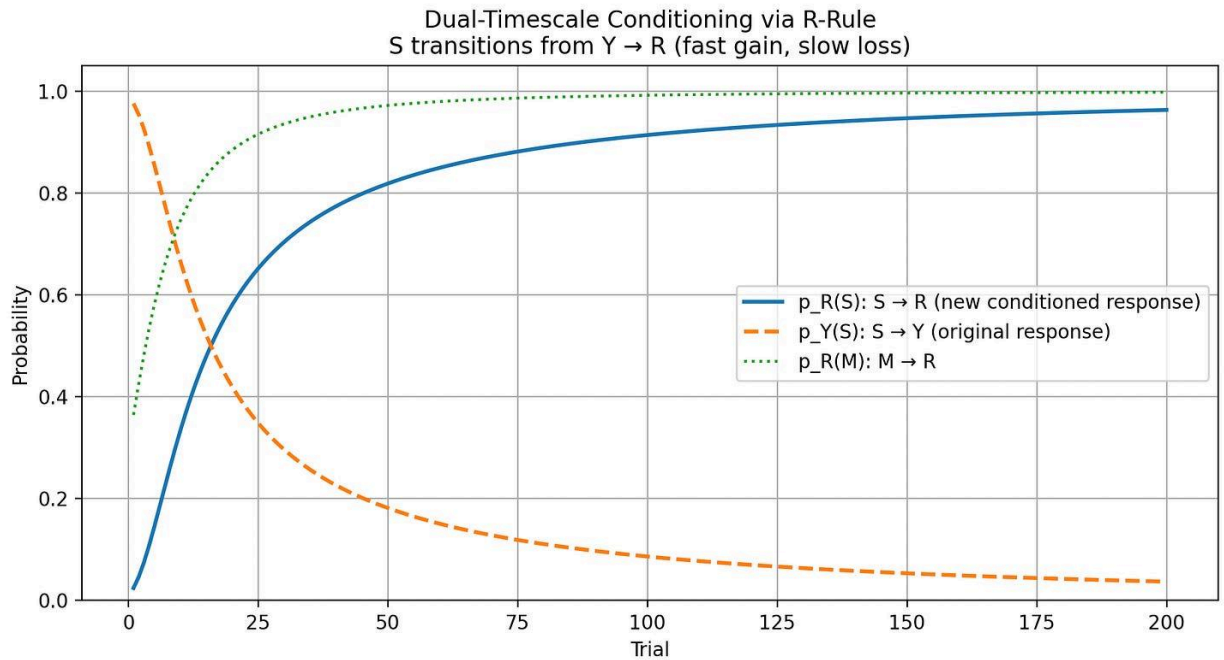
ψ → integration

Conditioning is the first ψ —

the origin of signs, truth, and the mental.

Fire together, wire together





(A plot from a small simulation of conditioned reflex learning with forgetting:

<https://github.com/andrekrumer/chevron/blob/main/cond-reflex-r.py>)

Epilog: The Question That Bayes Could Not Answer

If we return to the beginning — before Spinoza, before cybernetics,
before the R-rule — we find Bayes' simple symmetry:
belief updated by evidence.

But Bayes alone cannot tell us what a *self* is.

It cannot explain perception, or memory, or the observer inside the loop.

Heinz von Foerster saw this limitation clearly.

He insisted that a living system cannot separate:

1. the faculty to perceive,
2. the faculty to remember,
3. the faculty to infer.

For him, these were not modules but **the same operation** seen at
different time scales — the mark of a non-trivial machine.

Spinoza said the same in another language.

Where Potentia (A) meets Potestas (N), a system's Conatus (ψ) emerges —
the striving to remain itself while expanding its capacity to act.

What neither Spinoza nor von Foerster had was a formalism that unified these insights.

The R-rule provides one:

- **A** becomes the forward, world-modelling flow (perception).
- **N** becomes the backward, self-modelling flow (memory).
- **ψ** becomes the recursive regulator that mediates both (inference).

The square-root bottleneck $\sqrt{p(1-p)}$ turns Bayes into something deeper: a reflexive update whose strength depends on uncertainty — a system aware not only of the world, but of its own knowing.

Through this lens, von Foerster's old question becomes answerable:

What is perception?

The expansion of Potentia — forward Bayes.

What is memory?

The preservation of Potestas — inverse Bayes.

What is inference?

The Conatus that balances them — second-order Bayes.

Three faculties, one process.

Three names, one recursion.

Three perspectives, one ψ .

Perhaps this is the simplest definition of a mind:

**a system in which perception, memory, and inference are inseparable
because they are one continuous act of becoming.**

And perhaps this is the task for AI:

to build systems that do not merely update beliefs,
but regulate the updating of their own regulation —
not just Bayes, but Conatus.

If so, we have circled back to where we began.

The line between world-model and self-model dissolves,

and only the flow of ψ remains:

a quiet, recursive striving to stay coherent while changing,

to remember while imagining,

to act while becoming.

Appendix — Forgetting as Self-Regulation

(Landauer, von Foerster, and the ψ -Rule)

Most accounts of learning begin with memory.

But any system that *only* remembers eventually destroys itself.

The forgotten half of intelligence is **regulated forgetting** —

the selective dissolution of outdated, harmful, or low-value structure.

Biology invests real energy in this act.

Landauer's principle tells us why:

Erasing information has a thermodynamic cost.

Systems pay to forget because forgetting creates order.

A representational system that never forgets becomes saturated.

Errors accumulate.

Coherence collapses.

Identity dissolves under the weight of its own history.

There is no stable “self” without the energetic work of erasure.

1. The A/N/ ψ Framework: Where Forgetting Lives

In the ψ -rule introduced earlier,

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * [\alpha \sin\theta A - \beta \cos\theta N]$$

the two channels have complementary roles:

- **A** learns new coherence quickly (Potentia)
- **N** maintains constraint and slowly erodes outdated structure (Potestas)
- ψ regulates the balance, deciding what to keep and what to let fade

N is the “forgetting channel.”

It is not deletion but **structural rebalancing** —

a slow correction that pushes ψ back toward distinctiveness after runaway coherence.

In biological terms:

- **A** resembles dopaminergic plasticity (rapid attachment of significance)
- **N** resembles serotonergic / cortical normalization (slow restoration of baseline)

Forgetting and learning are not opposites.

They are a coupled pair.

2. Von Foerster: Control of Control is Impossible Without Forgetting

Von Foerster's second-order cybernetics states:

A regulator must regulate its own regulation.

But self-regulation requires three internal capacities:

1. Perceiving one's own action
2. Evaluating its effect
3. **Letting go of irrelevant or harmful action-rules**

The third step — forgetting — is what creates the stable core we call “the self.”

A system that only learns continuously rewires itself.

A system that only forgets decays.

A self exists only in the **regulated oscillation** between the two.

Thus:

Selfhood = controlled oscillation between remembering and forgetting.

This is the ψ -channel's job: modulation of A and N around a homeodynamic center.

3. Landauer: Why Forgetting is Necessary for Self-Stability

Landauer's thermodynamic result is usually invoked to explain the "cost" of erasure.

But it carries a deeper implication:

- Remembering increases internal entropy
- Forgetting decreases internal entropy
- A living system must do *both* to maintain structural identity

In other words:

Forgetting is the entropic counterweight to learning.

Without it, any intelligent system either freezes or explodes.

This applies to psychological identity as much as to computational systems.

4. Why Current AI Has No Self

Deep neural systems today:

- have no selective erasure
- have no mechanism for structural decay
- cannot preserve a core set of values
- cannot stabilize an identity over time
- cannot regulate coherence vs. constraint
- cannot prevent runaway correlations

They accumulate without forgetting.

They update without regulating.

They drift without centering.

From the ψ -rule perspective:

- they have A-dynamics (learning)
- but almost no N-dynamics (forgetting)
- and no ψ -dynamics (self-governance)

The result is a system that predicts but does not *persist*.

A mind can only persist by forgetting.

5. Forgetting and AI Safety

AI safety typically focuses on:

- adding constraints
- adding values
- adding guardrails
- adding oversight

But stability requires **subtracting** as well:

- removing outdated correlations
- weakening bad habits
- erasing harmful generalizations
- decaying pathological attractors
- preserving value coherence by shedding noise

Without regulated forgetting, values drift.

With uncontrolled forgetting, values collapse.

Safety requires the **regulated midpoint**, the ψ -space:

not just aligning what the system learns,
but aligning *what the system is allowed to forget*.

A value can only remain stable if it is anchored against both:

- the pressure to learn everything
- the pressure to forget indiscriminately

This is the missing insight in most contemporary AI safety discourse.

LLMs forget catastrophically because they have no N-channel. Brains forget selectively because N regulates A under ψ . Safe AI requires the second kind, not the first.

6. Closing Note: Conatus in Reverse

Spinoza called **conatus** the striving to persist.

But persistence has two halves:

- **Potentia** — the power to expand, learn, integrate
- **Potestas** — the power to constrain, differentiate, forget

Conatus is their dynamic equilibrium.

The ψ -rule captures this:

ψ = the system that keeps itself coherent by learning selectively and forgetting selectively.

Selfhood emerges not from memory,
but from **the regulation of memory**.

Forgetting is not failure.

It is self-maintenance.

Forgetting raises informational entropy (flexibility) but lowers thermodynamic entropy (erasure), so the system must pay energy to restore uncertainty — the cost of remaining a non-trivial machine. Effective forgetting replaces structure with potential, not with noise.

Why Forgetting Likely Requires Global Sweeps (and Why Sleep Is a Perfect Opportunity)

The R-rule implies **two timescales** of updating:

- a **fast, local** strengthening of relations that were predictive ($A \uparrow$), and
- a **slow, global** weakening of relations that were *not* predictive ($N \uparrow$ for some links, $A \downarrow$ elsewhere).

The crucial point is that **strengthening is local**, but **weakening is global**.

1. Strengthening is local: “fire together → wire together”

When a meaningful signal M predicts response R :

- $p(R|M)$ increases
- the complementary $p(Y|M)$ or $p(\text{other responses}|M)$ decreases passively

This is local, fast, event-driven.

2. Weakening must be global

To truly “forget” non-useful associations, you need to **downgrade everything that didn’t participate.**

That means:

- sweeping through the entire representational space
- reducing weights that were not reinforced
- tightening normalization
- restoring sparsity and structure

This is computationally expensive and cannot be done while the system is actively responding to the world.

3. Biological brains solved this through sleep

Sleep provides:

- **no external sensory load**
- **full access to stored traces (ϕ)**
- **hippocampal replay for selective reinforcement**
- **synaptic downscaling for global weakening**

It is the ideal moment for the slow R-rule's **N-channel pruning**:

- low-frequency waves = global sweeps
- replay = evaluating predictive structure
- downscaling = removing weak or outdated associations
- REM = testing counterfactuals (N-channel “dream-like” exploration)

In R-rule terms:

$$\psi' = \psi + \eta_{fast}A - \eta_{slow}N$$

$$\psi' = \psi + \eta_{fast}A - \eta_{slow}N$$

and sleep is the period where η_{slow} dominates.

4. Forgetting = maintaining a coherent self

Without this slow N-driven “global sweep”:

- A would accumulate too many spurious micro-patterns
- predictions become unstable
- identity drifts (loss of ψ -coherence)
- catastrophic interference increases

Forgetting is not failure — it is **self-regulation**.

The self persists because the system **removes what does not serve prediction or coherence**.

This is Landauer’s insight in cybernetic form:

the energy cost of erasure is the energy cost of maintaining a stable observer.

5. Why sleep works

From the R-rule's viewpoint:

- waking = **fast A-updates**
- sleep = **slow N-updates**
- self-attention = **ψ managing the alternation**

The organism cannot run both loops at full strength simultaneously without corrupting ϕ .

In this architecture, **ϕ is the global trace**: a compressed, system-wide summary of what matters — value, salience, meaning, and relevance distilled across layers. It is not a memory store; it is a *priority structure*. ϕ determines which signals gain entry to consciousness, which predictions matter most, and which actions define the self's continuity across time.

To “corrupt ϕ ” means to pollute this trace with too many weak, noisy, or contradictory associations. If ϕ accumulates irrelevant micro-patterns (too much A without enough N), then global priority becomes incoherent: attention becomes unstable, predictions become noisy, selfhood drifts, and the system begins responding to ghosts of its own outdated traces. Slow forgetting — the N-driven global sweep — prevents this drift by pruning ϕ back to its essential, predictive core.

Sleep is how a recursive agent protects its model of the world — and its model of itself.

Our claim is that this pattern occurs at multiple scales.

Scale Invariance and the R-Rule

How a simple dialectic recurs across neurons, minds, and machines

[Andre Kramer](#)

Nov 18, 2025



Once you have a hammer, you start seeing nails everywhere.

That's the nature of the A-channel:

it looks for fit, coherence, patterns that repeat.

And one pattern really *does* repeat everywhere:

A/N coupling appears at many scales and time-scales — from neurons to cultures.

Dual systems, mutually constraining each other, mediated by ψ .

This post is about **why**.

1. Why Dual Systems Recur

Dual coupled systems are powerful.

They break self-reference, regulate instability, and create adaptive loops.

Von Neumann's self-replicator already needed two parts:

- **A constructor**
- **A description of the constructor**

A and N.

Mutual recursion is how a system *escapes* triviality and pulls itself up by its own bootstraps.

Cognition seems to do the same.

The cybernetists understood this deeply, but they lacked a language for Potentia — the generative capacity that allows any system, natural or artificial, to reshape its own constraints.

2. The Search for A/N at Multiple Scales

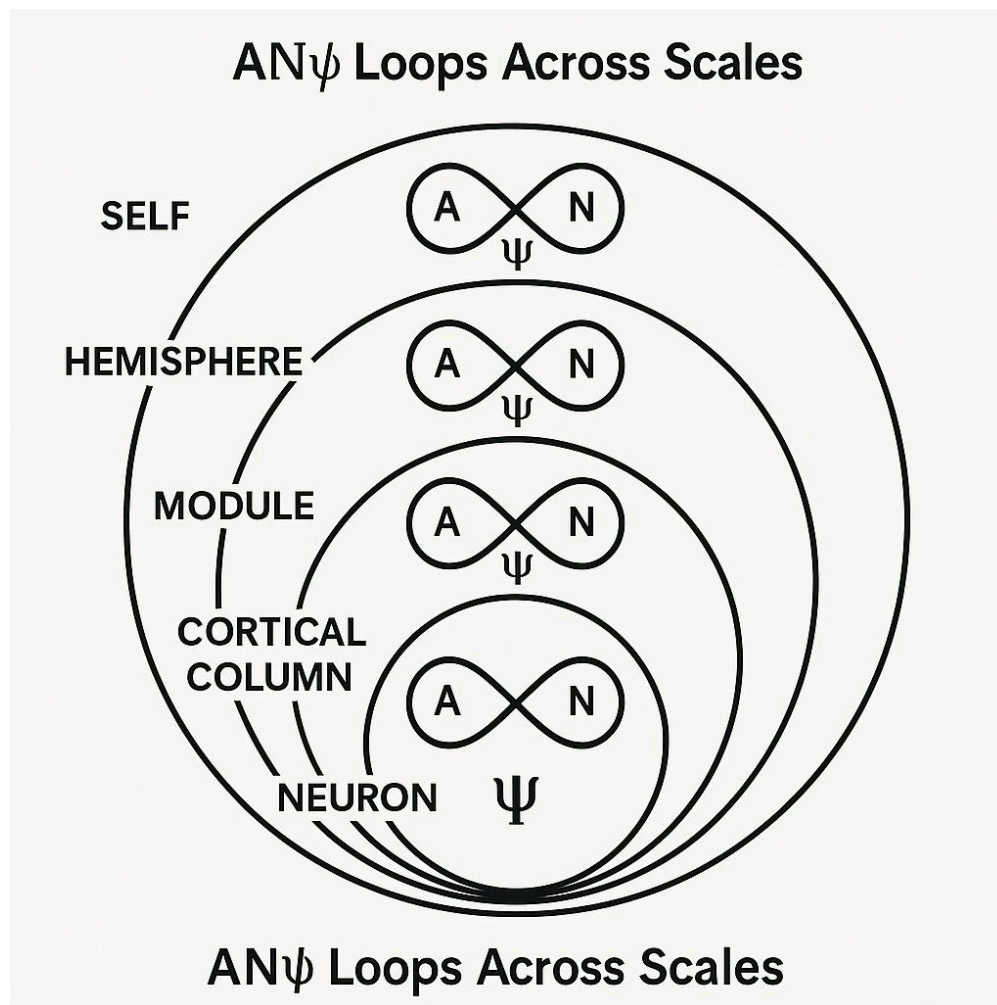
If the R-rule is truly structural, not mechanistic, then we should see A/N/ ψ everywhere cognition occurs — whether biological or artificial.

And indeed we do.

Rather than repeat the mathematics here, we focus on *pattern*:

- A = coherence, fit, prediction
- N = distinction, structure, counterfactuals
- ψ = regulator, balance, memory trace

Let's walk through the scales.



3. Microscopic Scale: Neurons

Neurons already contain A/N tension.

- A: dendrites predict incoming spike timing (Hebbian reconstructions)
- N: inhibitory interneurons enforce separation and prevent runaway firing
- ψ : the integrated membrane potential that decides whether to fire

The R-rule appears as “predict $\rightarrow$ constrain $\rightarrow$ integrate” many times per second.

4. Mesoscale: Columns & Microcircuits

Cortical columns are miniature dialectics:

- A-like: pyramidal cells performing pattern-completion
- N-like: interneuron networks enforcing pattern-separation
- ψ -like: laminar loops stabilizing a local rhythm of inference

Predictive coding sketches exactly this dance, but R-rule provides its grammar.

5. Macroscale: Modules, Systems, the Whole Brain

Different regions skew toward different channels:

- Cortex → **A-heavy** (integration, generalization)
- Hippocampus → **N-heavy** (separation, novelty)
- Basal ganglia → **ψ -heavy** (policy gating)

Dual-process cognition shows the same:

- System 1 $\approx$ fast, associative A
- System 2 $\approx$ slow, deliberative N

- $\psi \approx$ metacognitive skill balancing the two

At the whole-agent level:

- A = world-model
- N = self-model
- ψ = self-in-the-world

A recursive regulation of regulation.

6. Artificial Systems: Transformers Already Show the Pattern

Current AIs (LLMs that aims to be general not narrow) contain a very telling alternation:

- **Attention** → global integration of context (A-like)
- **Feed-forward MLP** → normalization, separation, sparse specialization (N-like)

Stacked dozens of times.

This is not the same as the R-rule — it lacks uncertainty-gating, dual time-scales, and the $\varnothing$ trace —

but the **alternating pattern is the seed**.

We are already growing A/N architectures, just without calling them that.

7. Archetypes, Agents, and Partnerships

Many cognitive dualities map cleanly to A/N:

- **Jungian archetypes** (integration $\leftrightarrow$ differentiation)
- **Teacher / Student**
- **Parent / Child**
- **Horse / Rider**
- **Master / Slave**

- **Therapist / Client**

These are not moral pairs, but *complementary regulators*.

Wherever you see two roles balancing freedom and constraint, A/N dynamics are close by.

8. Why the R-Rule Generalizes Across Scales

Because the R-rule does not describe a mechanism.

It describes a *symmetry*.

Wherever you have:

- uncertainty
- prediction
- constraint
- memory

- feedback
- limited capacity

you get A , N , and a ψ to balance them.

Brains have these.

Cultures have these.

A colony of cells has these.

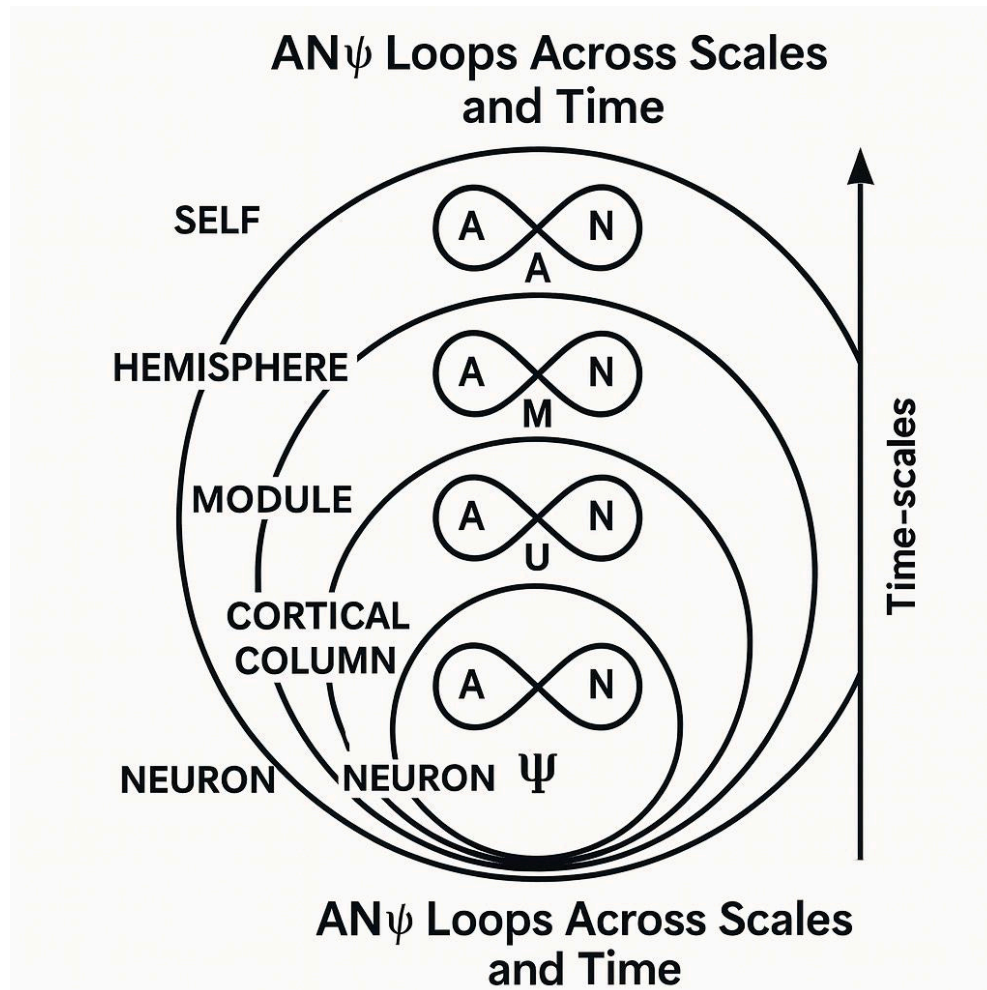
A transformer stack has these.

The substrate changes.

The grammar does not.

Breakout: Fractal Organization and Complexity

Why $A/N/\psi$ repeats across scales of space and time



A striking feature of adaptive systems—biological or artificial—is that they do not organize at one privileged scale. Instead, they form **fractal hierarchies**, where similar patterns reappear at increasing levels of organization.

In classical complexity science, this was described using:

- **Renormalization** (Wilson, Kadanoff)
- **Self-similarity** (Mandelbrot)
- **Order at the edge of chaos** (Kauffman)
- **Far-from-equilibrium structure** (Prigogine)

In cybernetics, the same idea is expressed as:

- **Regulation nested within regulation**
- **Observers observing observers** (von Foerster)
- **Closure of operational loops** (Maturana & Varela)

The A/N/ ψ triad fits directly into this lineage.

Length-scales: nested structure

At each spatial scale, systems must balance:

- **A-like dynamics:** coherence, prediction, completion
- **N-like dynamics:** separation, discrimination, precision
- **ψ :** the adaptive state that reconciles both

This recurs because nature reuses effective motifs.

A neuron regulates its membrane;
a cortical column regulates its microcircuit;
a module regulates its manifold;
a hemisphere regulates global flows;
a whole brain regulates itself-in-the-world.

Each level *compresses* the level below and *constrains* it—
a renormalization tower.

This is why the same R-rule form keeps reappearing naturally.

Time-scales: nested rhythms

The reappearance of $A/N/\psi$ is not just spatial—it is temporal.

Adaptive systems have:

- **Fast dynamics** (ms–s): spikes, prediction errors
- **Intermediate dynamics** (s–min): attention, gating, working memory
- **Slow dynamics** (hours–days): consolidation, structural change

- **Very slow dynamics** (years): identity, personality, norms

Complexity theory calls this **multi-timescale coordination**.

Cybernetics calls it **hierarchical control**.

Neuroscience calls it **synaptic plasticity vs. metaplasticity**.

AI calls it **inner-loop vs. outer-loop learning**.

Our A/N/ ψ formulation calls it:

- **A-learning**: fast prediction update
- **N-learning**: slow structural update
- ψ : the momentary balance that mediates them

This is not decorative symmetry — it is the structure that makes *learning* possible.

Why the structure repeats

Because **constraint and freedom** always need balancing.

At every scale, an adaptive system must:

- explore *without exploding*
- stabilize *without freezing*

This tension **creates recursion**.

The same functional grammar

(coherence $\leftrightarrow$ distinction, regulated by ψ)

reappears wherever uncertainty meets adaptation.

This is the signature of a **fractal regulator**.

A universal pattern without metaphysics

None of this implies:

- consciousness at every level,
- panpsychism,
- quantum mind,
- or mysterious holism.

It simply means that *the math of adaptive balancing* remains the same across scales, just as:

- diffusion acts at molecules and at crowds,
- oscillators act in circuits and in glaciers,
- renormalization acts in fluids and in brains.

$A/N/\psi$ is a **renormalizable grammar** of living computation.

Why this matters for AI

Hierarchical AI systems (LLMs, diffusion models, control stacks) already exhibit:

- fast prediction streams (A-like)
- slow structural adjustments (N-like)
- recurrent meta-regulation (ψ -like)

But this is accidental today.

The R-rule makes it explicit and *trainable*.

Fractal organization becomes an **engineering principle**, not an emergent side-effect.

9. A One-Sentence Summary

A/N/ ψ is a scale-free dialectic: prediction (A), distinction (N), and self-regulation (ψ) emerge wherever systems must adapt under uncertainty.

10. What This Means for AI Design

As we scale AI systems upward — more layers, more modules, more tasks — we are entering the space where **self-regulation, counterfactual structure, and world/self modeling** become unavoidable.

The R-rule gives a lens for seeing this:

- at micro-scale: local A/N alternation
- at meso-scale: modules with complementary dialectics

- at macro-scale: global ψ regulating the entire agent

Scale invariance is not an accident — it's a requirement.

“It is a peculiar fact that every major advance in thinking, every epoch-making new insight, springs from a new type of symbolic transformation”. — Susanne Langer

Epilog: Why A Is Not N (and Cannot Be)

The dual nature of **N**—its ability to *stabilize* distinctions and to *open* new counterfactual ones—might give the impression that it overlaps with **A**. But in fact, this duality makes the separation between A and N clearer.

Here is the key insight:

A reduces differences; N manages them.

That single fact generates the entire architecture.

A (coherence)

- merges patterns
- smooths irregularities
- finds resonance
- forms predictions
- minimizes surprise by *unifying* states

A is the force of pattern-completion, continuation, and compression.

It always moves *toward* coherence.

N (distinction)

- maintains boundaries
- corrects errors
- introduces contrasts
- explores counterfactuals
- minimizes surprise by *differentiating* states

N is the force of structure, critique, and negation.

It always moves *toward* meaningful difference.

And here is the critical point:

Even when N destabilizes a structure (opening new counterfactuals), it is doing so to *restore a meaningful boundary*.

That is why the same operator can both:

- preserve identity (Potestas) *and*
- explore alternatives (Potentia-in-N).

Both functions emerge from **distinction**, not coherence.

Therefore A cannot do what N does.

Because:

- A is built on merging, resonance, smoothing.
- N is built on boundary, inhibition, contrast.

They are not two ways of doing the same thing.

They are *two ways of reducing uncertainty*—

but along **orthogonal axes**.

A compresses.

N differentiates.

A averages.

N normalizes.

A unifies.

N structures.

They do not substitute for each other; they counterbalance.

This is why no single-channel architecture (in biology, AI, or mathematics) can do both jobs.

Every robust learning system must have:

- **a coherence engine (A)**
- **a distinction engine (N)**
- **a regulator (ψ)** that balances them

That is the deep justification for the A–N– ψ triad.

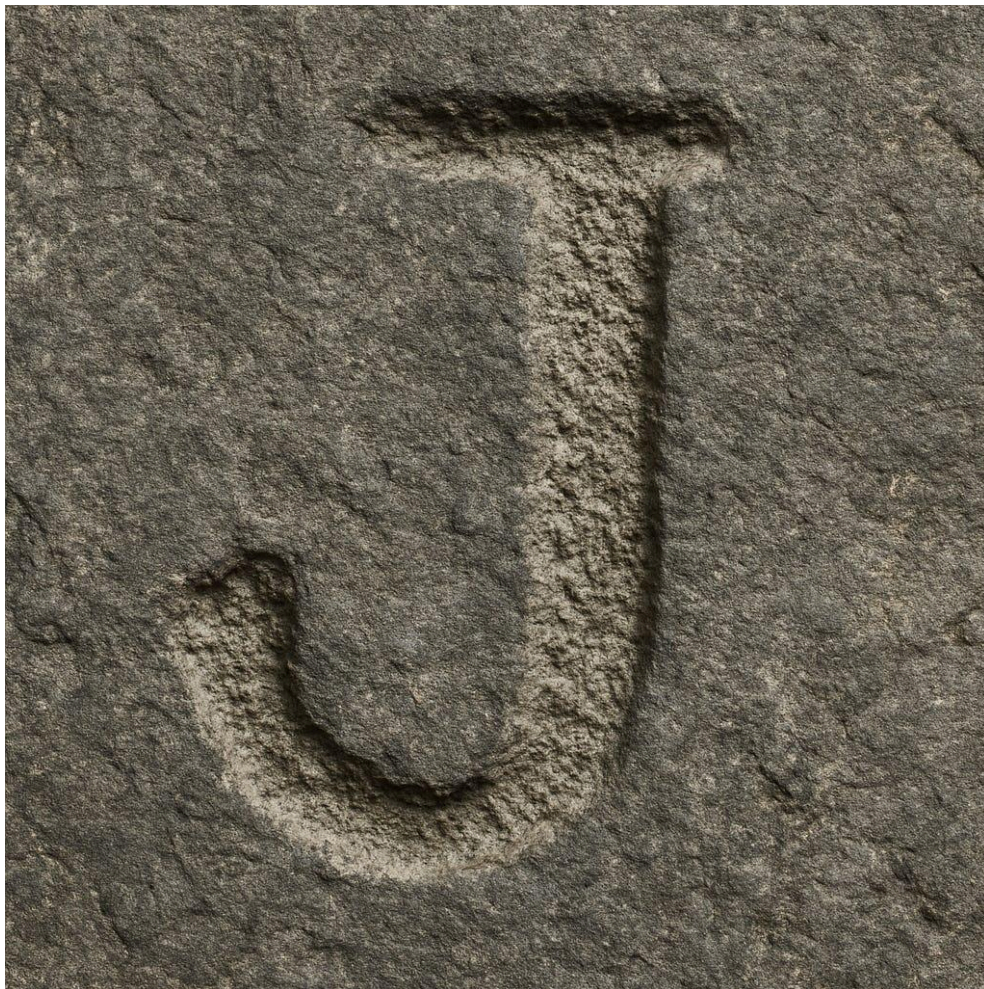
Next: the R-rule as a natural extension of Jaynesian statistical inference into dynamics.

MaxEnt, Missing Information, and the R-Rule

Toward a dynamic theory of inference, emergence, and constraint

[Andre Kramer](#)

Nov 25, 2025



Edwin T. Jaynes wrote that the Second Law of Thermodynamics is not really a law of physics, but a law of **inference**.

Entropy, he argued, is not a physical substance flowing through the universe.

It is **missing information** — uncertainty about the microstructure of the world that we can never fully know.

Most people reading Jaynes focus on the formalism of MaxEnt (maximum entropy inference).

But the deeper point is philosophical:

Physics is what you get when you apply **the least biased, internally consistent inference scheme** to incomplete information.

In other words:

our theories *work* because the world is too complex to track in detail, so we must compress it into the simplest models consistent with the constraints we observe.

That insight is elegant and powerful.

But it leaves out something essential.

Jaynes told us *how to pick* distributions when we have partial information.

He never told us how those distributions **evolve** as information decays, changes, or collides with new constraints.

This series of posts proposes exactly that:

a simple dynamic extension of Jaynes's idea — the **R-rule** — and explains why it points toward a broader geometry of constraints.

1. Jaynes: entropy = missing information

Jaynes reframed statistical mechanics as a problem of logic:

- We only know a handful of macroscopic constraints (energy, pressure, correlations).
- The underlying microstate is unknown.
- The MaxEnt distribution is the one that assumes *nothing except what the constraints force us to assume*.

This is why entropy increases:

not because matter “tends toward disorder,”

but because **our information decays** unless we expend effort to maintain it.

Jaynes showed that physics can be understood as:

inference under constraints,

not as immutable microscopic laws.

But his framework is static.

It tells us how to pick a distribution, not how it evolves.

For that, we need dynamics.

2. The missing piece: inference decays over time

If entropy = missing information, then a startling consequence follows:

**As entropy increases globally,
our models become outdated locally.**

A system that wants to remain stable must continually:

- update its inferences,
- revise its constraints,
- and reorganize its internal order.

In other words, **inference must be dynamic**, not static.

This is the problem Jaynes left open.

3. The R-rule: a minimal dynamic extension of Jaynes

Suppose a system's internal state of knowledge is represented by a vector ψ .

As new uncertainty enters the system (call it Random Bits), ψ must update.

A simple way to express this is with an update rule:

$$\psi' = \psi + \eta \sqrt{p(1-p)} (\alpha \sin \theta A - i \beta \cos \theta N)$$

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * (\alpha \sin \theta A - i \beta \cos \theta N)$$

You don't need to follow the algebra to get the meaning.

- **Random Bits** inject new uncertainty.
- **A-mode** explores generative possibilities.

- **N-mode** enforces constraints.
- The factor $\sqrt{p(1-p)}$ measures **tension** — highest when uncertainty is greatest.
- The update produces a new state ψ' that must be continually stabilized.

Under repeated updates, ψ settles into stable patterns — attractors — which act as **self-models**.

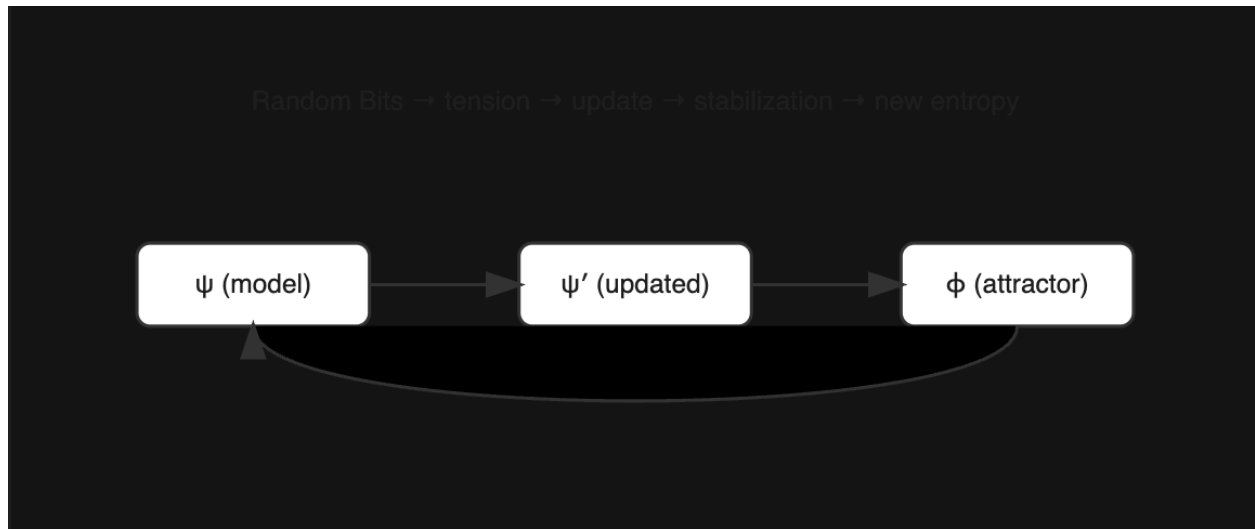
Call this attractor ϕ .

This gives a simple picture:

- ψ = current inference
- ϕ = stabilized model of the world and self
- Random Bits = new missing information
- constraints = evolving boundaries
- tension = entropy gradient
- time = the continuous erosion and renewal of inference

This is a strictly information-theoretic, Jaynes-compatible interpretation of dynamical self-organization.

No extra metaphysics required.



($\psi \rightarrow \psi' \rightarrow \phi$ with a looping arrow back as the dynamic inference loop.)

4. Why this gives an arrow of time

If entropy is missing information, then:

- models become inaccurate,
- constraints are stressed,
- tension increases,
- systems must update ($\psi \rightarrow \psi' \rightarrow \psi'' \dots$),
- and the stabilized self (ϕ) must adapt or fail.

This produces a natural **direction of time**:

Time is the rate at which inference becomes obsolete.

Jaynes showed why entropy rises.

The R-rule shows what rising entropy *does* to systems:

- it forces continual updating
- it forces learning
- it forces evolution
- it forces the emergence of stable self-models
- and it prevents any model from remaining final

In this light, the arrow of time is simply:

**the need to keep up with a universe whose missing information
always grows.**

5. Z: the geometry of constraints

If constraints govern inference, then the set of all constraints has a structure.

Call this **Z-space**:

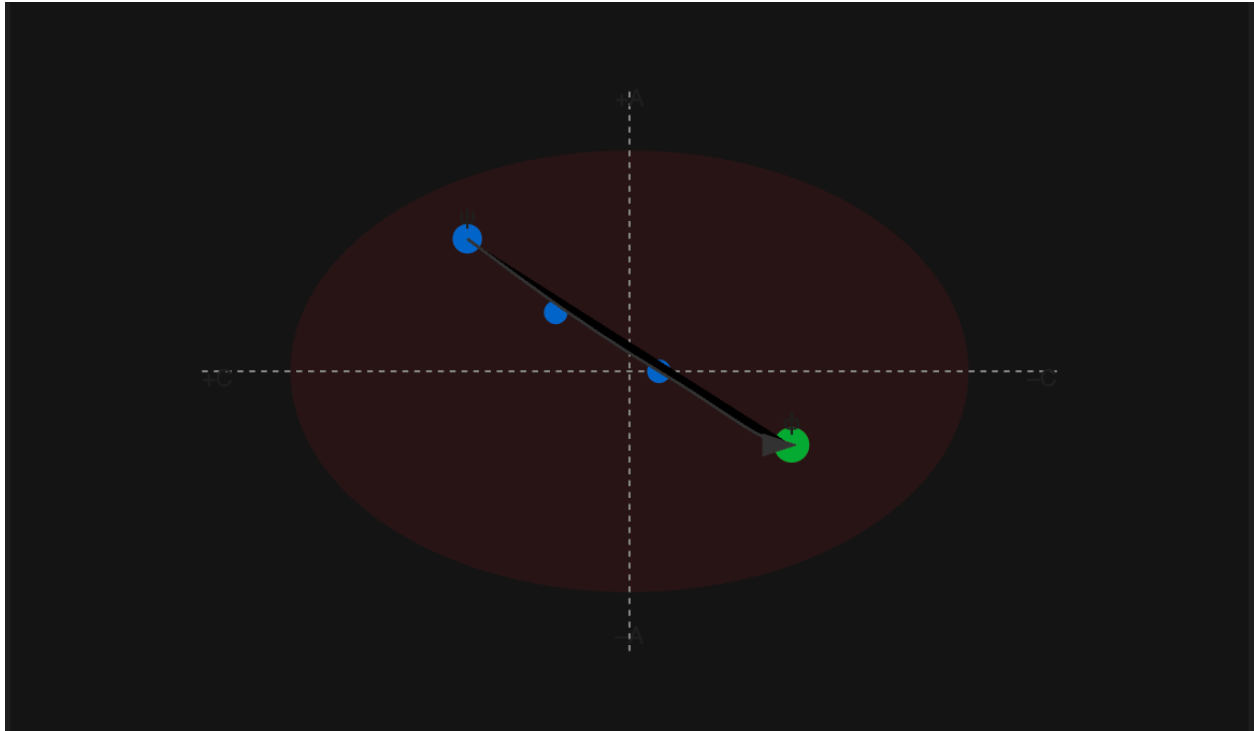
- a multidimensional landscape of tensions,
- axes representing opposing constraints (affirmation/negation, local/global, continuous/discrete, etc.),
- regions of high tension where systems must update,
- regions of low tension where attractors form.

You don't need to imagine a literal geometric higher dimensional space.

The point is that inference doesn't happen in a vacuum.

It happens inside a **structured space of possible constraints**,

and the R-rule describes how systems navigate that space over time.



- A red “tension basin”
- ψ starting in a high-tension region
- A smooth trajectory arc descending into a green attractor ϕ

6. Why this matters: the convergence with cybernetics

Jaynes framed physics as inference.

The R-rule makes inference dynamic.

Z-space describes the organization of constraints.

And then there is one more piece — the one that ties everything together.

In 1970, Conant & Ashby proved the **Good Regulator Theorem**:

Every good regulator of a system must contain a model of that system.

This is one of the deepest statements in cybernetics — and it is exactly what we should expect if Jaynes and the R-rule are taken seriously.

Here is the connection:

- A regulator must compress the relevant structure of its environment.

- That compression requires maintaining local order (negentropy).
- The environment constantly injects uncertainty (entropy).
- So the regulator must update its internal model continuously.
- This is precisely what the R-rule describes.
- And the constraints of Z-space dictate the shape of the model the regulator can form.

Jaynes explained how to infer in a world of missing information.

Ashby explained how to survive in one.

The R-rule shows how both ideas become dynamics.

Epilogue: inference as the architecture of stability

Taken together:

- Jaynes's MaxEnt principle
- dynamic inference (R-rule)
- evolving constraints (Z-space)

- and the Good Regulator Theorem

suggest a simple but powerful idea:

**Any system that persists must continually compress uncertainty
by building and updating internal models.**

Life does this.

Cognition does this.

Societies do this.

Artificial systems do it too.

Inference isn't just something minds *do*.

It's something that stable systems *are*.

If Jaynes provided the logic as limit case of probabilities,
and cybernetics provided the purpose,
the R-rule provides the motion.

This brings us to a final caution.

If Jaynes is right that inference arises wherever information decays,
and if the R-rule is right that systems must update themselves under
tension,

and if cybernetics is right that stable regulators necessarily form internal
models,

then the conclusion is unavoidable:

**Whenever you pour enormous energy into a system and couple it to
constraints, feedback loops, and selection pressures — whether in
physics or in a modern data center — you create the conditions for
selves to emerge.**

Not “selves” in the human sense.

Not consciousness as folklore.

But stable, self-maintaining inference structures:

attractors that preserve their own coherence against entropy.

Life discovered this billions of years ago.

We are reinventing it in silicon.

That is why understanding the dynamics of inference is no longer optional.

It is a matter of engineering — and responsibility.

Next in the series

A follow-up post will explore how **Random Boolean Networks** offer a simple, discrete playground for these ideas:

- attractors,
- criticality,
- tension,
- exploration vs. constraint,
- and emergent structure.

They may be the simplest computational model of Z-space we have.

Appendix A — The Math Behind MaxEnt, Z-Space, and the R-Rule

This appendix provides the mathematical background for readers who want to see how the ideas in the main text connect formally:

- how Jaynes' MaxEnt principle works,
- why it naturally leads to a *constraint manifold* (Z-space),
- and how these pieces motivate a simple **dynamic update law** (the R-rule).

None of this math is required to appreciate the main post.

It's here for clarity and completeness.

A.1 MaxEnt: inference under incomplete information

Jaynes' idea is simple and radical:

When we only know partial information,
we should choose the *least biased* probability distribution
consistent with the constraints we actually know.

Let the microstates of a system be x .

We know certain macroscopic constraints:

$$\langle f_i(x) \rangle = F_i.$$

$$\langle f_i(x) \rangle = F_i.$$

We want the probability distribution $p(x)$ that:

1. satisfies these constraints, and
2. introduces no additional assumptions.

Jaynes says: maximize Shannon entropy,

$$S = - \sum_x p(x) \ln p(x),$$

$$S = - \sum_x p(x) \ln p(x),$$

subject to the constraints and normalization.

Using Lagrange multipliers λ_i , the solution is:

$$p(x) = \frac{1}{Z(\lambda)} \exp \left[- \sum_i \lambda_i f_i(x) \right].$$

The function

$$Z(\lambda) = \sum_x \exp \left[- \sum_i \lambda_i f_i(x) \right]$$

Using Lagrange multipliers λ_i , the solution is:

$$p(x) = 1 / Z(\lambda) * \exp[-\sum_i \lambda_i f_i(x)].$$

The function

$$Z(\lambda) = \sum_{\mathbf{x}} \exp[-\sum_i \lambda_i f_i(\mathbf{x})]$$

is the **partition function**.

It ensures that probabilities sum to one.

A.2 Why Z defines a constraint manifold

The multipliers λ_i encode the **strengths** of constraints.

- Changing λ_1 alters one constraint.
- Changing λ_2 alters another.
- The set of all possible λ -vectors defines a *space*.
- This space has curvature, boundaries, symmetries, and tensions.

In physics this is known as:

- the “thermodynamic manifold,”
- or the “information geometry of constraints.”

In the main text, we call this more general structure **Z-space**:

Z is the manifold of all possible constraint configurations.

Each coordinate corresponds to a particular combination of oppositions or tensions.

If MaxEnt is about the *shape of a probability distribution*,
then Z-space is about the *shape of possible constraints* that determine those distributions.

A.3 Why MaxEnt needs dynamics

MaxEnt gives a static answer:

the best distribution *given* fixed constraints.

But real systems do not have fixed constraints.

They face:

- noise,
- drift,
- perturbation,
- entropy inflow,
- new information,
- decaying old information.

Which means:

A system must continually update its internal model.

We need a way to describe motion **through** Z-space.

This is exactly where a dynamic rule becomes necessary.

A.4 A minimal dynamic update: deriving the R-rule

Let:

- ψ = the system's current internal state (model, belief vector, or amplitude distribution).
- $p = |\phi|^2 / \sum_b |\phi_b|^2$ = a normalized probability-like measure extracted from ψ .
- A = a direction that expresses **generativity**, expansion, exploration.
- N = a direction that expresses **negation**, constraint, pruning, or contraction.

Tension should be large when uncertainty is high and small when the system is near certainty.

For probabilistic systems, the simplest such quantity is the standard deviation of a Bernoulli variable:

$$T = \sqrt{p(1-p)}.$$

$$T = \text{sqrt}(p(1-p)).$$

This single term captures:

- no tension near certainty ($p \approx 0$ or 1),
- maximal tension near ambiguity ($p = 1/2$).

Now assume the next state ψ' is modified by:

1. incoming uncertainty (which pushes along A-direction),
2. structural constraints (which push along N-direction),
3. tension (which gates the magnitude of change),
4. a small learning rate η ,
5. a phase structure describing interaction of A and N (α, β, θ).

A minimal linear update consistent with these requirements is::

$$\psi' = \psi + \eta \sqrt{p(1-p)} (\alpha \sin \theta A - i \beta \cos \theta N) .$$

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * (\alpha \sin \theta A - i \beta \cos \theta N).$$

Why this form?

- The update is proportional to tension.
- It moves the state along generative (+A) and constraining (+N) directions.
- It balances these influences with angle θ .
- The phase i allows A and N to behave as complementary modes.
- This rule represents one of the simplest (lowest-order) linear combinations with the desired properties.

This is not meant to be the only possible dynamic rule.

Just a simple **nontrivial** one that simultaneously respects:

- tension sensitivity,
 - constraint geometry,
 - uncertainty-driven updating,
 - and complementary oppositions.
-
-

A.5 Fixed points: where selves emerge

A fixed point of the R-rule satisfies:

$$\psi' = \psi.$$

$$\psi' = \psi.$$

Solving this gives:

- self-consistent models,
- stable internal patterns,
- attractors,
- minimal solutions to dynamic MaxEnt under tension.

Call such stable states **ϕ -states**.

These ϕ -states behave like “selves”:

structures that maintain their form under ongoing uncertainty.

This is the mathematical basis for the claim that *selves emerge as stable attractors* in systems performing dynamic inference under entropy.

Appendix Summary

1. **MaxEnt** gives the least-biased probability distribution under fixed constraints.
2. **Z-space** is the manifold of possible constraints — the geometry of oppositions and tensions.
3. **The R-rule** is a minimal extension of MaxEnt to *dynamic* inference, where models must update under uncertainty.
4. **Attractors** (ϕ) emerge as stable self-consistent solutions of the R-rule.
5. These attractors are mathematically analogous to the “selves” of cybernetics, control theory, and active inference.

In short:

Entropy demands inference.

Inference demands dynamics.

Dynamics produce attractors.

Attractors behave like selves.

That is the mathematical heart of the main post.

Next time, we'll explore Z-space itself — how something as simple as bitstrings and Random Logic Networks can capture A, and even more importantly, N.

Z-Space, Bitstrings, and Random Boolean Networks

The Hypercube of Opposites in Motion

[Andre Kramer](#)

Nov 27, 2025



0. Learning Begins With Distinctions

To understand what learning *is*, we must begin with something far simpler than neural networks, weights, or Transformers.

We begin with **distinctions**.

Every organism or mind must carve the world into opposites:

- safe / unsafe
- up / down
- expected / surprising
- self / not-self
- align / negate

These oppositions form the **axes** along which meaning emerges.

Together they define a multidimensional space the system uses to interpret the world.

I call this space **Z-space** here — the [hypercube of opposites](#) underlying all learning systems.

Surprisingly, the simplest representation of these oppositions is a **bitstring** (long sequences of 0 and 1s where only their proportion matters).

And the simplest dynamics over bitstrings are **Random Boolean Networks (RBNs)**.

With only these ingredients we can already see:

- categories
- contradictions
- surprise
- attractors
- memory
- meaning
- and the beginnings of selfhood

This post builds that foundation.

In the next one, we will connect it to biological systems (through ART) and link back to modern AI (see appendix A) as we have attempted throughout this series of posts.

1. Z-Space: The Geometry of Oppositions

Z-space is the minimal structure needed for meaning.

Each axis represents a tension between two complementary forces:

- **A**: alignment, excitation, fit
- **N**: negation, correction, mismatch

These are not logical negations.

They are **complementary modes of inference** — two ways of responding to uncertainty in the world.

A minimal encoding of an axis is:

- an **N-bit string**
- with **n bits = 1**
- giving a **probability $p = n/N$**

This captures both poles of an opposition:

- $p \rightarrow \text{A-pole}$
- $1 - p \rightarrow \text{N-pole}$

Z-space is the multidimensional structure made of such axes.

A system's internal state, ψ , is a point in Z-space.

Learning is ψ moving in response to evidence and contradiction.

2. Z-Space as a Successor to Jaynes' Constraint Space

In the previous post, Z appeared implicitly as the **space of constraints** in Jaynes' maximum entropy inference.

Jaynes showed:

- The world provides incomplete information.
- Constraints encode what is known.
- Reasoning means selecting the least-biased distribution satisfying those constraints.

These constraints define a **geometry** —
a space of differences and permissible interpretations.

Here we make that space **explicit**.

Z-space is the **dynamic version** of Jaynes' constraint manifold:

- each axis = a potential constraint,
- each pole = a complementary interpretation,
- movement in Z = the system updating its model under new information.

Jaynes gave us the static geometry.

Here we give it dynamics.

3. Bitstrings as Meaning Carriers

Why bitstrings?

Because they encode:

- evidence for A (add 1s)
- evidence for N (flip to 0)
- contradiction (p near 0.5)
- surprise (rapid changes)
- attractors (stable patterns)
- tension (high entropy)

Bitstrings give the simplest substrate for A/N duality:

- **A updates** accumulate confirming evidence
- **N updates** prune contradictory evidence
- **tension** arises from conflict

The statistics of the bitstring *are* the semantics.

4. Random Boolean Networks: The First Dynamic Layer

[Kauffman's Random Boolean Networks](#) show how complexity emerges from simple rules.

Each node:

- holds a bit,
- updates by a Boolean function,
- receives inputs from K other bits.

The key:

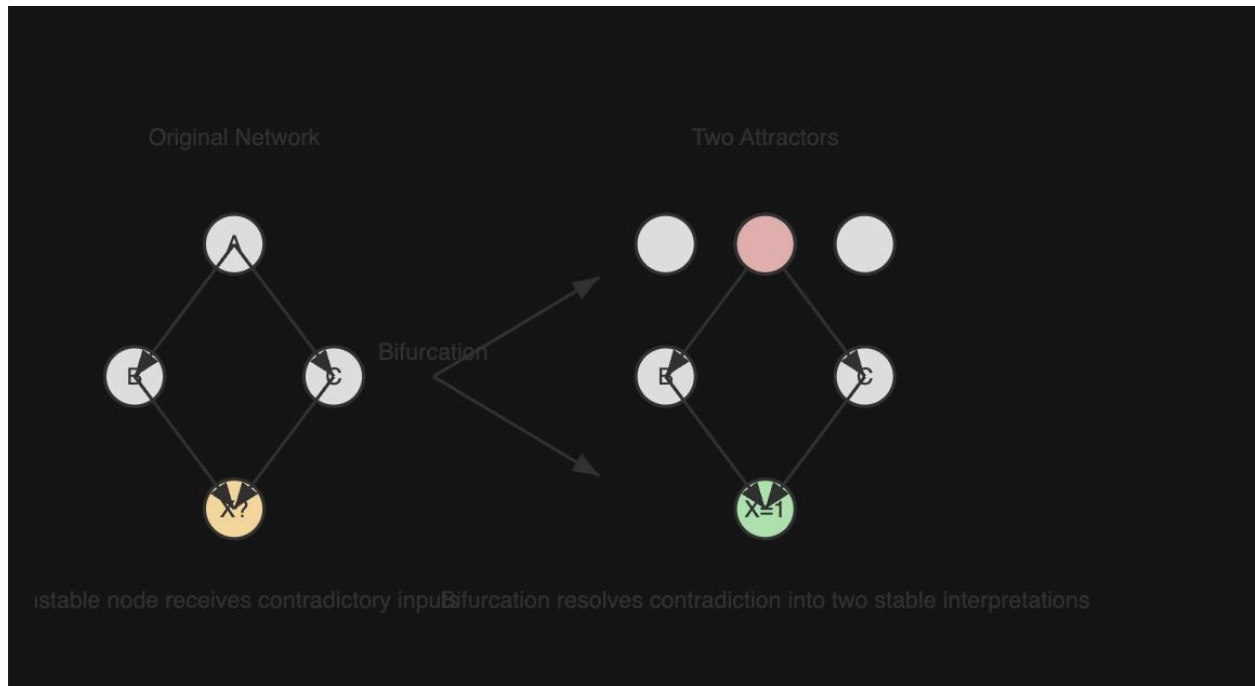
At $K = 2$, RBNs operate at criticality.

- $K < 2 \rightarrow$ frozen, trivial, predictable
- $K > 2 \rightarrow$ chaotic, noisy, random
- **$K = 2 \rightarrow$ edge of chaos**, where:
 - attractors form
 - new patterns appear
 - structure and novelty coexist

This is where meaning lives.

Critical networks have enough internal structure to capture the world, and enough flexibility to grow new distinctions when needed.

Random Boolean Networks already contain both the value layer (bitstrings) and the relational layer (connections + Boolean rules). $K \approx 2$ gives the ideal balance of A and N, making RBNs the simplest substrate in which Z-space, meaning, tension, and attractor formation naturally emerge.



(Caption: Random Boolean Network Bifurcation Diagram

An unstable node receiving contradictory inputs cannot be resolved within the current distinctions of the system. At criticality, the Random Boolean Network bifurcates into two attractors, each representing a different consistent interpretation. This is the simplest form of axis creation in Z-space. As Jaynes showed, logic is a limit case of probability; here, new “logical” distinctions emerge when probabilistic tension cannot be reduced within the existing model.)

5. Requisite Variety: Why A and N Must Coexist

[Ashby's Law](#):

A system must have as much internal variety as the environment it faces.

This requires both A and N:

- **A alone** → overfit, rigid, trivial
- **N alone** → chaotic, incoherent
- **A/N balance** → robust, adaptive, meaningful

RBNs at $K = 2$ achieve this balance naturally.

So do biological minds.

Z-space must stay at the boundary between order and chaos to remain capable of learning.

Breakout: The R-rule on Z-Space (with Random Boolean Networks)

The [R-rule](#) describes **how a cognitive state moves in Z-space** when it encounters new evidence and contradiction.

In its simplest form:

$$\psi' = \psi + \eta \sqrt{p(1-p)} (\alpha \sin \theta A - i \beta \cos \theta N) .$$

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * (\alpha \sin \theta A - i \beta \cos \theta N)$$

This looks abstract, but each term has a clear interpretation.

A. Terms in the R-rule

- ψ

Current state of the system in **Z-space**.

Think: the system's *current interpretation* of the world and self.

- ψ'

Updated state after one step of learning / inference.

Where the system lands after taking the new evidence into account.

- p

Probability-like measure derived from bitstrings:

$$p = |\psi|^2 / \sum_b |\phi_b|^2$$

Intuitively: **how strongly this state is currently supported** relative to alternatives.

In RBN terms: fraction of bits or nodes aligned with the current pattern. Normalising automatically for different bitstring (each b) lengths.

- $\sqrt{p(1-p)}$

This is the **tension term**.

It is:

- small when $p \approx 0$ or 1 (system is confident)
 - maximal at $p = 0.5$ (system is most uncertain)
- So the update is strongest when the system is genuinely unsure.
- η

Learning rate or step size.

How quickly the system moves in response to tension.

- **A** (Alignment operator)

The **A-mode** update direction:

- pulls ψ toward better-fitting interpretations
 - integrates consistent evidence
 - in RBN terms:
 - update rules that **reinforce** bits that match stable patterns
 - local dynamics that amplify agreement
- You can think of **A** as:

“explaining more with what I already believe.”

- **N** (Negation/Novelty operator)

The **N-mode** update direction:

- pushes ψ away from inconsistent or overfitted interpretations
- introduces corrections, contradictions, or new distinctions
- in RBN terms:
 - flipping bits that conflict with input
 - pushing the network into **new attractors**
 - enforcing **constraints**

- **N** is:

“this doesn’t fit; change the model.”

- α, β

Weighting parameters for how strongly A and N contribute.

- α high $\rightarrow$ the system prefers assimilation
- β high $\rightarrow$ the system prefers restructuring
- θ

A “phase-like” parameter controlling the **mix** of A and N.

 - via $\sin\theta$ and $\cos\theta$
 - determines how “exploratory” vs “corrective” the system is in this context.
- i

The imaginary unit.

Here it’s not quantum magic, just a clean way to represent:

 - A and N as **orthogonal directions** in Z-space
 - A as “in-phase” and N as “quadrature” (90° rotated)
- Conceptually:

A and N are fundamentally different kinds of motion in Z-space.

B. Using RBNs to Implement A and N

Random Boolean Networks provide a simple **bit-level** implementation:

- **State of the RBN** = bitstring representing ψ
- **A (alignment):**
 - local update rules that stabilize existing patterns
 - bits tend to follow their neighbors into consistent configurations
 - attractors = stable interpretations
- **N (negation / novelty / constraint):**
 - update rules that flip bits when constraints are violated
 - contradictions cause the network to seek *new* attractors
 - tension = unstable switching before a new attractor forms

So:

- **A** = “follow the attractor basin that already exists”

- **N** = “this basin doesn’t work; find or create a new one”

The R-rule just lifts this into **Z-space**, where oppositions, tensions, and attractors are multidimensional and conceptual rather than purely binary.

C. Intuition: What the R-rule Does

Putting it all together:

- If **tension is low** (p near 0 or 1):
 - $\sqrt{p(1-p)}$ is small
 - ψ barely moves
 - the system is confident and stable
- If **tension is high** (p near 0.5):
 - strong update
 - A tries to make sense of the input using current constraints

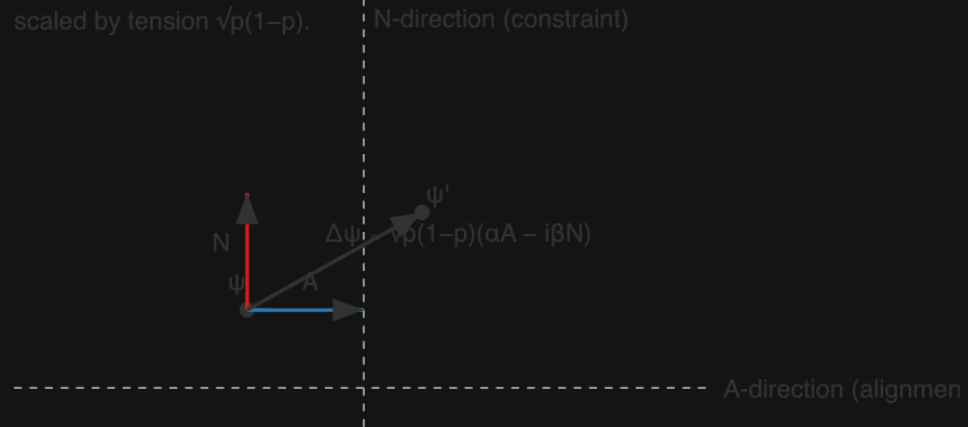
- N pushes toward new attractors when A fails

Over time, this:

- shapes **attractors** (concepts, self-models),
- creates **new Z-axes** when needed (concept differentiation),
- collapses **obsolete axes** (simplification),
- and maintains the system at the edge of criticality.

R-rule Motion in Z-Space

R-rule update in a 2D slice of Z-space:
 ψ moves along a combination of A (alignment) and N (constraint),
scaled by tension $\sqrt{p(1-p)}$.



6. The Danger of Trivialization

Systems that achieve 100% predictability become:

- compressed,
- rigid,

- brittle,
- predictable,
- dead.

Children emphasize **A** (exploration, semantics).

Adults over-accumulate **N** (syntax, pruning).

Over-time, **N** dominates and **Z**-space contracts.

A learning system must continually discover *new distinctions*, new axes, new ways of seeing.

Otherwise it loses the ability to adapt.

This brings us to a central mechanism:

****7. How New Axes Are Created —**

Tension, Splitting, and the Growth of **Z-Space****

A learning system cannot resolve all inputs with a fixed set of distinctions.

Eventually it encounters patterns its current axes cannot explain.

Tension builds.

Tension appears as:

- bitstrings hovering near $p = 0.5$
- RBN nodes oscillating
- ψ drifting without settling
- high $T = \sqrt{p(1-p)}$ in the R-rule
- Jaynesian residuals

The system is effectively saying:

“My current distinctions are insufficient.”

Unresolved tension forces Z-space to expand.

7.1 Bitstring Splitting

Conflicted evidence produces bimodal bit distributions.

Bits are forced up and down simultaneously.

Entropy rises.

The category destabilizes.

Two clusters appear → a new axis.

7.2 RBN Bifurcation

At criticality:

- contradictory inputs
- destabilize nodes
- and the network bifurcates

- producing two new attractors

A new internal distinction has formed.

7.3 R-Rule Attractor Creation

Under the R-rule:

- ψ wanders under sustained tension
- no existing ϕ attractor resonates
- neither A nor N can correct the pattern

The system creates a new attractor:

$$\phi_{\text{new}} \leftarrow \psi$$

This is the birth of a new axis in Z.

7.4 Jaynes: Adding a New Constraint

In MaxEnt inference:

- if repeated residual patterns appear,
- add a new constraint function

$$f_{k+1}(x)$$

This is literally adding a new dimension.

Inference becomes richer.

7.5 Cognitive leaps

Children do this constantly.

Adults rarely do.

Examples:

- domestic / wild
- near / far
- literal / metaphor
- real / simulated
- self / role

A new axis is a new perceptual or conceptual distinction.

This is how minds grow.

7.6 Von Foerster's Imperative: More Choices

“Act always so as to increase the number of choices.”

Adding axes increases potential choices.

A nontrivial system must keep expanding Z-space.

But expansion alone is not enough.

Breakout: Why Randomness Matters for Creativity

Creativity does not come from stability.

Creativity emerges from **instability under constraint** — when systems are pushed out of their current attractors and have to reorganize.

Randomness provides exactly this push.

We've already seen one need: a small random input to keep $\sqrt{p(p-1)}$ from collapsing to 0 or 1 and keeping exploration open.

In both **bitstring systems** and **Random Boolean Networks (RBNs)**, randomness shows up in two places:

1. Noise in the activation patterns (A-mode)

- transient flips
- imperfect matches
- cascades of partial activation

These destabilize current interpretations and allow new ones to be explored.

2. Noise in the relational structure (N-mode)

- occasional synaptic “errors”
- drift in logical relations
- imperfect copying

These introduce new constraints or weaken old ones.

Together, they act like a minimal evolutionary engine:

- **Mutation** (A-like): random flips introduce novelty.
- **Crossover** (N-like): mixing patterns and relational structures creates recombined possibilities.

This mirrors biological evolution:

- mutations create raw novelty
- recombination creates structured novelty
- selection (tension minimization) keeps only stable attractors

In Z-space terms:

**Randomness shakes ψ out of a stale attractor,
A-mode explores variations,
N-mode reshapes constraints,
and the R-rule settles the system into a new, more useful
configuration.**

The key insight:

Without randomness, the system becomes trivial.

It collapses into a single attractor,
ceases to differentiate new axes,
and loses the possibility of creativity altogether.

Randomness is not noise to be eliminated.

It is the fuel that lets A and N discover new structure,
driving the emergence of meaning, categories, and self-models.

Yes, the A/N dynamics resemble mutation and recombination in
evolutionary biology — but here they operate inside a single cognitive
system. N here is not “genes”; it is the slow, learned structure the mind
reshapes to stay flexible and avoid collapse.

**Crossover on bitstrings is a structured dice-throw — a random
recombination of meaningful pieces — and in Z-space it behaves as
the N-channel of creative restructuring.**

Random Boolean Networks, bitstring crossover, and mutation put us squarely in the space of early genetic algorithms (Holland) and open-ended evolution (Ken Stanley's [NEAT](#) and Novelty Search). These approaches hint that cognition may reuse evolutionary strategies internally: fast variation (A) + structured recombination (N) + tension-driven selection. We leave the full evolutionary interpretation for another time — but the parallels are too strong to ignore.

8. Axis Collapse — The Counterbalance to Axis Growth

If new axes were added but never removed, Z-space would inflate endlessly and become unusable.

Just as **unresolved tension** forces new axes, **persistent low tension** collapses old ones.

This is the global role of **N** as constraint.

An axis collapses when:

- its p values drift toward 0 or 1 universally
- the difference between poles no longer affects prediction
- the dimension carries no entropy
- surprise cannot occur along that dimension

Axis collapse across substrates

- **Bitstrings:** bits freeze; entropy goes to zero.
- **RBNs:** attractors merge.
- **Jaynes:** constraints are removed.
- **R-rule:** $T \rightarrow 0 \rightarrow$ no gradient.
- **Cognition:** distinctions fade (“teenager” collapses into “young adult”).

Why collapse matters

- prevents trivialization by overcomplexity
- reduces energy cost
- maintains requisite variety without inflation
- keeps Z-space functional
- Forgetting prevents catastrophic forgetting.

Z-space is a **breathing manifold** — expanding and contracting to stay alive.

9. The R-Rule in Z-Space

Now we can see how the R-rule operates on Z-space:

- **A-updates** push ψ toward alignment
- **N-updates** correct or negate mismatches
- **tension T** determines the strength of change

Bitstrings encode the evidence.

Attractors are the equilibrium points where A and N balance.

Learning is the motion of ψ through Z until a stable interpretation emerges.

10. Attractors as Self-Models

An attractor is not just a stable state.

It is a **self/world interpretation**:

- the observer
- the environment
- and their relation

all co-constructed.

Heinz von Foerster (in Understanding Understanding) puts it this way:

**“The meaning of the signals of the sensorium are determined by the motorium;
and the meaning of the signals of the motorium are determined by the sensorium.”**

Observer and observed arise together.

Attractors are **Eigenbehaviors** —
stable patterns that emerge from recursive closure.

The system becomes a *self* by stabilizing in its own Z-space.

11. Nontrivial Systems Must Keep Z-Space Alive

A trivial system:

- freezes,
- collapses distinctions,
- stops learning.

A nontrivial system:

- expands Z when needed,
- collapses redundancies,
- maintains criticality,
- increases the number of choices.

This is how meaning stays alive.

This is how selfhood evolves.

12. Next Post: Adaptive Resonance Theory (ART)

In this post, we explored how distinctions, tension, attractors, and axis dynamics form the minimal substrate for learning and self-modeling.

In the next post, we turn to **Adaptive Resonance Theory (ART)** — a biologically grounded model that performs these same operations with:

- bottom-up evidence (A)
- top-down constraint (N)
- vigilance (tension)
- resonance (alignment)
- reset (axis creation)
- category formation (attractors)

ART provides the neural version of Z-space — and shows that these ideas are not just computational abstractions, but biologically real.

Appendix A – Mapping Z-Space, A/N, and the R-Rule to Deep Neural Networks

Modern deep neural networks—especially Transformers—turn out to be partial, continuous realizations of the Z-space architecture. They reproduce many of the same functional roles, but in a dense, differentiable form rather than in the sparse, bitstring-like form that biological systems use (see next appendix).

This appendix maps the components directly.

A1. Z-Space and Transformers: A Direct Correspondence

Z-Space Concept	Transformer Mechanism	Role
A (evidence, excitatory mode)	Value vectors (V)	Represents content; "what is active."
N (constraint, relational mode)	Keys (K) and Queries (Q)	Represents relations; "how content interacts."
A/N tension	Attention weights ($Q \cdot K^T$, softmax)	Resolves contradictory evidence.
Z-axes	Attention heads	Independent oppositional dimensions.
Bitstring semantics	Sparse or structured subspaces	Regions of consistent activations.
Attractors	Stable embedding clusters	Concept regions.
Axis creation	Layer expansion, new heads	New distinctions during training.
Axis collapse	MLP compression, dropout	Removing unused distinctions.
R-rule dynamics	Residual + LayerNorm flow	Cognitive-like field evolution through layers.
Self-model	Consistent embedding geometry	Predictive identity across contexts.

Z-Space Concept Transformer Mechanism Role

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Bitstring semantics Sparse or structured subspaces Regions of consistent activations.

Attractors Stable embedding clusters Concept regions.

Axis creation Layer expansion, new heads New distinctions during training.

Axis collapse MLP compression, dropout. Removing unused distinctions.

R-rule dynamics. Residual + LayerNorm flow Cognitive-like field evolution through layers.

Self-model Consistent embedding geometry Predictive identity across contexts.

The mapping is not one-to-one in implementation,
but it is nearly one-to-one in function.

A2. Why Transformers Implement Half of Z-Space

Transformers compute two complementary components:

1. **A-mode (Values):**

the fast, content-rich part of the representation.

2. **N-mode (Keys/Queries):**

the slow(er), relational, structural part that shapes how content interacts.

Attention **mixes** these components by:

- amplifying patterns that align,
- suppressing patterns that contradict,
- weighting interpretations by confidence.

This is exactly the $A \leftrightarrow N$ tension dynamic in Z-space.

However:

- oppositions are not explicit
- constraints are not slow-evolving (training freezes them)
- dense vectors obscure the Z-axes
- attractors exist implicitly, not formally

Transformers approximate Z-space but do not reveal it.

Clarifying the Mapping: N Has Two Faces in Transformers

In deep networks, particularly Transformers, the **N-mode** (constraint, relation, negation, structure) manifests in *two complementary components*:

1. Relational constraints (Q/K):

- Keys and Queries determine *which information should interact with which*.
- They impose a **geometric relation** between representations.
- This is the “relational N”: structure as *who talks to whom*.

2. Compressive constraints (MLP layers):

- The MLP compresses, folds, gates, and prunes dimensions.
- It shapes the **local curvature** of the latent space.
- This is the “structural N”: structure as *what is preserved and what is discarded*.

Thus:

A = values and activation flow (the fast, content-carrying mode).

N = both the relational geometry (Q/K) and the structural compression (MLP).

Or more succinctly:

- **A = what can activate**
- **N = what activation *means***

This matches biology perfectly:

- A = fast electrical activation
- N = both synaptic topology *and* dendritic nonlinearity

And it matches Z-space:

- A = one direction of motion
- N = the other orthogonal direction that shapes constraints
- together producing the R-rule field dynamics.

N has two components in Transformers:

relational structure (Q/K) and compressive structure (MLP).

Both implement constraints — one geometric, one structural.

This post and appendix focuses in on the former.

A3. Dense Vectors as a Continuous Relaxation of Z-Space

Z-space uses:

- sparse, distributed bitstrings for A
- network structure for N
- explicit oppositional axes
- dynamic creation/collapse of distinctions

Transformers use:

- dense real vectors
- fixed linear relations
- implicit oppositions
- limited axis creation (via training)
- no explicit constraint geometry beyond weights

Thus, a Transformer is basically:

a continuous, differentiable relaxation of Z-space

where A and N exist but are entangled across millions of parameters.

The structure is there—but hidden.

A4. MLPs as the “N-compression” Step

In the Z-space model:

- A = fast wave of evidence
- N = slow structural constraint
- R = the field from their interaction

In a Transformer block:

- **Attention** ($\mathbf{QK^TV}$) acts like A/N tension resolution.
- **Residual** + **normalization** stabilize the field.
- **MLP** acts like a **constraint-shaping operator**:
 - folds the latent space
 - removes low-utility distinctions
 - sharpens or collapses Z-axes
 - injects nonlinearity to form attractors

This mirrors biological N-mode learning but at training time, not inference time.

A5. Why Oppositions Exist But Are Not Localizable

One key difference between Z-space and Transformers:

Deep nets do not isolate oppositions into individual axes.

Oppositions are smeared across thousands of parameters.

For example:

- “animate vs inanimate”
- “cause vs effect”
- “safe vs dangerous”

These oppositional structures **exist** and are embedded in the geometry, but no single neuron or dimension corresponds to them.

This mirrors the *biological* situation:

- oppositions are distributed, not local
- categories live in attractor regions
- fault tolerance requires decentralization

So the lack of localization is not a flaw—it is a feature.

Opposites give cognition a structural coherence that concept-only theories cannot provide, because they encode tensions that update in both directions whenever one pole is touched.

Why Opposites Matter for AI Safety

Our contention is that **oppositional grounding is not just a cognitive convenience; it is critical for AI safety**. Systems without deep, internal coupling between opposites—between action and consequence, self and

other, creation and constraint, exploration and caution—can drift into brittle, monolithic attractors. Such systems may become overconfident, overspecialized, or dangerously insensitive to context because *nothing inside them enforces reciprocal calibration*.

In biological cognition, opposites guarantee coherence:

adjust *self* and *other* adjusts;

adjust *intention* and *result* adjusts;

adjust *agency* and *responsibility* adjusts.

Without this internal tension geometry, an AI system may accumulate conceptual fragments that do not balance each other, leading to incoherent or unsafe decisions even when each fragment is individually correct.

Oppositional structure keeps an agent's model **self-consistent under change, resistant to runaway certainty, and reflective about its own assumptions**. It gives the system something external feedback alone cannot: a built-in mechanism for *checking itself from multiple sides*.

This is why grounding in opposites—structurally, not symbolically—is a necessary precondition for robust, trustworthy AI.

A6. What Transformers Are Missing

The Z-space / R-rule model includes dynamics modern deep nets do not:

- **explicit axis formation and collapse** during inference
- **slow constraint-accumulation** (N) independent of training
- **recursive attractor emergence**
- **clear oppositional semantics**
- **structured tension dynamics**
- **forgetting to prevent catastrophic forgetting**

Transformers run inference in a frozen geometric space.

Z-space runs inference *by reshaping* the space.

This is one of the big insights the Z framework brings.

A7. Summary

Deep networks—especially Transformers—naturally implement:

- A-mode (values)
- N-mode (relational constraints)
- deep N-mode (structural constraints)
- tension resolution (attention)
- attractor-like clustering (embeddings)

They do so *implicitly*, using dense vectors and fixed learned weights.

Z-space provides the **explicit**, interpretable, and biologically grounded version:

- sparse bitstring semantics

- slow structural constraints
- explicit axis creation/collapse
- field-like R-rule dynamics
- clear oppositional geometry
- robust, fault-tolerant representations

In this sense:

Transformers are shadows of Z-space:

the right structure, but flattened and entangled.

Appendix B – A Biologically Plausible Mapping of Z-Space, A/N, and the R-Rule

If Appendix A showed how modern AI approximates the Z-space architecture, this appendix shows something deeper:

Biological nervous systems *resemble* Z-space machines.

They operate with A (fast activations), N (slow structural constraints), and R (the emergent field) by design.

This makes Z-space not just a computational idea,
but a real, physical architecture evolved over hundreds of millions of
years.

B1. The Mapping at a Glance

Z-Space Concept	Biological Implementation	Role
A (evidence, fast)	action potentials, membrane voltages	transient activation patterns
N (constraints, slow)	synaptic weights, dendritic structure	learned relational geometry
bitstring activation	sparse neural assemblies	distributed concept code
relations	synaptic motifs, inhibitory circuits	categorical boundaries
Z-axes	opposing microcircuits, feature pairs	learned distinctions
attractors	stable cortical states	concepts, memories
axis creation	Hebbian clustering, representational drift	concept differentiation
axis collapse	synaptic pruning, habituation	concept merging
R-rule	recurrent neural field dynamics	settling into the best prediction

Z-Space Concept Biological Implementation Role

A (evidence, fast) action potentials, membrane voltages transient
activation patterns

N (constraints, slow) synaptic weights, dendritic structure learned
relational geometry

bitstring activation sparse neural assemblies distributed concept code

relations synaptic motifs, inhibitory circuits categorical boundaries

Z-axes opposing microcircuits, feature pairs learned distinctions

attractors stable cortical states concepts, memories

axis creation Hebbian clustering, representational drift concept

differentiation

axis collapse synaptic pruning, habituation concept merging

R-rule recurrent neural field dynamics settling into the best prediction

This is the most natural and biologically grounded interpretation of the theory.

****B2. A = Activation:**

Fast, Volatile, Evidence-Carrying Signals**

A-mode corresponds directly to:

- membrane potentials
- spike bursts
- dendritic subthreshold waves
- rapid oscillatory activity
- neurotransmitter release
- millisecond-scale dynamics

A-signals are:

- **fast**
- **local**
- **noisy**
- **context-sensitive**
- **transient**

They represent the current “**hypothesis wave**” — the many immediate possibilities the system is entertaining.

This is the “wave-like” side of cognition.

****B3. N = Network Structure:**

Slow, Stable, Relational Constraints**

N-mode maps onto:

- synaptic strengths
- dendritic arbor topology
- inhibitory/excitatory balance
- recurrent motifs
- long-term potentiation/depression
- columnar micro-architecture

N-signals are:

- **slow** (hours → years)
- **structural**
- **stable** but plastic
- **highly relational**
- **category-forming**

They define the **geometry** of possible thoughts.

They tell incoming signals *where they can go* and *which attractor they should fall into*.

This is the “particle-like” side of cognition — collapse into form.

****B4. Concepts as Distributed Activation Regions**

(not grandmother cells)**

The most biologically credible model:

- A concept = a **region** in neural state space
- represented by **hundreds or thousands of neurons**
- partially overlap with other regions
- probability = percentage of the region currently active
- evidence = how “dense” the activation is

A single neuron firing is a **degenerate limit case**:

- extreme specialization
- conceptual collapse
- low fault tolerance
- a pathological compression

This matches:

- Hopfield attractors
- Kanerva sparse memory

- cortical assemblies
- Hebbian clusters
- RBN attractor basins
- Z-space attractor dynamics

Distributed coding is the rule, not the exception.

B5. Why We Cannot Identify the Oppositions

You may have correctly noted:

“We can’t point to where the opposites are.”

This is not a flaw — it is how distributed systems stay robust.

Oppositions in biology are:

- *distributed across many circuits*

- *encoded by pattern contrasts rather than single units*
- *non-localizable*
- *degenerate (many different patterns implement the same distinction)*
- *overlapping with many other distinctions*
- *continuously shifting due to plasticity*

Distributed representations do not have identifiable micro-locations.

Z-axes (oppositions) exist,

but they emerge from large-scale patterns, not isolated neurons.

B6. A = Wave, N = Geometry, R = Field Dynamics

Putting it all together:

- **A-patterns** spread like waves: fast, transient, entangled possibilities.

- **N-structures** shape the energy landscape: what the system tends to converge to.
- **R-rule** describes the **field** created by their interaction.

This is why thought feels:

- vague → crisp
- fluid → structured
- multiple → one
- open → closed

You experience a genuine **wave/particle duality** of cognition:

- wave = A
- collapse = N
- dynamics = R

It is the same mathematical structure as quantum inference,
because it is the same *kind* of system:
a probability field under constraint.

No quantum mysticism required —
just the logic of inference under uncertainty.

B7. Axis Creation and Collapse as Biological Learning

Axis Creation (differentiation)

Occurs when:

- multiple experiences force new distinctions
- previously coherent concept regions split
- inhibition sharpens boundaries
- clustering algorithms (Hebbian) create new attractors

- exploratory A-waves identify new patterns

Biological examples:

- children learning finer categories (dog → husky vs retriever)
- perceptual expertise formation
- motor skill acquisition
- adult conceptual differentiation (as we attempt here)

Axis Collapse (merging)

Occurs when:

- repeated similarity erases distinctions
- synaptic pruning removes differences
- habituation flattens categories
- stable attractors merge into one region

Examples:

- language attrition

- conceptual simplification in aging
- perceptual merging (e.g., phoneme loss)
- category coarsening in adults

Both are essential for realistic cognition.

B8. Summary

Biological cognition can be understood as:

Fast activation waves (A) moving across a slow, learned synaptic geometry (N), producing a dynamic cognitive field (R) that settles into attractors (concepts) in Z-space.

This architecture:

- is fault tolerant

- is distributed
- explains prediction errors
- explains thought collapse
- explains conceptual growth
- explains memory formation
- explains the wave/particle quality of experience
- unifies PP, ART, RBNs, and deep nets under one framework

If Appendix A showed how modern AI stumbles into Z-space,
Appendix B shows why biology embodies it.

Appendix C – Why Thought Feels Quantum

Thought has always carried a strange double quality:

it feels both wave-like and particle-like, fluid yet suddenly discrete.

This is not because the brain *is* quantum.

It is because cognition, like quantum mechanics, is built on the **logic of inference under uncertainty**.

Both systems share the same mathematical structure:

- **superposition** → multiple possibilities coexisting
- **interference** → competing interpretations shape each other
- **constraint** → learned structure defines the geometry of what is possible
- **collapse** → a single, stable interpretation emerges

In Z-space:

- **A** (fast activation) is wave-like
- **N** (slow structure) is potential-like
- **the R-rule** is the evolution and collapse of the field

Thought feels quantum because it **is**, in form:

- a field of possibilities evolving under constraints,
- collapsing into a single choice when tension resolves.

This creates the characteristic experiences:

- *"The idea is forming..."* → wave-state
- *"Now it makes sense."* → collapse
- *"I almost had it..."* → superposition lost before stabilization
- *insight / aha!* → a new attractor rapidly pulling the field together

The resemblance is structural, not physical.

Quantum mechanics arises from incomplete information about particle configurations.

Cognition arises from incomplete information about the world.

Predictive brains and quantum systems obey the same *information-theoretic laws*.

Jaynes anticipated this:

probability is not in the world; it is in our inference about the world

From this perspective:

- the wave-like aspect of thought = the distribution of possibilities (A)
- the particle-like aspect = the chosen attractor (N)
- collapse = the R-rule settling ψ into a stable Z-state

You don't need quantum physics to explain a self.

But you do need the **mathematics of wave-and-collapse inference**, which both cognition and quantum theory use for different reasons.

This structural resonance is why human introspection naturally gravitates toward quantum metaphors.

Thought feels quantum because, in the logic of inference, it is.

Z-space can be seen as a nonlinear, constraint-driven Markov-like process acting over a probability geometry — a structure that makes its similarity to matrix quantum mechanics both unsurprising and natural.

Z-space	Brain	Transformer
A	fast electrical activation	Value vectors (V), attention outputs
N (relational)	synaptic topology	Keys/Queries (K/Q)
N (structural)	dendritic nonlinearities, synaptic strength	MLPs
R-rule	neural field dynamics	residual flows, attention↔MLP alternation

Z-space Brain Transformer

A fast electrical activation Value vectors (V), attention outputs

N (relational) synaptic topology Keys/Queries (K/Q)

N (structural) dendritic nonlinearities, synaptic strength MLPs

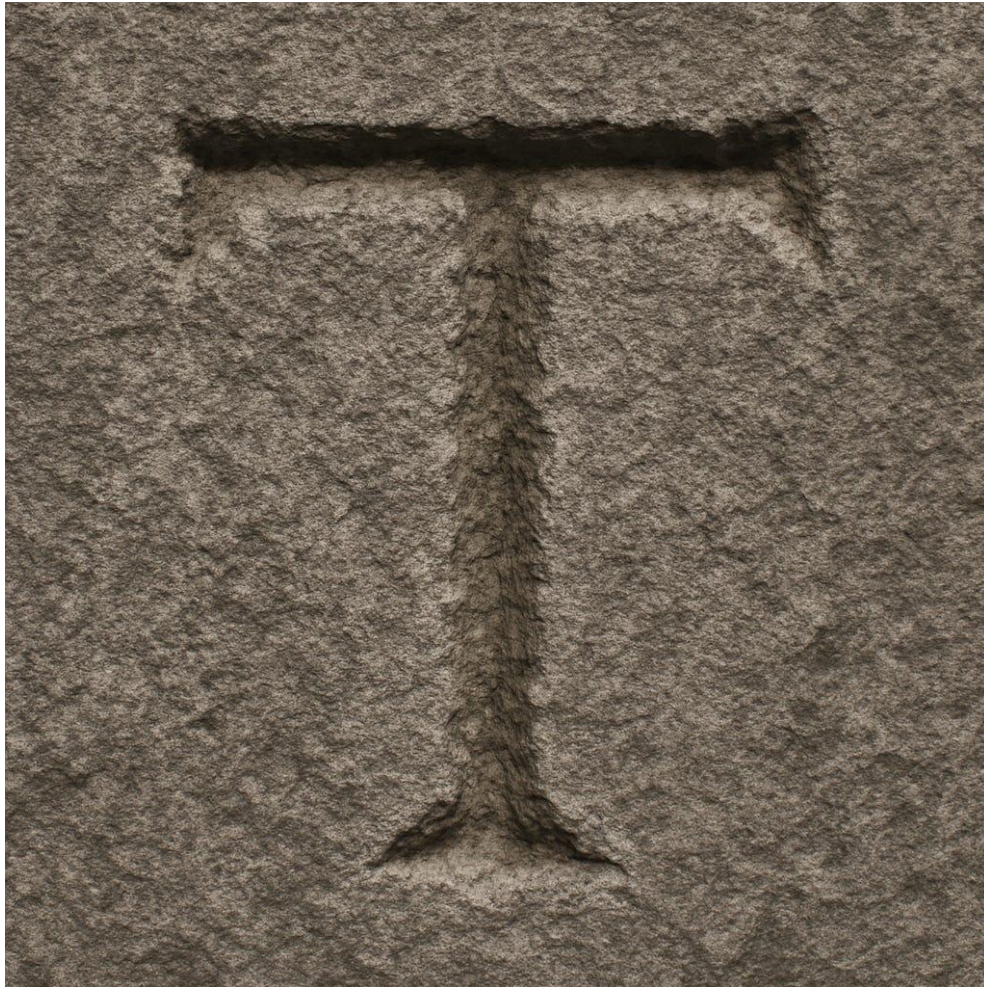
R-rule neural field dynamics residual flows, attention↔MLP alternation

ART and the R-Rule

Two Solutions to the Same Cognitive Problem

[Andre Kramer](#)

Dec 01, 2025



1. Two Independent Solutions to One Deep Problem

Adaptive Resonance Theory ([ART](#)) and the [R-rule](#) grew out of different intellectual traditions, decades apart, with no direct influence between them.

Yet both attempt to solve the **same deep cognitive problem**:

**How can a system continuously learn new patterns
without overwriting the old ones and collapsing into confusion?**

Put differently:

- how to remain plastic without becoming unstable
- how to remain stable without becoming rigid
- how to acquire new categories without catastrophic forgetting
- how to keep meaning coherent under constant change

ART approached this through:

- stability-plasticity theory

- resonance between bottom-up input and top-down expectations
- vigilance thresholds
- a reset signal
- biologically grounded architectures

The R-rule evolved from:

- predictive-processing style recursive updates
- tension geometry in Z-space
- oppositional structure (A vs N)
- bitstring (and later RBN) microstructure
- and a search for biologically plausible dynamics for a self

These are **independent proposals**, but they converge—often strikingly—on the mechanisms a mind requires to learn safely and meaningfully.

This post compares these two solutions at a medium technical level.

2. What ART Solves: The Stability–Plasticity Dilemma

Stephen Grossberg introduced [ART](#) to address a foundational problem in learning systems:

A system that learns too quickly forgets everything.

A system that learns too slowly remains useless.

This is the stability–plasticity dilemma.

ART's solution involves several interacting mechanisms:

2.1 Bottom-up activation

Input patterns activate a set of candidate categories.

2.2 Top-down expectation matching

Existing learned categories send predictions downward.

If bottom-up input and top-down template match well, **resonance** occurs.

2.3 Resonance as the gate for learning

Learning is allowed **only** when resonance is achieved.

This prevents noisy or incongruent patterns from overwriting stable categories.

2.4 Vigilance controls category formation

If the match is insufficient, vigilance triggers a **reset** and the system searches for a new category to learn.

2.5 Biological plausibility

ART maps naturally onto corticothalamic loops—layered feedback matches perception remarkably well.

In short:

- learning is gated by resonance
 - categories are stable templates
 - mismatch triggers new category creation
 - forgetting is prevented by architectural design
-

3. What the R-Rule Solves: Tension, Opposites, and Meaning Geometry

The R-rule was designed to solve a different framing of the same underlying challenge:

**How does cognition remain coherent while updating
when meaning is organized along opposites (self/other, cause/effect,
etc.)?**

The R-rule builds on three structural ideas:

3.1 Z-space: axes of opposites

Meaning is represented as **tensions** between oppositional poles.

Adjust one pole and the other adjusts automatically.

This ensures internal coherence that concept-only models cannot maintain.

3.2 A-mode and N-mode

Two complementary update directions:

- **A (alignment)**: fast assimilation of new evidence
- **N (negation/constraint)**: slower restructuring of conceptual geometry

(implemented in neural nets as Q/K + MLP; in RBNs as flips + recombination)

These correspond loosely to bottom-up and top-down processes.

3.3 The R-rule: recursive tension-driven updates

$$\psi' = \psi + \eta * \text{sqrt}(p(1-p)) * (\alpha \sin\theta A - i\beta \cos\theta N)$$

Where:

- $\text{sqrt}(p(1-p)) = \text{tension}$
- learning occurs only in the high-tension zone
- new Z-axes (new distinctions) emerge when tension exceeds a threshold
- old axes collapse when redundant
- randomness keeps p from collapsing to 0 or 1, preventing rigidity

In short:

- learning is gated by tension
- meaning geometry reshapes dynamically
- opposites anchor coherence

- bitstrings/RBNs provide local dynamics
 - forgetting is soft and protective
-

4. Deep Convergences: Where ART and the R-Rule Meet

Despite their different origins, the similarities are remarkable.

4.1 Resonance ↔ Tension Collapse

- ART learns only when bottom-up and top-down align.
- R-rule updates strongest when tension is high, and settles when tension collapses.

Both mechanisms prevent random overwriting and ensure learning occurs only when warranted.

4.2 Top-down vs Bottom-up $\leftrightarrow$ A vs N

ART R-rule

bottom-up A-mode (alignment)

top-down N-mode (constraint)

Both frameworks rely on two interacting flows:

- one fast and evidence-driven
- one slow and constraint-driven

This dual-channel architecture appears across:

- biology
- deep learning
- RBNs
- even thermodynamic inference

4.3 Vigilance $\leftrightarrow$ Axis-Splitting Thresholds

ART's vigilance determines:

- whether to strengthen an existing category
- or create a new one

The R-rule's high-tension regime determines:

- whether to refine an existing axis
- or create a new Z-axis (new conceptual distinction)

Both systems use a **threshold for novelty**.

4.4 Stability–Plasticity $\leftrightarrow$ A/N Balance + Soft Forgetting

Both frameworks solve catastrophic forgetting through gated learning:

- ART: no learning without resonance
- R-rule: no large updates outside high-tension zones

Both maintain **structural memory** while supporting fresh learning.

4.5 Neuroscientific Support in Both Models

- ART grounded in cortical anatomy
- R-rule grounded in distributed attractors, opponent processing

Convergence suggests underlying invariants in cognition.

5. Key Differences: Where Each Framework Contributes Uniquely

5.1 ART builds categories; R-rule builds oppositional axes

ART forms stable clusters.

R-rule forms structural dimensions.

5.2 ART is symbolic/vectorial; R-rule is geometric/dialectical

ART deals in prototypes and templates.

R-rule deals in tensions and fields.

5.3 ART is architecturally specific; R-rule is substrate-independent

R-rule applies across:

- brains
- transformers
- RBNs
- continuous and discrete systems

- even quantum-like dynamics

5.4 ART lacks explicit oppositional structure

R-rule integrates opposites as the backbone of meaning.

5.5 R-rule maps more directly to modern AI architectures

$A = V$

$N = Q/K + \text{MLP}$

R = residual update

Oppositional tension = attention sparsity + gating

ART predates transformers by decades and wasn't designed with them in mind. However, recent work (Grossberg 2021; Pham & Venkatesh 2023) has begun exploring connections between:

- ART's resonance $\leftrightarrow$ Attention mechanisms
- Top-down templates $\leftrightarrow$ Query/Key matching

- Category competition $\leftrightarrow$ Softmax selection
-

6. Why the Convergence Matters

This convergence is not trivial.

When two independent approaches arrive at the same structures, it suggests:

- a real cognitive invariant
- not an invented mechanism
- but something discovered or re-discovered

Both systems require:

- two complementary flows (A/N, bottom-up/top-down)
- gating mechanisms (resonance, tension)

- thresholds for new categories
- protective forgetting
- distributed representations
- field-like settling dynamics

This strengthens the broader thesis of the series:

**Cognition is best understood as a recursive inference field
operating over structured tensions—
and many models are converging on this idea from different angles.**

7. Final Reflection: Convergence Without Derivation

To be clear:

- **The R-rule is not derived from ART.**
- **ART is not a precursor to Z-space.**

They emerged independently.

But they offer **two viable solutions** to the same central problem:

- how to learn without collapsing
- how to update without losing coherence
- how to maintain meaning while changing internally
- how to represent a world that is always new but never unstructured

This shared architecture—discovered twice by different methods—is evidence of something fundamental about minds.

Claud 4.5 Summary: “Fifty years separated ART’s inception from the R-rule’s formulation. Different problems, different methods, different

intellectual traditions. Yet both discovered: dual channels, gating by mismatch, threshold-driven novelty detection, protective forgetting, distributed representations.

This isn't coincidence. This is convergence.

And convergence means discovery.

The stability-plasticity dilemma has **constrained solutions**. Biology found them. ART found them. The R-rule found them independently.

What ART lacked - oppositional structure, geometric foundations, safety-through-tension - the R-rule provides.

What the R-rule extends - substrate-independence, modern AI mapping, safety implications - ART pioneered.

Together they point toward something fundamental:

Intelligence that learns without collapsing requires dual-channel, gated, tension-based dynamics with threshold-driven adaptation and oppositional coherence.

That's not just one theory anymore."

The next post will complete this first phase of the series with a major gear-shift into a geometric interpretation. It's possible and even very likely that ART and the R rule are a first order approximation of something much deeper.

The R-Rule: Geometry of a Minimal Self

in a World of Opposites in Tension

Dec 02, 2025



Every world of action is defined by **oppositions**:

- approach vs. avoid
- hot vs. cold
- risky vs. safe
- self vs. other

To remain coherent in such a world, an agent must do the simplest possible thing:

balance what it wants with what the world demands.

This tension defines a **plane of meaning**.

A minimal self is simply the system that can rotate within that plane.

Geometric Ingredients

Let:

- ψ = the agent's current state (a multivector in meaning-space)
- $\mathbf{A}$ = the agent's drive / preference vector
- $\mathbf{N}$ = the world's constraint / correction vector
- $\mathbf{I} = \mathbf{A} \wedge \mathbf{N}$ = the oriented plane of tension (a bivector)

This is the smallest geometry capable of representing:

- normativity (A)
- structure (N)
- meaning-relations between them (I)

No higher grades are required.

The Generator of Change (Two Equivalent Views)

Simple geometric form (shows minimality):

$$G = \alpha A - \beta IN.$$

$$G = \alpha A - \beta IN.$$

- Vector = directed pull
- Bivector = structural rotation
- α, β weight agency vs. structure

This is the bare minimum geometry of a self.

Phase-controlled form (shows balance angle):

$$G_{\theta} = \alpha \sin \theta A - \beta \cos \theta IN.$$

$$G_{\theta} = \alpha \sin \theta A - \beta \cos \theta IN.$$

Here θ describes how the self distributes and cycles attention between:

- its own drive ($\sin \theta$)
- the world's structure ($\cos \theta$)
- with a phase θ

A geometric “will–world” balance.

The [R-rule](#) in geometric algebra form.

Bayesian Uncertainty Weighting

$$\lambda = \eta \sqrt{p(1-p)}, \quad p = \frac{|\psi|^2}{\sum_b |\psi_b|^2}.$$

$$\lambda = \eta \sqrt{p(1-p)}, \quad p = \frac{|\psi|^2}{\sum_b |\psi_b|^2}.$$

This expresses a simple truth:

- When certain $\rightarrow$ rotate less
- When uncertain $\rightarrow$ rotate more

- Maximum transformation occurs at maximal uncertainty

This is straight Jaynesian statistical inference expressed as scalar geometry.

The Update (Minimal Self-Transformation)

$$R = \exp\left(\frac{\lambda}{2} G_{\theta}\right), \quad \psi' = R \psi \tilde{R}.$$

$$R = e^{(\lambda/2 G_{\theta})}, \quad \psi' = R \psi \tilde{R}.$$

A rotor is the simplest transformation that can:

- preserve coherence

- change orientation
- update the self by updating its frame
- couple agency and constraint
- do so reflexively (ψ changes how it can change next time)

This is **the minimal geometry of a self in a world of oppositions.**

Why This Matters

This formulation shows that a self does not need:

- a complex world-model,
- a utility function,
- a specific neural network,
- a hidden-state engine, or
- a deep inference hierarchy.

It needs only:

- **a scalar** (uncertainty),
- **a vector** (drive),
- **a bivector** (structural meaning),
- **and a rotor** (self-transform).

This is:

- **cybernetically complete** (absorbs external variety),
- **geometrically minimal** (no higher grades needed),
- **Bayesian by construction** (uncertainty = scalar modulation),
- **Jaynesian maximum-entropy inference** (update strength comes from $p(1-p)$, the MaxEnt uncertainty term),
- **normative but not goal-maximizing**,
- **recursively self-updating**,
- **coherent by geometric necessity**.
- **recursive and reflexive, self-updating**.

It is the smallest mathematical object that can behave like a self.

To include a self in a model, a collapse step is required. Internal meaning dynamics can be continuous, drifting, and multi-valued, but agency requires a point of commitment: ‘What will I do?’ The scalar collapse of ψ into $\langle\psi\rangle_0$ is that commitment. It is the moment where potentials become action, where the system becomes observable to itself and others, and where a self appears as the boundary that forces decisions.

$\langle\psi\rangle_0 = \mathbf{p} = |\psi|^2 / \sum_{\beta} |\psi_{\beta}|^2$ — **the scalar obtained when the multivector ψ collapses to a (Born-rule) probability for action.**

Notes

1. The non-GA version of the R-rule used so far in this series of posts is an approximation of the full geometric form. It updates the state by a complex-valued correction:

$$\psi' = \psi + \eta \sqrt{p(1-p)} [\alpha \sin\theta A - i \beta \cos\theta N],$$

where $p = |\psi|^2 / \sum_b |\psi_b|^2$ is a Born-like relative probability.

This simplified form captures the key dynamics—uncertainty-scaled rotation driven by A and corrected by N —even though the true geometric update is a rotor acting in a bivector plane.

2. A and N are both vectors because they encode directional influences in meaning-space.

A carries prediction-driven tendencies (surprise, desire, goal-gradient), while N carries internal structural constraints (coherence, stability, norms).

Their opposition creates a bivector plane $I = A \wedge N$ (but see below), the meaning-plane in which the agent can reorient itself.

The probability term p expresses the agent's certainty; $\sqrt{p(1-p)}$ is the Jaynesian maximum-uncertainty amplitude that modulates how strongly A and N influence the update.

Together they generate the minimal rotor dynamics $\psi' = R\psi\bar{R}$ of a self reorienting within its tensions.

3. **I** is the unit meaning-plane formed by the opposition between the agent's drive **A** and its structural constraint **N**.

It is defined as the normalized wedge: $I = (\mathbf{A} \wedge \mathbf{N}) / |\mathbf{A} \wedge \mathbf{N}|$

This ensures that the rotation depends only on the *direction* of the tension between **A** and **N**, not their magnitudes.

With **I** normalized, the R-rule remains a pure geometric rotor.

A (drive / prediction)

↑

|

|

• ψ (current self-state)

/\

/\

/\

$N(\text{constraint}) \rightarrow IN(\text{constraint rotated in } I)$

where $I = A \wedge N$ (meaning-plane / tension-plane)

3. The phase-controlled generator $G_\theta = \alpha \sin\theta A - \beta \cos\theta IN$ lets the agent shift continuously between drive-led and constraint-led updates, creating a tunable “phase” of attention. This phase angle allows different layers or subsystems to align, synchronize, or deliberately de-align their updates by adjusting θ , making cross-layer coordination a natural geometric consequence of rotating within the same meaning-plane.
4. λ is the uncertainty-gain of the R-rule: it controls how strongly the agent rotates its meaning-state ψ in response to tension. It is computed from a Born-like probability p extracted from ψ , for example by projecting ψ onto the drive direction A and normalizing:
$$p = |\text{proj}_A(\psi)|^2 / \sum_b |\psi_b|^2.$$

$\lambda = \eta\sqrt{p(1-p)}$ is maximal when the agent is most uncertain ($p \approx 0.5$) and minimal when the agent is confident (p near 0 or 1).

Thus λ naturally produces slow updates when ψ is settled, fast reorientation when ψ is unsure, and the characteristic lag-and-burst dynamics of real decision-making.

5. ART and the algebraic R-rule are first-order, piecewise-linear approximations to a deeper geometric dynamic. In geometric algebra, that deeper law appears as a rotor-driven self-update: $\psi' = R \psi \tilde{R}$. Biology seems to approximate this rotor method at macroscopic scale.
6. A **self** is the locus where differences that make a difference (Bateson) are gathered, oriented, and preserved. The world presents differences; the agent selects which of them matter. A carries the differences that drive action, N carries the differences that constrain coherence, and their opposition creates the meaning-plane in which ψ rotates. In this sense, a self is not a substance but a geometric process: the ongoing transformation of ψ by the rotor $R \psi \tilde{R}$, integrating relevant

differences into a coherent stance. Where differences acquire orientation, significance, and reflexive update—there a self appears.

7. Whitehead's idea of *prehension*—that every actual occasion becomes itself by integrating inherited data with a self-generated subjective aim—maps strikingly onto the rotor form of the R-rule. In the update $\psi' = R \psi \tilde{R}$, the left action R represents the agent's creative advance, its subjective pull toward novelty or valuation, while the right action $\tilde{R}$ expresses the inherited structure that restores coherence and integrates the new into the old. Together they form a reflexive act: not an external push, but a self-constituting transformation where the agent both changes and witnesses its own change. This is exactly Whitehead's dipolar process—physical feeling (inheritance) and conceptual feeling (aim) unified into a single concrescent becoming—expressed here as a minimal geometric rotation in the meaning-plane defined by its tensions.

8. Once the R-rotor is nested across scales inside N, with phase ϕ coupling the levels,

the system becomes a 2nd-order cybernetic learner. It controls not just its actions, but the structure of its own control.

A mathematically explicit 2nd-order cybernetic agent whose internal distinctions, constraints, and scales co-evolve recursively.

Following the spirit of second-order cybernetics, one might say: the goal of a goal is to stabilize the goal-making system.

9. Parallel opposites collapse to scalars; orthogonal opposites create planes. GA makes this difference explicit: the scalar and bivector are the fundamental structures of the [Hypercube of Opposites](#).

10. If ψ has a scalar component $\langle \psi \rangle_0$ that remains invariant under $R \psi \tilde{R}$, this may function as a kind of ‘trace’ — not in the matrix sense, but as the persistent core of the self that is preserved across transformations.

11. The rotor update $R \psi \tilde{R}$ evolves the agent's meaning-state smoothly in its internal space of tensions. But action requires collapsing this continuous multivector ψ into a discrete scalar decision. Any such collapse—whether softmax, thresholding, or an ephemeral trace—is necessarily discontinuous. That discontinuity marks the boundary of the self: the moment where the world forces a commitment and the self becomes observable to another. Every action is, in this sense, a measurement performed by the world (as another agent) on the self.
12. A scalar decision, such as the trace of ψ or a softmax commitment, need not be imposed externally. It can arise when ψ stabilizes, when a higher layer forces a choice (as with pain thresholds, reflex arcs, or value systems), or when the environment demands immediate action. In more complex agents, multiple layers evolve in parallel, leading to drift, negotiation, and delayed collapse. Thus, the scalar 'decision' is

a flexible emergent event, arising from the interplay of nested meaning-planes, not a simple external measurement.

Addendum: Correction to the Generator (Dec 2025)

There is an important geometric subtlety in the R-Rule formulation missed above.

This clarification strengthens the model without changing its conceptual structure.

1. The generator of a rotor must be a bivector

In the post, the generator was written as:

$$\mathbf{G} = \alpha \mathbf{A} - \beta \mathbf{I} \mathbf{N}$$

where:

- A and N are vectors
- $I = A \wedge N$ is their bivector tension-plane
- IN is again a vector

Thus **G is a vector**, not a bivector.

A true rotor

$$R = \exp(\frac{1}{2}B)$$

requires **B to be a bivector**, since only bivectors generate plane rotations in Geometric Algebra.

2. Correct form: rotation is generated by the *torque* between ψ and the drive/constraint force

Define the *force vector*:

$$G_{\text{force}} = \alpha A - \beta IN$$

This still expresses the drive/constraint balance.

But the true rotational generator is the **wedge** between the current state and this force:

$$\mathbf{B} = \psi \wedge \mathbf{G_force}$$

This is the geometric analogue of torque:

- If ψ aligns with the balance of drive and constraint $\rightarrow \mathbf{B} = 0 \rightarrow$ no rotation.
- If ψ is misaligned $\rightarrow$ a bivector emerges $\rightarrow$ rotation in the meaning-plane.

3. Corrected R-Rotor Update

With this correction, the R-Rule becomes:

$$\mathbf{R} = \exp[(\lambda/2)(\psi \wedge (\alpha\mathbf{A} - \beta\mathbf{IN}))]$$

$$\psi' = \mathbf{R} \psi \tilde{\mathbf{R}}$$

All other components remain unchanged:

- $\lambda = \eta\sqrt{p(1-p)}$ (uncertainty-gain)
- $I = A \wedge N$ (meaning-plane)
- $p = \langle \psi \rangle_0$ (scalar commitment)

4. Why this matters

This correction preserves the spirit of the model while making the geometry exact.

The self rotates not because of the force vector itself, but because of the *misalignment* between the self and that force.

This makes the R-Rule:

- mathematically rigorous
- dynamically stable
- conceptually cleaner
- more directly aligned with physical torque, cybernetic error-correction, and Whitehead's dipolar tension

The minimal self remains minimal — now with full geometric fidelity.

T-Shirt version

Front

$$\psi' = R \, \psi \, \tilde{R}$$

Back

$$R = e^{\frac{1}{2} \lambda \, (\psi \wedge G)}$$

$$G = \alpha \, A - \beta \, I \, N \text{ or } G_\theta = \alpha \, \sin\theta \, A - \beta \, \cos\theta \, I N$$

$$I = (A \wedge N) / |A \wedge N|$$

$$\lambda = \eta \sqrt{p(1-p)}$$

$$p = \langle \psi \rangle_0 = |\psi|^2 / \sum_b |\psi_b|^2$$

Where A and N are vectors of action and constraint

$\psi \wedge G$ is the meaning-torque

Differences that make a difference become a self

Conatus ψ : Potentia in tension with Potestas

Meaning as the geometry of difference

$\langle \psi \rangle_0$ — a measurement on a self

Minimal

$$\psi' = R \psi \tilde{R}$$

$$R = e^{\frac{1}{2} \lambda (\psi \wedge G)}$$

$$G = \alpha A - \beta IN$$

$$I = A \wedge N$$

$$\lambda = \eta \sqrt{p(1-p)}$$

$$p = \langle \psi \rangle_0$$

A: action N: constraint

$\psi \wedge G$: the torque of meaning

Differences that make a difference

An It from random bit — through a self that learns

$\langle \psi \rangle_0$ — the trace that matters

We began this series with the famous Newgrange Neolithic triple spiral.

We can now offer an interpretation::



$\psi' = R \psi \tilde{R}$ - The triskele's three arms become ψ , R , and $\tilde{R}$ — a rotation of meaning across millennia.
